

Instance Optimal Sparse Recovery from Nonlinear Observations: A Unified Framework

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September 2, 2026

Abstract

Sparse recovery has found numerous applications. As most real-world signals and parameters are only approximately sparse, a fundamental property for sparse recovery algorithms is the robustness to model error. This is characterized by the instance optimality in compressed sensing, which concerns sparse recovery from *linear* measurements. While restricted isometry property (RIP) yields instance optimality in compressed sensing, it remains much less clear how to achieve instance optimal sparse recovery from *nonlinear* observations. This paper develops a unified framework to this end. The main ingredient is a signal-dependent restricted approximate invertibility condition (RAIC) of some “gradient”, which leads to the instance optimality of iterative hard thresholding. Under Gaussian designs, we apply the proposed framework to phaseless, one-bit, and ReLU measurements, which correspond to the problems of sparse phase retrieval, one-bit compressed sensing, and sparse ReLU regression, respectively. For sparse phase retrieval, we propose a variant of thresholded amplitude flow and show its instance optimality under $O(s^3)$ measurements (up to logarithmic factors), where s is the sparsity level. To our best knowledge, this is the first instance optimal efficient algorithm for sparse phase retrieval and complements [33] that achieved this via a computationally intractable program. In one-bit compressed sensing, we establish the instance optimality of normalized binary iterative hard thresholding and strengthen the recent result of [47]. In sparse ReLU regression, it is shown that a slight variant of the algorithm in [61] is instance optimal. Moreover, (ℓ_2, ℓ_2) non-uniform instance optimal guarantees are obtained for these problems. The analysis is built upon a number of high-dimensional concentration bounds, including bounds on restricted eigenvalues and a novel instance-dependent hyperplane tessellation result.

Mathematics Subject Classification (2020). Primary 94A12; Secondary 94A20, 60B20, 60D05.

Keywords. instance optimality; compressed sensing; nonlinear observations; iterative hard thresholding; sparse phase retrieval; hyperplane tessellation; oracle inequality

1 Introduction

In compressed sensing that concerns sparse recovery from linear measurements, instance optimality is a basic notion that characterizes the robustness of an algorithm to model error, where model error refers to the error arising when the true signal is not exactly sparse. Let the ℓ_q ($q \geq 1$) norm of $\mathbf{u}$ be $\|\mathbf{u}\|_q = (\sum_i |u_i|^q)^{1/q}$, an estimator $\hat{\mathbf{x}}$ constructed from the sensing matrix and measurements is said to be (ℓ_q, ℓ_p) -instance optimal of order s if [31]

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_q \leq \frac{C}{s^{1/p-1/q}} \|\mathbf{x} - \mathbf{x}_{[s]}\|_p, \quad \forall \mathbf{x} \in \mathbb{R}^n \quad (1.1)$$

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holds for some constant $C > 0$. Throughout this paper, $\mathbf{u}_{[s]}$ denotes the best s -sparse approximation of $\mathbf{u}$; ties are broken by selecting the support with the smallest indices. In a nutshell, an instance optimal algorithm is capable of recovering all signals to an error proportional to a certain distance to $\Sigma_s^n = \{\mathbf{u} \in \mathbb{R}^n : \|\mathbf{u}\|_0 \leq s\}$.

To start, we consider the sparse recovery of an n -dimensional vector $\mathbf{x}$ from the linear measurements $\mathbf{y} = \mathbf{A}\mathbf{x}$ for some measurement matrix $\mathbf{A}$. It is well known that a number of algorithms—such as basis pursuit denoising [10, 64], iterative hard thresholding [5], CoSaMP [51], hard thresholding pursuit [29]—achieve the (ℓ_2, ℓ_1) -instance optimality

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \frac{C}{\sqrt{s}} \|\mathbf{x} - \mathbf{x}_{[s]}\|_1, \quad \forall \mathbf{x} \in \mathbb{R}^n, \quad (1.2)$$

as long as $\mathbf{A}$ satisfies a certain restricted isometry property (RIP). Also, there is a negative result that the (ℓ_2, ℓ_2) -instance optimality

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq C \|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

is not possible unless $m \gtrsim n$ [22, 8]. Nonetheless, the non-uniform (ℓ_2, ℓ_2) -instance optimality, which only concerns the recovery of a fixed $\mathbf{x} \in \mathbb{R}^n$, is possible in high dimensions (i.e., $m \ll n$). It was proved by [69] that, under $O(s \log \frac{en}{s})$ Gaussian linear measurements, basis pursuit denoising achieves

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq C \|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \text{with high probability (w.h.p.)}, \quad (1.3)$$

for any $\mathbf{x} \in \mathbb{R}^n$ oblivious to the sensing matrix. This is also known as instance optimality in probability.

Nonlinear measurements, however, are ubiquitous in signal processing, statistics, and machine learning. The first such example is one-bit quantization, which helps increase the sampling rates in signal processing and overcome the communication cost bottleneck in distributed machine learning. In the setting of sparse recovery, this leads to the extensively studied problem of one-bit compressed sensing [7]. Another prominent example is that of phase retrieval [60], inspired by many imaging problems where the sensors unavoidably lose the phase information and capture the magnitude only. In high dimensions, it is standard to leverage the sparsity of the signals, which then inspires the problem of sparse phase retrieval [38]. Further examples, to name just a few, include ReLU regression [61], modulo measurements [3], saturated measurements [30], phase-only measurements [52], one-bit phaseless measurements [17], and more general unknown link functions [58].

Since most real-world signals and parameters are only approximately sparse rather than exactly sparse [45], we ask in this paper the following question:

How to achieve instance optimal sparse recovery from nonlinear observations?

Observe that (1.1) from compressed sensing exhibits the two defining features: (i) a model error term that vanishes for $\mathbf{x} \in \Sigma_s^n$, and (ii) the uniformity for all signals in $\mathbb{R}^n$. Yet, defining instance optimality in general nonlinear sparse recovery problems already involves some subtleties. First of all, the entire signal space may not be $\mathbb{R}^n$. For instance, in the recovery of $\mathbf{x}$ from $\mathbf{y} = \text{sign}(\mathbf{A}\mathbf{x})$ where $\text{sign}(a) = \mathbb{1}(a \geq 0) - \mathbb{1}(a < 0)$, i.e., the one-bit sensing problem, the signal norm is completely lost and it is standard to assume the signals live in the unit Euclidean sphere $\mathbb{S}^{n-1}$. In this paper, we shall introduce a generic notation $\mathcal{X}$ to denote the signal space. Perhaps more importantly, while (1.1) guarantees exact recovery for $\mathbf{x} \in \Sigma_s^n$, this is not possible for some nonlinear problems (again like one-bit sensing), where exact recovery is not possible even when the signal is s -sparse. In these problems, the best we can hope for is to achieve the statistically optimal error rate for $\mathbf{x} \in \Sigma_s^n$. We will formalize these discussions in Section 2.

1.1 Main Contributions

We develop a unified framework for proving instance optimality of iterative hard thresholding (IHT) in sparse recovery from nonlinear measurements. Given a sparsity level k and a problem-dependent gradient map $\mathbf{h}_{\mathbf{x}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, IHT refers to the iteration

$$\mathbf{x}_{t+1} = H_k(\mathbf{x}_t - \eta \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)),$$

where $H_k(\mathbf{u}) = \mathbf{u}_{[k]}$ is the hard-thresholding operator. Our approach first decomposes the signal space $\bar{\mathcal{X}}$ into $\mathcal{X}$ and $\mathcal{X}^c = \bar{\mathcal{X}} \setminus \mathcal{X}$, where $\mathcal{X}$ collects signals that are suitably close to Σ_s^n , and particularly contains $\Sigma_s^n \cap \bar{\mathcal{X}}$.

The main challenge lies in establishing the instance optimal recovery for $\mathbf{x} \in \mathcal{X}$. To this end, our key technical ingredient is a structured condition on $\mathbf{h}_{\mathbf{x}}$ —a *signal-dependent* variant of the *restricted approximate invertibility condition* (RAIC)—which directly yields instance optimality over $\mathcal{X}$. In prior works, the RAIC uniformly controls the deviation of $\mathbf{h}_{\mathbf{x}}(\mathbf{u})$ from the ideal invertibility step $\mathbf{u} - \mathbf{x}$ for exactly sparse pairs $(\mathbf{u}, \mathbf{x})$. Our variant extends this control to $\mathbf{x} \in \mathcal{X}$ at the cost of an additional signal-dependent error term. Further details are given in Section 2.2.

As applications, we establish instance optimality of IHT-type algorithms for sparse phase retrieval, one-bit compressed sensing and ReLU regression with Gaussian designs. We highlight two representative results below.

- In sparse phase retrieval (i.e., recovery of $\mathbf{x}$ with a sparse prior from $\mathbf{y} = |\mathbf{A}\mathbf{x}|$), our main result (Theorem 3.1) shows that a variant of thresholded amplitude flow, viewed as IHT with the amplitude-based ℓ_2 loss, produces a sequence $\{\mathbf{x}_t\}_{t=0}^\infty$ satisfying

$$\lim_{t \rightarrow \infty} \text{dist}(\mathbf{x}_t, \mathbf{x}) \leq \frac{C \|\mathbf{x} - \mathbf{x}_{[s]}\|_1}{\sqrt{s}}, \quad \forall \mathbf{x} \in \mathbb{R}^n \quad (1.4)$$

w.h.p. using $\tilde{O}(s^3)$ measurements. Here, $\text{dist}(\mathbf{u}, \mathbf{v}) = \min\{\|\mathbf{u} + \mathbf{v}\|_2, \|\mathbf{u} - \mathbf{v}\|_2\}$ is the standard error metric in phase retrieval.

We also obtain a non-uniform version (Theorem 3.2): for any fixed $\mathbf{x} \in \mathbb{R}^n$, it holds w.h.p. that

$$\lim_{t \rightarrow \infty} \text{dist}(\mathbf{x}_t, \mathbf{x}) \leq C \|\mathbf{x} - \mathbf{x}_{[s]}\|_2. \quad (1.5)$$

- In one-bit compressed sensing (i.e., recovery of $\mathbf{x}$ with a sparse prior from $\mathbf{y} = \text{sign}(\mathbf{A}\mathbf{x})$), our main result (Theorem 4.1) establishes the instance optimality of normalized binary iterative hard thresholding (NBIHT), which is essentially a procedure of normalized IHT with the ReLU loss:

$$\sup_{t \geq 2 \log(m/s)} \|\mathbf{x}_t - \mathbf{x}\|_2 \leq \tilde{O}\left(\frac{s}{m}\right) + O\left(\frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_1}{\sqrt{s}} \log\left(\frac{1}{\|\mathbf{x} - \mathbf{x}_{[s]}\|_1 / \sqrt{s} \wedge 1/2}\right)\right), \quad \forall \mathbf{x} \in \mathbb{S}^{n-1}. \quad (1.6)$$

Here, our techniques generate a logarithmic factor of $\log\left(\frac{1}{\|\mathbf{x} - \mathbf{x}_{[s]}\|_1 / \sqrt{s} \wedge 1/2}\right)$, where $a \wedge b = \min\{a, b\}$, in the model error term.

The non-uniform counterpart (Theorem 4.2) gives that, for any fixed $\mathbf{x} \in \mathbb{S}^{n-1}$, w.h.p.,

$$\sup_{t \geq 2 \log(m/s)} \|\mathbf{x}_t - \mathbf{x}\|_2 \leq \tilde{O}\left(\frac{s}{m}\right) + O(\|\mathbf{x} - \mathbf{x}_{[s]}\|_2). \quad (1.7)$$

Overall, our framework yields instance optimal guarantees for nonlinear sparse recovery that are comparable to the classical compressed sensing results in (1.2) and (1.3). The analysis involves several high-dimensional probability tools, including concentration bounds, embedding guarantees, and restricted eigenvalue bounds. Some of these ingredients, especially Lemma D.2, appear to be new and may be of independent interest.

1.2 Related Works

This paper is related to the classical works of instance optimality in compressed sensing, which we briefly recap in Equations (1.1)–(1.3); see [31, Sections 6 & 11] and references therein for further details. For more recent progress, see [54] for instance. These developments are restricted to linear measurements. In the following, we shall review several lines of works that are most relevant to our paper.

High-dimensional estimation via RAIC. Provably optimal and computationally efficient algorithms have recently been obtained for several statistical estimation problems by establishing the restricted approximate invertibility condition (RAIC) for suitable gradient map. Such an RAIC requires that the gradient well approximates the ideal descent step over a low-dimensional set of structured signals (such as Σ_s^n), and implies both the convergence and the statistical performance of some nonconvex optimization algorithm. Worked examples include one-bit compressed sensing [47, 32, 16], binary generalized linear models [48], nonlinear tensor estimation [18], logistic regression [15], one-bit phase retrieval [17], among others.

Sparse phase retrieval (SPR). A variety of efficient solvers have been proposed for recovering $\mathbf{x} \in \Sigma_s^n$ from $\mathbf{y} = |\mathbf{Ax}|$ (i.e., the SPR problem), including a convex program [44] and several nonconvex algorithms [10, 68, 62], typically adapted from their counterparts for unstructured signals; see, e.g., [12, 13, 67, 72]. The best known guarantees for these algorithms ensure the *non-uniform exact recovery*: if $m = \tilde{\Omega}(s^2)$, then for any fixed $\mathbf{x} \in \Sigma_s^n$, w.h.p., some algorithm recovers $\pm\mathbf{x}$ exactly. These exact recovery guarantees are not instance optimal, as they provide no recovery guarantee for $\mathbf{x} \notin \Sigma_s^n$. To the best of our knowledge, no previous efficient sparse phase retrieval solver comes with guarantees for non-sparse (particularly approximately sparse) signals. On the other hand, [33] used a strong RIP to show that the constrained ℓ_1 minimization program

$$\hat{\mathbf{x}}_{\ell_1} = \arg \min_{\mathbf{u}} \|\mathbf{u}\|_1, \quad \text{subject to } |\mathbf{Au}| = \mathbf{y}, \quad (1.8)$$

is (ℓ_2, ℓ_1) -instance optimal. In particular, when $m \gtrsim s \log(en/s)$, w.h.p.,

$$\text{dist}(\hat{\mathbf{x}}_{\ell_1}, \mathbf{x}) \leq \frac{C \|\mathbf{x} - \mathbf{x}_{[s]}\|_1}{\sqrt{s}}, \quad \forall \mathbf{x} \in \mathbb{R}^n.$$

This result was recently extended to the complex-valued setting in [70]. However, due to the nonconvex constraint, the program (1.8) is in general computationally intractable.

To our best knowledge, Theorems 3.1 and 3.2 in this paper provide the first instance optimal guarantees for an efficient SPR solver, uniformly for all $\mathbf{x} \in \mathbb{R}^n$ under $m = \tilde{\Omega}(s^3)$, and for a fixed $\mathbf{x} \in \mathbb{R}^n$ under $m = \tilde{\Omega}(s^2)$.

One-bit compressed sensing (1bCS). Most existing 1bCS results concern the recovery of exactly sparse signals $\mathbf{x} \in \Sigma_s^{n,*} := \Sigma_s^n \cap \mathbb{S}^{n-1}$ from $\mathbf{y} = \text{sign}(\mathbf{A}\mathbf{x})$, in which the information-theoretic optimal uniform error rate reads

$$\inf_{\hat{\mathbf{x}}=\hat{\mathbf{x}}(\mathbf{A},\mathbf{y})} \sup_{\mathbf{x} \in \Sigma_s^{n,*}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 = \tilde{\Theta}(s/m) \quad (1.9)$$

provided $m \gtrsim s \log(en/s)$ [37]. It was recently shown by [47] that this rate is achieved by the normalized binary iterative hard thresholding (NBIHT), that is,

$$\sup_{\mathbf{x} \in \Sigma_s^{n,*}} \|\hat{\mathbf{x}}_{\text{nbiht}} - \mathbf{x}\|_2 = \tilde{O}(s/m), \quad (\text{w.h.p.}),$$

where $\hat{\mathbf{x}}_{\text{nbiht}}$ denotes the output of NBIHT with sufficiently many iterations.

To account for model error, prior works considered the recovery of *approximately sparse* signals in $\sqrt{s}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$. Under $m \gtrsim s \log(en/s)$, the information-theoretic optimal uniform error rate is

$$\inf_{\hat{\mathbf{x}}=\hat{\mathbf{x}}(\mathbf{A},\mathbf{y})} \sup_{\mathbf{x} \in \sqrt{s}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 = \tilde{\Theta}\left(\left(\frac{s}{m}\right)^{1/3}\right), \quad (1.10)$$

where the upper bound is achieved by projected gradient descent [16] and Adaboost [21], and the lower bound is established recently in [19]. However, (1.10) is not instance optimal as it gives the same worst-case error rate for all signals. On the other hand, a desirable instance optimal guarantee should provide signal-dependent error bounds that recover the information-theoretic optimal rate $\tilde{O}(s/m)$ over $\mathbf{x} \in \Sigma_s^{n,*}$, and deteriorate continuously as the distance of $\mathbf{x}$ from Σ_s^n increases. Our Theorem 4.1 is the first such result for 1bCS, to the best of our knowledge.

1.3 Additional Related Works

We review several lines of additional related works.

Instance optimality under nonlinear measurements. [42] pursued a similar goal of seeking instance optimality in sparse recovery problems with nonlinear observations. However, the decoder is typically intractable and no non-asymptotic instance optimal guarantees are provided. [33, 70] established instance optimality for a computationally intractable program in sparse phase retrieval, while [20] proved (ℓ_2, ℓ_1) instance optimality for an efficient algorithm in phase-only compressed sensing. These results are tailored to specific nonlinear models and do not provide a unified approach.

Recovery of approximately sparse signals. As with (1.10), robustness to model error has indeed been studied under nonlinear observations by considering approximately sparse signals living in a scaled ℓ_1 ball. Beyond quantization, [58, 59] developed efficient algorithms for single-index models and established the rate $\tilde{O}(\sqrt{s/m})$ for $\mathbf{x} \in \Sigma_s^{n,*}$, while a slower rate $\tilde{O}((s/m)^{1/4})$ for $\mathbf{x} \in \sqrt{s}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$. These results concern the worst-case recovery error and are not instance-dependent. The present paper provides a different treatment to model error under nonlinear observations and typically yields improved error rate for signals being sufficiently close to Σ_s^n ; see, e.g., Remark 4.3.

Embedding and hyperplane tessellation. Most prior analyses of nonlinear measurements relied on different high-dimensional embedding phenomena [55, 56, 57, 71, 24, 36], and our work is no exception in this regard. Here we focus on hyperplane tessellations, or binary embeddings. Particularly, one asks how well a map from a set $\mathcal{U}$ to the binary cube $\{-1, 1\}^m$, induced by m random hyperplanes, preserves Euclidean distances. The seminal work of [57] showed that a number of Gaussian hyperplanes proportional to the Gaussian width of $\mathcal{U}$ suffices for uniform tessellation over $\mathcal{U}$. This result was sharpened and extended to local embeddings by [53]; see also recent developments in [24, 26, 40, 27, 17, 25]. In this work, a key component of our 1bCS analysis is a new signal-dependent hyperplane tessellation result that may be of independent interest.

Oracle inequality. Our non-uniform instance optimal guarantees are closely related to oracle inequalities in high-dimensional statistics, which decompose the prediction/estimation error into statistical error and model error. Consider the sparse linear regression model $\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon}$ with $\mathbf{A}$ satisfying order- s restricted eigenvalue condition and $\boldsymbol{\epsilon} \sim N(0, \sigma^2 \mathbf{I}_m)$. A representative oracle inequality states that the Lasso estimator satisfies, w.h.p.,

$$\frac{1}{m} \|\mathbf{A}(\hat{\mathbf{x}}_{\text{Lasso}} - \mathbf{x})\|_2^2 \leq \inf_{\mathbf{u} \in \Sigma_s^n} \frac{1}{m} \|\mathbf{A}(\mathbf{u} - \mathbf{x})\|_2^2 + \frac{C\sigma^2 s \log n}{m}. \quad (1.11)$$

See, e.g., [43, 2]. In statistics, this shows that Lasso tolerates model misspecification: the true parameter $\mathbf{x}$ need not be s -sparse, and the price to pay is simply $\inf_{\mathbf{u} \in \Sigma_s^n} \frac{1}{m} \|\mathbf{A}(\mathbf{u} - \mathbf{x})\|_2^2$. Mathematically, it is possible to write our results in a similar form. For instance, let $\{\mathbf{x}_t\}_{t \geq 0}$ be the iterates of NBIHT, our 1bCS result in (1.7) already gives

$$\sup_{t \geq 2 \log(m/s)} \|\mathbf{x}_t - \mathbf{x}\|_2 \leq O\left(\inf_{\mathbf{u} \in \Sigma_s^n} \|\mathbf{u} - \mathbf{x}\|_2\right) + \tilde{O}\left(\frac{s}{m}\right);$$

combining with the local binary embedding result from [53], this further implies that

$$\sup_{t \geq 2 \log(m/s)} \frac{1}{m} \|\text{sign}(\mathbf{A}\mathbf{x}_t) - \text{sign}(\mathbf{A}\mathbf{x})\|_2^2 \lesssim \inf_{\mathbf{u} \in \Sigma_s^{n,*}} \frac{1}{m} \|\text{sign}(\mathbf{A}\mathbf{u}) - \text{sign}(\mathbf{A}\mathbf{x})\|_2^2 + \frac{s}{m}$$

up to logarithmic factors.

Yet we point out some important distinctions. First, most oracle inequalities in statistics are for regularized empirical risk minimizers (or M-estimators) in sparse *linear* regression, and technically, they are often established by optimality conditions and certain deterministic assumptions on the design. Also, the oracle inequalities may not be immediately achieved by a polynomial-time algorithm, for instance, when nonconvex regularizer is used. See [4, 43, 9, 28, 63, 2], and see also [66, Sections 9–10] and references therein. In contrast, our framework is developed for *nonlinear* regression problems, and the guarantees are achieved by the computationally fast procedure of IHT, whose analysis is based on the RAIC. On the other hand, some of these works (e.g. [43, 2, 63]) established *sharp* oracle inequalities with the leading constant of the model error term being exactly 1 (e.g., the above Equation (1.11)), which is beyond the scope of this work.

1.4 Notation and Overview

We use C, C_i, c, c_i to denote universal constants whose values vary from line to line. For $T_1, T_2 > 0$, we write $T_1 \lesssim T_2$ or $T_1 = O(T_2)$ to denote that $T_1 \leq CT_2$ for some constant $C > 0$, and write $T_1 \gtrsim T_2$ or $T_1 = \Omega(T_2)$ to denote that $T_1 \geq CT_2$ for some $C > 0$; if both hold, then we write $T_1 \asymp T_2$ or $T_1 = \Theta(T_2)$. We shall use $\tilde{O}(\cdot), \tilde{\Omega}(\cdot), \tilde{\Theta}(\cdot)$ to further hide logarithmic factors in m, n in

$O(\cdot)$, $\Omega(\cdot)$, $\Theta(\cdot)$, respectively. The ψ_2 norm of a random variable X is defined as $\|X\|_{\psi_2} = \inf\{t > 0 : \mathbb{E} \exp(X^2/t^2) \leq 2\}$, and of a random vector $\mathbf{X} \in \mathbb{R}^n$ is $\|\mathbf{X}\|_{\psi_2} = \sup_{\mathbf{v} \in \mathbb{S}^{n-1}} \|\mathbf{v}^\top \mathbf{X}\|_{\psi_2}$. For some sparsity level $s \in [n] := \{1, 2, \dots, n\}$, $H_s : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the hard thresholding operator defined as $H_s(\mathbf{u}) = \mathbf{u}_{[s]}$, where ties are broken by selecting the support with the smallest indices. We use $\mathbb{B}_2^n$ to denote the unit ℓ_2 ball in $\mathbb{R}^n$, and write $\mathbb{B}_2^n(\lambda) := \lambda \mathbb{B}_2^n$, $\mathbb{B}_2^n(\mathbf{u}; \lambda) = \mathbf{u} + \lambda \mathbb{B}_2^n$. More notation will be introduced when appropriate.

Overview. In Section 2, we develop a unified framework for instance optimal sparse recovery via (normalized) IHT. The major applications of the framework appear in Sections 3–5, where we show that (normalized) IHT is instance optimal for SPR, 1bCS and sparse ReLU regression. We close the paper with concluding remarks in Section 6. The complete proofs of our results appear in the appendices.

2 Unified Framework

2.1 RAIC and (Normalized) IHT

For the recovery of $\mathbf{x}$, we construct a gradient map $\mathbf{h}_{\mathbf{x}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, so that $\mathbf{h}_{\mathbf{x}}(\mathbf{u})$ serves as the gradient at $\mathbf{u}$. We focus on the algorithm of iterative hard thresholding (IHT)

$$\mathbf{x}_{t+1} = H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)), \quad t = 0, 1, \dots, \quad (\text{IHT})$$

for some sparsity level s .

If the signal lives in the unit sphere $\mathbb{S}^{n-1}$,¹ we may utilize an additional normalization step, yielding normalized IHT (NIHT)

$$\mathbf{x}_{t+1} = \frac{H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t))}{\|H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t))\|_2}, \quad t = 0, 1, \dots. \quad (\text{NIHT})$$

We define the top- k ℓ_2 norm as

$$\|\mathbf{v}\|_{2,k} = \|\mathbf{v}_{[k]}\|_2 = \sup_{\mathbf{u} \in \Sigma_k^{n,*}} \langle \mathbf{u}, \mathbf{v} \rangle,$$

and then define the RAIC in the following.

Definition 2.1. For a fixed target signal $\mathbf{x}$, we say that the gradient $\mathbf{h}_{\mathbf{x}}(\mathbf{u})$ satisfies the RAIC of order k over a constraint set $\mathcal{U} \subset \Sigma_k^n$ with an error function $R_{\mathbf{x}}(\mathbf{u})$, if it holds that

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2,2k} \leq R_{\mathbf{x}}(\mathbf{u}), \quad \forall \mathbf{u} \in \mathcal{U}.$$

We write this as

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}(\Sigma_{2k}^{n,*}; \mathcal{U}, \mathbf{x}, R_{\mathbf{x}}(\mathbf{u})).$$

The following results state that an RAIC with large enough constraint set (in the sense of (2.1), (2.3)) and error function $\mu_1 \|\mathbf{u} - \mathbf{x}\|_2 + \mu_2$ yields the linear convergence of (N)IHT with a proper initialization. Their proofs are provided in Section A.

¹This can be generalized to $\mathbf{x} \in \beta \mathbb{S}^{n-1}$ for some known $\beta > 0$, i.e., the setting where the signal norm is known a priori.

Theorem 2.1 (Convergence of [IHT](#)). For a fixed $\mathbf{x} \in \mathbb{R}^n$, suppose that

$$\eta \mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}(\Sigma_{2s}^{n,*}; \mathcal{U}, \mathbf{x}, \mu_1 \|\mathbf{u} - \mathbf{x}\|_2 + \mu_2).$$

If $\mu_1 < \frac{1}{2}$ and for some $0 < R_{\mathbf{x}}^{\text{loc}} \leq \infty$

$$\mathcal{U} \supset \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}}), \quad \text{where } R_{\mathbf{x}}^{\text{loc}} > \frac{2\mu_2 + 3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 2\mu_1}, \quad (2.1)$$

then $\{\mathbf{x}_t\}_{t \geq 0}$ generated by running [IHT](#)

$$\mathbf{x}_{t+1} = H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)), \quad t \geq 0$$

with $\mathbf{x}_0 \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}})$ satisfies

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq (2\mu_1)^t \|\mathbf{x}_0 - \mathbf{x}\|_2 + \frac{2\mu_2 + 3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 2\mu_1}, \quad t \geq 0. \quad (2.2)$$

Theorem 2.2 (Convergence of [NIHT](#)). For a target signal $\mathbf{x} \in \mathbb{S}^{n-1}$, suppose that

$$\eta \mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}(\Sigma_{2s}^{n,*}; \mathcal{U}, \mathbf{x}, \mu_1 \|\mathbf{u} - \mathbf{x}\|_2 + \mu_2).$$

If $\mu_1 < \frac{1}{4}$ and for some $0 < R_{\mathbf{x}}^{\text{loc}} \leq \infty$

$$\mathcal{U} \supset \Sigma_s^{n,*} \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}}), \quad \text{where } R_{\mathbf{x}}^{\text{loc}} > \frac{4\mu_2 + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 4\mu_1}, \quad (2.3)$$

then $\{\mathbf{x}_t\}_{t \geq 0}$ generated by running [NIHT](#)

$$\mathbf{x}_{t+1} = \frac{H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t))}{\|H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t))\|_2}, \quad t \geq 0$$

with $\mathbf{x}_0 \in \Sigma_s^{n,*} \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}})$ satisfies

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq (4\mu_1)^t \|\mathbf{x}_0 - \mathbf{x}\|_2 + \frac{4\mu_2 + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 4\mu_1}, \quad t \geq 0. \quad (2.4)$$

2.2 From Signal-Dependent RAIC to Instance Optimality

Our goal is to use (N)IHT to achieve instance optimality

$$\|\hat{\mathbf{x}}_{(\text{N})\text{IHT}} - \mathbf{x}\|_2 \leq \mathcal{E}_{\text{opt}}(t; \mathbf{x}) + \mathcal{E}_{\text{stat}} + \mathcal{E}_{\text{iop},s}(\mathbf{x}), \quad \forall \mathbf{x} \in \bar{\mathcal{X}}, \quad (2.5)$$

where $\mathcal{E}_{\text{opt}}(t; \mathbf{x})$ denotes the optimization error that exponentially decays to 0 when $t \rightarrow \infty$, $\mathcal{E}_{\text{stat}}$ is the optimal statistical error that does not depend on the signal, and $\mathcal{E}_{\text{iop},s}(\mathbf{x})$ characterizes the instance optimality of the algorithm. To be more concrete, in this paper, we establish $\mathcal{E}_{\text{opt}}(t; \mathbf{x}) = O(\rho^t \|\mathbf{x}\|_2)$ for some $\rho \in (0, 1)$, indicating the linear convergence of (N)IHT. The statistical error, $\mathcal{E}_{\text{stat}}$, vanishes in SPR and ReLU regression, while is at the order $\tilde{O}(\frac{s}{m})$ in 1bCS, matching the uniform recovery lower bound in (1.9). Moreover, $\mathcal{E}_{\text{iop},s}(\mathbf{x})$ typically takes the form of

$$\tau_s(\mathbf{x}) := \sum_{j \geq 1} \|\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}\|_2. \quad (2.6)$$

2.2.1 (Uniform) Instance Optimality

For each $\mathbf{x} \in \bar{\mathcal{X}}$ suppose that we work with the gradient map $\mathbf{h}_{\mathbf{x}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Based on RAIC and its implications, we propose a unified framework for establishing instance optimal guarantee as in (2.5), which consists of the following steps:

1. **(Step 1: Decomposition of $\bar{\mathcal{X}}$)** Choose $\mathcal{X} \subset \bar{\mathcal{X}}$ and separately treat $\mathbf{x} \in \mathcal{X}$ and $\mathbf{x} \in \mathcal{X}^c := \bar{\mathcal{X}} \setminus \mathcal{X}$. In general, $\mathcal{X}$ is the set of signals that are closer to Σ_s^n and have small $\mathcal{E}_{\text{iop},s}(\mathbf{x})$. More concretely, in the subsequent examples, we set

$$\mathcal{X} = \mathcal{X}_{\text{unif}} := \{\mathbf{u} \in \bar{\mathcal{X}} : \tau_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\} \quad (2.7)$$

for some small enough universal constant $c_* > 0$.

2. **(Step 2: Instance optimality for $\mathbf{x} \in \mathcal{X}$)** The analysis of $\mathcal{X}$ is based on establishing a signal-dependent RAIC for the gradients of all $\mathbf{x} \in \mathcal{X}$:

$$\eta \mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}(\Sigma_{2s}^{n,*}; \mathcal{U}_{\mathbf{x}}, \mathbf{x}, \mu_1 \|\mathbf{u} - \mathbf{x}\|_2 + \mu_2 + e(\mathbf{x})), \quad \forall \mathbf{x} \in \mathcal{X}, \quad (2.8)$$

where the error function is signal-dependent because of the term $e(\mathbf{x})$.

If $\bar{\mathcal{X}} = \mathbb{R}^n$ and **IHT** is considered, then we require that

$$\mu_1 < \frac{1}{2}, \quad (2.9)$$

$$\mathcal{U}_{\mathbf{x}} \supset \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}}) \text{ for some } R_{\mathbf{x}}^{\text{loc}} \in (0, \infty], \quad (2.10)$$

$$R_{\mathbf{x}}^{\text{loc}} > \frac{2\mu_2 + 2e(\mathbf{x}) + 3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 2\mu_1}, \quad (2.11)$$

$$\mathbf{x}_0 \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}}) \quad (2.12)$$

hold for all $\mathbf{x} \in \mathcal{X}$.

If $\bar{\mathcal{X}} = \mathbb{S}^{n-1}$ and **NIHT** is considered, then we require that

$$\mu_1 < \frac{1}{4}, \quad (2.13)$$

$$\mathcal{U}_{\mathbf{x}} \supset \Sigma_s^{n,*} \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}}) \text{ for some } R_{\mathbf{x}}^{\text{loc}} \in (0, \infty], \quad (2.14)$$

$$R_{\mathbf{x}}^{\text{loc}} > \frac{4\mu_2 + 4e(\mathbf{x}) + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 4\mu_1}, \quad (2.15)$$

$$\mathbf{x}_0 \in \Sigma_s^{n,*} \cap \mathbb{B}_2^n(\mathbf{x}; R_{\mathbf{x}}^{\text{loc}}). \quad (2.16)$$

hold for all $\mathbf{x} \in \mathcal{X}$.

3. **(Step 3: Instance optimality for $\mathbf{x} \in \mathcal{X}^c$)** The analysis of $\mathcal{X}^c$ is based on the observation that the desired bound (2.5) becomes a crude one due to the large $\mathcal{E}_{\text{iop},s}(\mathbf{x})$. As such, a simple analysis, together with algorithmic regularization if needed, is sufficient.

To see why Step 2 leads to the instance optimality over $\mathbf{x} \in \mathcal{X}$, we provide the following two statements.

Theorem 2.3 (**IHT** is instance optimal for $\mathbf{x} \in \mathcal{X}$). *Assume that (2.8) and (2.9)–(2.12) hold for all $\mathbf{x} \in \mathcal{X}$, then **IHT** satisfies*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq (2\mu_1)^t \|\mathbf{x}_0 - \mathbf{x}\|_2 + \frac{2\mu_2 + 2e(\mathbf{x}) + 3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 2\mu_1}, \quad \forall t \geq 0, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (2.17)$$

Proof. The statement follows by applying Theorem 2.1 to every $\mathbf{x} \in \mathcal{X}$. $\square$

Theorem 2.4 (NIHT is instance optimal for $\mathbf{x} \in \mathcal{X}$). Assume that (2.8) and (2.13)–(2.16) hold for all $\mathbf{x} \in \mathcal{X}$, then NIHT satisfies

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq (4\mu_1)^t \|\mathbf{x}_0 - \mathbf{x}\|_2 + \frac{4\mu_2 + 4e(\mathbf{x}) + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{1 - 4\mu_1}, \quad \forall t \geq 0, \forall \mathbf{x} \in \mathcal{X}. \quad (2.18)$$

Proof. The statement follows by applying Theorem 2.2 to every $\mathbf{x} \in \mathcal{X}$. $\square$

Remark 2.1. In short, both algorithms linearly converge to ℓ_2 error at the order of $O(\mu_2 + e(\mathbf{x}) + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2)$, uniformly for all $\mathbf{x} \in \mathcal{X}$. Our hope is that $O(\mu_2)$ captures the (optimal) statistical error as with $\mathcal{E}_{\text{stat}}$ in (2.5), while $O(e(\mathbf{x}) + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2)$ captures the instance optimality as with the approximation error $\mathcal{E}_{\text{iop},s}(\mathbf{x})$ in (2.5). For the latter, we point out that $O(\|\mathbf{x} - \mathbf{x}_{[s]}\|_2)$ is intrinsic to (N)IHT since the s -sparse iterates always satisfy $\|\mathbf{x}_t - \mathbf{x}\|_2 \geq \|\mathbf{x} - \mathbf{x}_{[s]}\|_2$.

Remark 2.2. If $R_{\mathbf{x}}^{\text{loc}} < \infty$, then we have to separately obtain a warm initializer obeying (2.12) or (2.16). This may require a separate procedure and independent analysis.

Since $\mathbf{x} \in \mathcal{X}^c$ is settled by a separate argument, we then arrive at (2.5). See also Figure 1 for an intuitive explanation of our proof scheme.

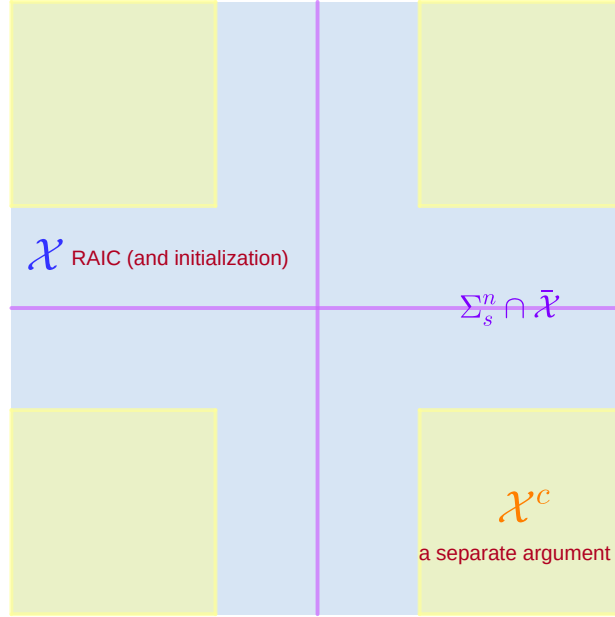


Figure 1: Our approach to instance optimal sparse recovery. The figure considers $\bar{\mathcal{X}} = \mathbb{R}^2$ and shows $\mathcal{X} = \{\mathbf{u} : \|\mathbf{u} - \mathbf{u}_{[1]}\|_2 \leq c_*\}$ (the blue region) as a “relaxation” of $\Sigma_1^2 = \{\mathbf{u} : \|\mathbf{u}\|_0 \leq 1\}$ (the purple lines). The proposed approach treats the blue region $\mathcal{X}$ by RAIC (and initialization analysis, if needed), then covers the yellow region $\mathcal{X}^c$ by a separate argument.

2.2.2 Non-Uniform Instance Optimality

We are also interested in the non-uniform instance optimality

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \mathcal{E}_{\text{opt}}(t; \mathbf{x}) + \mathcal{E}_{\text{stat}} + \mathcal{E}_{\text{iop},s}(\mathbf{x}), \quad \text{for a fixed } \mathbf{x} \in \bar{\mathcal{X}}. \quad (2.19)$$

This is weaker than (2.5) in the sense that it only guarantees the recovery of a fixed $\mathbf{x}$. On the other hand, the non-uniform instance optimality often allows for a sharper model error term $\mathcal{E}_{\text{iop},s}(\mathbf{x})$ at the order of

$$\delta_s(\mathbf{x}) := \|\mathbf{x} - \mathbf{x}_{[s]}\|_2. \quad (2.20)$$

Given the gradient map $\mathbf{h}_{\mathbf{x}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, we adapt our unified framework for (2.19) in the following:

1. **(Step 1: Decomposition of $\bar{\mathcal{X}}$)** Choose $\mathcal{X} \subset \bar{\mathcal{X}}$ as a class of approximately sparse signals. In subsequent concrete examples, we set

$$\mathcal{X} = \mathcal{X}_{\text{nonunif}} := \{\mathbf{u} \in \bar{\mathcal{X}} : \delta_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\} \quad (2.21)$$

for some small enough universal constant $c_* > 0$.

2. **(Step 2: Instance optimality if $\mathbf{x} \in \mathcal{X}$)** If $\mathbf{x} \in \mathcal{X}$, then we seek to establish

$$\eta \mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}(\Sigma_{2s}^{n,*}; \mathcal{U}_{\mathbf{x}}, \mathbf{x}, \mu_1 \|\mathbf{u} - \mathbf{x}\|_2 + \mu_2 + e(\mathbf{x})),$$

along with (2.9)–(2.12) for IHT, or along with (2.13)–(2.16) for NIHT.

3. **(Step 3: Instance optimality if $\mathbf{x} \in \mathcal{X}^c$)** If $\mathbf{x} \in \mathcal{X}^c$, we provide a separate argument.

While the framework is developed for nonlinear observations, as a proof-of-concept, readers may refer to [14, Section 1] to see that the framework applies to compressed sensing and recovers (1.2) and (1.3) for IHT.

3 Sparse Phase Retrieval

Sparse phase retrieval (SPR) concerns the recovery of $\mathbf{x}$ with a sparse prior² from $\mathbf{y} = |\mathbf{A}\mathbf{x}|$, or equivalently, $\{y_i = |\mathbf{a}_i^\top \mathbf{x}|\}_{i=1}^m$. We work with a Gaussian matrix $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_m]^\top \sim N^{m \times n}(0, 1)$. To our best knowledge, there is no known efficient and instance optimal SPR algorithm. The main aim of this section is to devise the first such algorithm.

3.1 Algorithm

Our algorithm is a variant of the thresholded amplitude flow [62, 68]. See also [72, 67] for the counterparts of unstructured signals.

IHT with amplitude-based loss. Note that a specific subgradient of the amplitude-based ℓ_2 loss

$$\mathcal{L}_{\text{amp}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{u}| - y_i)^2$$

is given by

$$\partial \mathcal{L}_{\text{amp}}(\mathbf{u}) = \frac{1}{m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{u}| - y_i) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i.$$

²This means that the signal is assumed to be sparse or approximately sparse, but need not belong exactly to Σ_s^n . Indeed, our desired instance-optimal guarantees apply to all $\mathbf{x} \in \bar{\mathcal{X}}$, for example all $\mathbf{x} \in \mathbb{R}^n$ in SPR. The same convention applies in the rest of the paper.

Substituting $y_i = |\mathbf{a}_i^\top \mathbf{x}|$, we set

$$\mathbf{h}_\mathbf{x}(\mathbf{u}) := \frac{1}{m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{u}| - |\mathbf{a}_i^\top \mathbf{x}|) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i. \quad (3.1)$$

We shall use a fixed step size 1, and thus the [IHT](#) is given by

$$\mathbf{x}_{t+1} = H_s(\mathbf{x}_t - \mathbf{h}_\mathbf{x}(\mathbf{x}_t)), \quad t = 0, 1, \dots. \quad (3.2)$$

Initialization. In light of $\mathbb{E}y_i = \mathbb{E}|\mathbf{a}_i^\top \mathbf{x}| = \sqrt{2/\pi} \|\mathbf{x}\|_2$, we let

$$\lambda_\mathbf{x} = \sqrt{\frac{\pi}{2}} \frac{1}{m} \sum_{i=1}^m y_i$$

be the norm estimate for $\|\mathbf{x}\|_2$. The mechanism for spectral initialization is to construct a data matrix whose leading eigenvector aligns well with the true signal. To this end, we define the truncation function as

$$\mathsf{T}_\xi(a) = \begin{cases} a, & \text{if } |a| \leq \xi \\ \xi \text{sign}(a), & \text{if } |a| > \xi \end{cases}$$

and consider the matrix

$$\hat{\mathbf{S}}_\mathbf{x} = \frac{1}{m} \sum_{i=1}^m \mathsf{T}_{2\lambda_\mathbf{x}}(y_i) \mathbf{a}_i \mathbf{a}_i^\top. \quad (3.3)$$

To find its sparse leading eigenvector, we need an estimate of the support set of $\mathbf{x}$; we shall use the size- s index set $\mathcal{I}_\mathbf{x}$ corresponding to the largest s diagonal entries of $\hat{\mathbf{S}}_\mathbf{x}$. Let $\mathcal{S}_1 \subset [m]$ and $\mathcal{S}_2 \subset [n]$, for $\mathbf{S} \in \mathbb{R}^{m \times n}$, we shall let $[\mathbf{S}]_{\mathcal{S}_1, \mathcal{S}_2}$ be the $m \times n$ matrix obtained from $\mathbf{S}$ by setting the rows not in $\mathcal{S}_1$ and the columns not in $\mathcal{S}_2$ to zero. We now find the (normalized) leading eigenvector of $[\hat{\mathbf{S}}_\mathbf{x}]_{\mathcal{I}_\mathbf{x}, \mathcal{I}_\mathbf{x}}$, which we denote by $\mathbf{v}_\mathbf{x}$, as the estimate of the signal direction $\mathbf{x}/\|\mathbf{x}\|_2$. Taken collectively, we obtain an initial guess of $\mathbf{x}$:

$$\mathbf{x}_0 := \lambda_\mathbf{x} \mathbf{v}_\mathbf{x}.$$

An additional projection. While the above [IHT](#) and initialization are well developed in prior works (e.g., [11, 39, 62]), we incorporate an additional projection into [IHT](#). Specifically, we utilize the projection onto the ball

$$\mathbb{B}_2^n(2\lambda_\mathbf{x}) := \{\mathbf{u} \in \mathbb{R}^n : \|\mathbf{u}\|_2 \leq 2\lambda_\mathbf{x}\},$$

to handle signals that are distant from Σ_s^n (precisely, signals in $\mathcal{X}^c$ in our framework). Therefore, our final update rule is

$$\mathbf{x}_{t+1} = P_{\mathbb{B}_2^n(2\lambda_\mathbf{x})}(H_s(\mathbf{x}_t - \mathbf{h}_\mathbf{x}(\mathbf{x}_t))), \quad t = 0, 1, \dots \quad (3.4)$$

instead of (3.2).

Combining these pieces, we are now ready to formally introduce our algorithm, referred to as bounded thresholded amplitude flow (BTAF), in Algorithm 1.

Algorithm 1 Bounded Thresholded Amplitude Flow (BTAF)

Input: Matrix $\mathbf{A}$, observations $y = |\mathbf{Ax}|$, sparsity level s

1: **Step 1: Initialization**

2: $\lambda_{\mathbf{x}} = \sqrt{\frac{\pi}{2}} \frac{1}{m} \sum_{i=1}^m y_i$

3: $\hat{\mathbf{S}}_{\mathbf{x}} = \frac{1}{m} \sum_{i=1}^m \mathsf{T}_{2\lambda_{\mathbf{x}}}(y_i) \mathbf{a}_i \mathbf{a}_i^\top$

4: $\mathcal{I}_{\mathbf{x}} \leftarrow$ indices of the s largest diagonal entries of $\hat{\mathbf{S}}_{\mathbf{x}}$

5: $\mathbf{v}_{\mathbf{x}} \leftarrow$ leading eigenvector of $[\hat{\mathbf{S}}_{\mathbf{x}}]_{\mathcal{I}_{\mathbf{x}}, \mathcal{I}_{\mathbf{x}}}$

6: $\mathbf{x}_0 = \lambda_{\mathbf{x}} \mathbf{v}_{\mathbf{x}}$

7: **Step 2: Refinement**

8: **for** $t = 0, 1, 2, \dots$ **do**

9: $\mathbf{x}_{t+1} = P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})} \left(H_s \left(\mathbf{x}_t - \frac{1}{m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{x}_t| - y_i) \text{sign}(\mathbf{a}_i^\top \mathbf{x}_t) \mathbf{a}_i \right) \right)$

10: **end for**

Output: Sequence $\{\mathbf{x}_t\}_{t \geq 0}$

3.2 Instance Optimality

Under $\tilde{O}(s^3)$ Gaussian measurements, our main result shows the instance optimality of BTAF.

Theorem 3.1. *Under Gaussian matrix $\mathbf{A}$, if*

$$m \gtrsim s^3 \log \left(\frac{en}{s} \right) \log^2 \left(\frac{em}{s \log(en/s)} \right)$$

then with probability at least $1 - C_1 \exp(-c_2 s \log \frac{en}{s})$, the $\{\mathbf{x}_t\}_{t \geq 0}$ generated by Algorithm 1 satisfies

$$\text{dist}(\mathbf{x}_t, \mathbf{x}) \leq \frac{\|\mathbf{x}\|_2}{2^t} + C_2 \tau_s(\mathbf{x}), \quad \forall t \geq 0, \mathbf{x} \in \mathbb{R}^n. \quad (3.5)$$

In particular, if s is even and $s = 2s_0$, then with the same probability,

$$\text{dist}(\mathbf{x}_t, \mathbf{x}) \leq \frac{\|\mathbf{x}\|_2}{2^t} + \frac{C_3 \|\mathbf{x} - \mathbf{x}_{[s_0]}\|_1}{\sqrt{s_0}}, \quad \forall t \geq 0, \mathbf{x} \in \mathbb{R}^n.$$

Proof. We give technical overview below and provide the complete proof in Appendix C.1. $\square$

Remark 3.1 (The s^3 complexity). *The sample complexity $\tilde{O}(s^3)$ in Theorem 3.1 exhibits an essential gap from the best known $\tilde{O}(s^2)$ for existing efficient algorithms to exactly recover any s -sparse signals. It is an open question whether $\tilde{O}(s^3)$ for efficient and instance optimal sparse phase retrieval is improvable.*

3.2.1 Technical Overview for Theorem 3.1

With no loss of generality, we can assume $\|\mathbf{x}_0 - \mathbf{x}\|_2 \leq \|\mathbf{x}_0 + \mathbf{x}\|_2$ and hence $\text{dist}(\mathbf{x}_0, \mathbf{x}) = \|\mathbf{x}_0 - \mathbf{x}\|_2$. (Otherwise, we can instead treat $-\mathbf{x}$ as the true signal and use the same argument with minor adaptation.) For $\mathbf{h}_{\mathbf{x}}$ chosen as in (3.1), the result is proved by following the three steps in the unified framework in Section 2.2.1:

1. **(Decomposition of $\mathbb{R}^n$)** As with (2.7), we set $\mathcal{X} := \{\mathbf{u} \in \mathbb{R}^n : \tau_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\}$ for some small enough c_* .

2. **(Instance optimality for $\mathbf{x} \in \mathcal{X}$)** We first analyze (3.2) and then transfer the result to (3.4) by showing that $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$ has no effect on the iterates associated with $\mathbf{x} \in \mathcal{X}$.

RAIC. For some universal constant $c > 0$, we show that the signal-dependent RAIC

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2), \mathbf{x}, \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 4\tau_s(\mathbf{x})\right), \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\} \quad (3.6)$$

holds (w.h.p.). It is not hard to check that (2.9)–(2.11) hold under sufficiently small c_* , with

$$\mu_1 = \frac{1}{4}, \quad \mu_2 = 0, \quad R_{\mathbf{x}}^{\text{loc}} = c\|\mathbf{x}\|_2, \quad \text{and} \quad e(\mathbf{x}) = 4\tau_s(\mathbf{x}).$$

Initialization. Moreover, we prove that under $m \gtrsim s^3$ (up to logarithmic factors), w.h.p.,

$$\mathbf{x}_0 \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2), \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\}. \quad (3.7)$$

By Theorem 2.3, IHT in (3.2) with $\mathbf{x}_0$ from Algorithm 1 satisfies (2.17), i.e.,

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{c\|\mathbf{x}\|_2}{2^t} + 22\tau_s(\mathbf{x}), \quad \forall t \geq 0, \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\} \quad (3.8)$$

as desired.

Incorporating $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$. We further show that $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$ has no effect on the iterates of $\mathbf{x} \in \mathcal{X}$. Therefore, $\{\mathbf{x}_t\}_{t \geq 0}$ corresponding to $\mathbf{x} \in \mathcal{X}$ satisfy (3.8).

3. **(Instance optimality for $\mathbf{x} \in \mathcal{X}^c$)** Note that the desired error bound (3.5) for $\mathbf{x} \in \mathcal{X}^c$ scales as a constant. Consequently, a simple argument based on the projection onto $\mathbb{B}_2^n(2\lambda_{\mathbf{x}})$ is sufficient for all these signals.

The two most technical parts lie in the proof of the RAIC and the analysis of the initialization.

Establishing the Uniform RAIC (3.6). The RAIC over $\mathbf{x} \in \Sigma_s^n$ has been established in [18, Theorem B.9], hence we only need an extension to $\mathbf{x} \in \mathcal{X}$. The major effect of the model error turns out to be characterized by the gradient mismatch term $\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$. Due to the Lipschitz continuity, standard sparse decomposition and sparse eigenvalue bound of $\mathbf{A}$ establish

$$\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \lesssim \tau_s(\mathbf{x}), \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\}. \quad (3.9)$$

Establishing the Uniform Initialization Bound (3.7). Existing analyses of the spectral initialization only concern a fixed $\mathbf{x} \in \Sigma_s^n$, while we have to establish (3.7) for all $\mathbf{x} \in \mathcal{X} \setminus \{0\}$. By adapting some existing technicalities (see, e.g., [39]) and accounting for the model error $\mathbf{x} - \mathbf{x}_{[s]}$, we establish a deterministic initialization error bound in Lemma C.4. The problem then boils down to establishing (C.10), (C.11), (C.12) for all $\mathbf{x} \in \mathcal{X} \setminus \{0\}$, where $\tilde{\mathbf{S}}_{\mathbf{x}}$ defined in (C.8) serves as a surrogate of $\hat{\mathbf{S}}_{\mathbf{x}}$.

In view of $\lambda_{\mathbf{x}} = \sqrt{\frac{\pi}{2}} \frac{\|\mathbf{A}\mathbf{x}\|_1}{m}$, it turns out that the uniform (C.10) follows from an ℓ_1 embedding result due to [57]. On the other hand, the analyses of uniform (C.11) and (C.12) appear more delicate, where our aims are to show

$$\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top \left(\frac{\tilde{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]}{\|\mathbf{x}\|_2} \right) \mathbf{u} \right| \lesssim 1, \quad (3.10)$$

$$\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \sup_{\mathbf{u} \in \{\mathbf{e}_j; j \in [n]\}} \left| \mathbf{u}^\top \left(\frac{\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]}{\|\mathbf{x}\|_2} \right) \mathbf{u} \right| \lesssim \frac{1}{s}. \quad (3.11)$$

Note that these concern the restricted eigenvalues of a set of random matrices $\{(\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}])/\|\mathbf{x}\|_2 : \mathbf{x} \in \mathcal{X} \setminus \{0\}\}$. By empirical process theory we establish in Lemma C.6 the following:

$$\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top \left(\frac{\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]}{\|\mathbf{x}\|_2} \right) \mathbf{u} \right| = \tilde{O}\left(\sqrt{\frac{s}{m}}\right); \quad (3.12)$$

As it turns out, (despite a smaller range of $\mathbf{u}$) we are not able to establish an essentially sharper bound on the empirical process in (3.11). Hence, we directly reuse the bound $\tilde{O}(\sqrt{s/m})$ for it. Therefore, ignoring logarithmic factors, (3.11) is ensured by $\sqrt{\frac{s}{m}} \lesssim \frac{1}{s}$, which precisely dictates the s^3 sample size.

3.3 Non-Uniform Instance Optimality

We further establish a non-uniform instance optimal guarantee for any fixed $\mathbf{x} \in \mathbb{R}^n$ under $\tilde{O}(s^2)$ measurements.

Theorem 3.2. *Let $\mathbf{x}$ be any fixed signal in $\mathbb{R}^n$. Under Gaussian matrix $\mathbf{A}$, if $m \gtrsim s^2 \log n$, then with probability at least $1 - C_1 n^{-1} - C_2 \exp(-c_3 s \log(en/s))$, $\{\mathbf{x}_t\}_{t \geq 0}$ generated by Algorithm 1 satisfies*

$$\text{dist}(\mathbf{x}_t, \mathbf{x}) \leq \frac{\|\mathbf{x}\|_2}{2^t} + C\delta_s(\mathbf{x}), \quad \forall t \geq 0. \quad (3.13)$$

Proof. The proof can be found in Appendix C.2. $\square$

Remark 3.2. *When transitioning to non-uniform instance optimality, we have an improvement on the model error term $\mathcal{E}_{\text{io},s}(\mathbf{x})$, that is, $O(\delta_s(\mathbf{x}))$ in (3.13) which is tighter than $O(\tau_s(\mathbf{x}))$ in (3.5). This is consistent with existing instance optimal guarantees for compressed sensing recalled in (1.2) and (1.3).*

Remark 3.3. *Note that $\tilde{O}(s^2)$ coincides with the best-known sample complexity of computationally efficient algorithms that guarantee exact recovery for arbitrary s -sparse signals. In this sense, instance optimality for a fixed signal comes essentially for free.*

The proof of Theorem 3.2 largely follows from that of Theorem 3.1, except that the RAIC and initialization bound only need to hold for a fixed $\mathbf{x}$. As such, we shall only discuss some main distinctions. In the RAIC, the effect of model error is again captured by $\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$, while for a fixed $\mathbf{x}$ this satisfies a sharper bound than (3.9):

$$\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \lesssim \|\mathbf{x} - \mathbf{x}_{[s]}\|_2.$$

This is exactly what improves $\tau_s(\mathbf{x})$ to $\delta_s(\mathbf{x})$. For the initialization, we only need to guarantee the following non-uniform counterpart of (3.10)–(3.11):

$$\sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top \left(\frac{\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]}{\|\mathbf{x}\|_2} \right) \mathbf{u} \right| \lesssim 1, \quad \sup_{\mathbf{u} \in \{\mathbf{e}_j; j \in [n]\}} \left| \mathbf{u}^\top \left(\frac{\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]}{\|\mathbf{x}\|_2} \right) \mathbf{u} \right| \lesssim \frac{1}{s}. \quad (3.14)$$

Due to the transition from “all $\mathbf{x}$ ” to “a fixed $\mathbf{x}$ ”, we establish substantially tighter bound

$$\sup_{\mathbf{u} \in \{\mathbf{e}_j; j \in [n]\}} \left| \mathbf{u}^\top \left(\frac{\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]}{\|\mathbf{x}\|_2} \right) \mathbf{u} \right| = \tilde{O}\left(\frac{1}{\sqrt{m}}\right).$$

As such, ignoring logarithmic factors, we need $\frac{1}{\sqrt{m}} \leq \frac{1}{s}$ to ensure the second condition in (3.14), which determines the s^2 sample complexity.

4 One-Bit Compressed Sensing

One-bit compressed sensing (1bCS) concerns the recovery of $\mathbf{x} \in \bar{\mathcal{X}} = \mathbb{S}^{n-1}$ with a sparse prior from $\mathbf{y} = \text{sign}(\mathbf{A}\mathbf{x})$, where $\mathbf{A} \sim N^{m \times n}(0, 1)$ is a Gaussian matrix. The most relevant developments of 1bCS have been reviewed in Section 1.2. Here, we establish the instance optimality of NBIHT.

4.1 Algorithm

We shall show that NBIHT is a specific instance of our NIHT algorithm. We adopt the ReLU loss function

$$\mathcal{L}_{\text{ReLU}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{u}| - y_i \mathbf{a}_i^\top \mathbf{u})$$

which is a convex relaxation of the hamming distance loss (or 0-1 loss). A subgradient of the ReLU loss is given by

$$\partial \mathcal{L}_{\text{ReLU}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{u}) - y_i) \mathbf{a}_i.$$

Combining with $y_i = \text{sign}(\mathbf{a}_i^\top \mathbf{x})$, we shall set

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{u}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x})) \mathbf{a}_i. \quad (4.1)$$

Taking the step size $\eta = \sqrt{2\pi}$, our NIHT specializes to the following algorithm, referred to as NBIHT in the literature.

Algorithm 2 Normalized Binary Iterative Hard Thresholding (NBIHT)

Input: Matrix $\mathbf{A}$, observations $\mathbf{y} = \text{sign}(\mathbf{A}\mathbf{x})$, sparsity level s

1: **Step 1: Initialization**

2: Let $\mathbf{x}_0 = \mathbf{e}_1$

3: **Step 2: Refinement**

4: **for** $t = 0, 1, 2, \dots$ **do**

5: $\mathbf{x}_{t+1} = \frac{H_s(\mathbf{x}_t - \sqrt{\frac{\pi}{2}} \frac{1}{m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}_t) - y_i) \mathbf{a}_i)}{\|H_s(\mathbf{x}_t - \sqrt{\frac{\pi}{2}} \frac{1}{m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}_t) - y_i) \mathbf{a}_i)\|_2},$

6: **end for**

Output: Sequence $\{\mathbf{x}_t\}_{t \geq 0}$

4.2 Instance Optimality

Our main result in this section shows the instance optimality of NBIHT.

Theorem 4.1. *Under Gaussian matrix $\mathbf{A}$, if $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - \exp(-cs \log \frac{en}{s})$, the output of Algorithm 2 satisfies*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2\left(\frac{2}{5}\right)^t + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) + C_2 \tau_s(\mathbf{x}) \log\left(\frac{1}{\tau_s(\mathbf{x}) \wedge \frac{1}{2}}\right), \quad \forall t \geq 0, \forall \mathbf{x} \in \mathbb{S}^{n-1}.$$

In particular, if s is even and $s = 2s_0$, then

$$\begin{aligned} \|\mathbf{x}_t - \mathbf{x}\|_2 &\leq 2\left(\frac{2}{5}\right)^t + \frac{C_3 s_0}{m} \log\left(\frac{mn}{s_0^2}\right) \log^{1/2}\left(\frac{m}{s_0}\right) \\ &\quad + \frac{C_4 \|\mathbf{x} - \mathbf{x}_{[s_0]}\|_1}{\sqrt{s_0}} \log\left(\frac{1}{s_0^{-1/2} \|\mathbf{x} - \mathbf{x}_{[s_0]}\|_1 \wedge \frac{1}{2}}\right), \quad \forall t \geq 0, \forall \mathbf{x} \in \mathbb{S}^{n-1}. \end{aligned}$$

Proof. We provide technical overview below and provide the complete proof in Appendix D.1. $\square$

Remark 4.1. Unlike in SPR, it is not possible to perfectly recover $\mathbf{x}$ from the one-bit measurements (even if $\mathbf{x} \in \Sigma_s^n$). Therefore, the statistical error term $O(\frac{s}{m} \log(\frac{mn}{s^2}) \log^{1/2}(\frac{m}{s}))$ appears in the error rate, which is optimal up to logarithmic factors [37]. To our best knowledge, NBIHT is the only known efficient algorithm that provably achieves the error rate $\tilde{O}(\frac{s}{m})$ over s -sparse signals, due to the recent work of Matsumoto and Mazumdar [47]. Our Theorem 4.1 upgrades their main result to an instance optimal one.

Remark 4.2. Note that the model error term is consistent with compressed sensing (1.2) and sparse phase retrieval (Theorem 3.1) but exhibits an extra logarithmic factor. In fact, some of the arguments in our proofs necessarily generate additional logarithmic factor. It is an open question whether this logarithmic factor can be avoided by a more refined argument.

4.2.1 Technical Overview for Theorem 4.1

For the gradient map in (4.1), the proof follows the three steps in the unified framework:

1. **(Decomposition of $\mathbb{S}^{n-1}$)** For some small enough universal constant $c_* > 0$, we set $\mathcal{X} := \{\mathbf{u} \in \mathbb{S}^{n-1} : \tau_s(\mathbf{u}) \leq c_*\}$ as in (2.7);
2. **(Instance optimality for $\mathbf{x} \in \mathcal{X}$)** For $\mathbf{x} \in \mathcal{X}$, we establish the signal-dependent RAIC

$$\begin{aligned} \sqrt{2\pi} \mathbf{h}_{\mathbf{x}}(\mathbf{u}) &\sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^{n,*}, \mathbf{x}, \frac{\|\mathbf{u} - \mathbf{x}\|_2}{10} + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) + e(\mathbf{x})\right), \quad \forall \mathbf{x} \in \mathcal{X}, \\ \text{where } e(\mathbf{x}) &\asymp \tau_s(\mathbf{x}) \log\left(\frac{1}{\tau_s(\mathbf{x})}\right). \end{aligned} \quad (4.2)$$

This satisfies (2.13)–(2.16) with $\mu_1 = \frac{1}{10}$, $R_{\mathbf{x}}^{\text{loc}} = \infty$, $\mathbf{x}_0 = \mathbf{e}_1$ for all $\mathbf{x} \in \mathcal{X}$, and therefore yields (2.18) for the NBIHT iterates $\{\mathbf{x}_t\}_{t \geq 0}$:

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2\left(\frac{2}{5}\right)^t + O\left(\frac{s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) + e(\mathbf{x})\right), \quad \forall t \geq 0, \mathbf{x} \in \mathcal{X}. \quad (4.3)$$

3. **(Instance optimality for $\mathbf{x} \in \mathcal{X}^c$)** For $\mathbf{x} \in \mathcal{X}^c$, the crude bound $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2$ is enough, in light of $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2 \leq \frac{2}{c_*} \tau_s(\mathbf{x})$.

The main bulk of techniques is devoted to establishing (4.2) for all $\mathbf{x} \in \mathcal{X}$. We shall build on the RAIC for all $\mathbf{x} \in \Sigma_s^{n,*}$ recently established in [47] to prove the optimality of NBIHT. Here, we quote a statement from [16] that is closer to our development (see Lemma D.1), which delivers

$$\|\mathbf{u} - \mathbf{x} - \sqrt{2\pi} \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{10} + \tilde{O}\left(\frac{s}{m}\right), \quad \forall \mathbf{u}, \mathbf{x} \in \Sigma_s^{n,*}. \quad (4.4)$$

It remains to seek an extension to $\mathbf{x} \in \mathcal{X}$. Like in the proof of Theorem 3.1, the effect of model error turns out to be mainly driven by the gradient mismatch term

$$\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} = \left\| \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i \right\|_{2s,2}. \quad (4.5)$$

Yet, the discontinuity of sign forbids a simple Lipschitzness-based argument as in SPR.

Observe that the i -th contributor is non-zero only if $\text{sign}(\mathbf{a}_i^\top \mathbf{x}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})$, our strategy is to bound

$$L_{\mathbf{x}} := |\{i \in [m] : \text{sign}(\mathbf{a}_i^\top \mathbf{x}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})\}|$$

This bound, combined with Lemma C.9, delivers a worst-case upper bound. Following this approach, we have to establish signal-dependent bound on $L_{\mathbf{x}}$. This seems plausible in light of

$$L_{\mathbf{x}} \sim \text{Binomial}(m, \pi^{-1} \arccos(\langle \mathbf{x}, \mathbf{x}_{[s]} / \|\mathbf{x}_{[s]}\|_2 \rangle)),$$

which concentrates about $m\pi^{-1} \arccos(\langle \mathbf{x}, \mathbf{x}_{[s]} / \|\mathbf{x}_{[s]}\|_2 \rangle)$.

Limitation of existing results. Geometrically, the study of $L_{\mathbf{x}}$ can be interpreted as a *hyperplane tessellation* problem: arrange m Gaussian hyperplanes, how many of them separate $\mathbf{x}$ and $\mathbf{x}_{[s]}$, uniformly over all $\mathbf{x} \in \mathcal{X}$? However, existing results (see, e.g., [57, 53, 24, 26]) are not instance-dependent but rather provide a uniform bound that is identical for all $\mathbf{x}$ of interest, and hence do not serve our purpose. For instance, Theorem 2.2 of [53] yields the following high-probability concentration bound on $S_{\mathbf{u}, \mathbf{v}} := |\{i \in [m] : \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{v})\}|$:

$$\left| \frac{S_{\mathbf{u}, \mathbf{v}}}{m} - \pi^{-1} \arccos(\langle \mathbf{u}, \mathbf{v} \rangle) \right| = \tilde{O}\left(\left(\frac{s}{m}\right)^{1/4}\right), \quad \forall \mathbf{u}, \mathbf{v} \in \mathcal{X}. \quad (4.6)$$

This implies

$$\frac{L_{\mathbf{x}}}{m} = \frac{S_{\mathbf{x}, \mathbf{x}_{[s]} / \|\mathbf{x}_{[s]}\|_2}}{m} \lesssim \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \tilde{O}\left(\left(\frac{s}{m}\right)^{1/4}\right), \quad \forall \mathbf{x} \in \mathcal{X} \quad (4.7)$$

in our context. However, this is not sufficient for establishing Theorem 4.1, since the concentration error term $\tilde{O}((s/m)^{1/4})$ carries over to the statistical error and compromises entirely our ability to attain the optimal statistical rate $\tilde{O}(s/m)$ for $\mathbf{x} \in \Sigma_s^{n,*}$.

Technical contribution. We establish a novel instance-dependent hyperplane tessellation bound

$$\frac{L_{\mathbf{x}}}{m} \leq \tilde{O}(\tau_s(\mathbf{x})) + \tilde{O}\left(\frac{s}{m}\right), \quad \forall \mathbf{x} \in \mathcal{X}; \quad (4.8)$$

see Lemma D.2. The constant term $\tilde{O}(s/m)$ is essentially sharper than $\tilde{O}((s/m)^{1/4})$ in (4.7) and allows us to retain the optimal statistical error for all $\mathbf{x} \in \mathcal{X}$. While the price to pay is the degradation from $O(\|\mathbf{x} - \mathbf{x}_{[s]}\|_2)$ to $\tilde{O}(\tau_s(\mathbf{x}))$ in the model error, this is nonetheless consistent with the model error terms in instance optimal compressed sensing and SPR, up to logarithmic factors.

In view of $L_{\mathbf{x}} = S_{\mathbf{x}, \mathbf{x}_{[s]} / \|\mathbf{x}_{[s]}\|_2}$, the main inspiration for our establishment of (4.8) is the *asymmetry* between $\mathbf{x}$ and $\mathbf{x}_{[s]} / \|\mathbf{x}_{[s]}\|_2$:

$$\mathbf{x} \in \mathcal{X}, \quad \mathbf{x}_{[s]} / \|\mathbf{x}_{[s]}\|_2 \in \Sigma_s^{n,*},$$

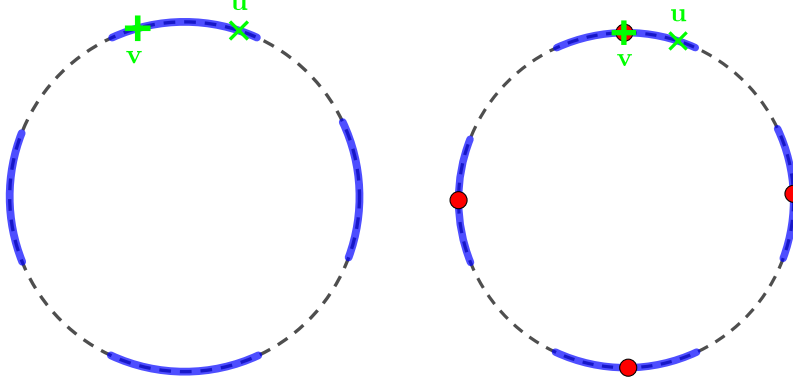


Figure 2: The blue set $\mathcal{W}$ represents approximately sparse vectors, while the red set $\mathcal{W}' := \Sigma_1^{2,*}$ is a smaller subset. Classical hyperplane tessellation results relate $S_{\mathbf{u},\mathbf{v}}$ to $\pi^{-1} \arccos(\langle \mathbf{u}, \mathbf{v} \rangle)$ when both $\mathbf{u}, \mathbf{v}$ lie in $\mathcal{W}$ (left), while our argument exploits the asymmetry that $\mathbf{v}$ lies in the smaller set $\mathcal{W}'$ (right).

and in a certain sense $\Sigma_s^{n,*}$ is a much thinner set than $\mathcal{X}$. Our developments generalize to $\{S_{\mathbf{u},\mathbf{v}} : \mathbf{u} \in \mathcal{W}, \mathbf{v} \in \mathcal{W}'\}$ for some $\mathcal{W}'$ being much thinner than $\mathcal{W}$, which is in contrast to $\{S_{\mathbf{u},\mathbf{v}} : \mathbf{u}, \mathbf{v} \in \mathcal{W}\}$ treated in prior works; see Figure 2 for an intuitive illustration.

The rest of this proof remains rather involved. In a nutshell, we note that

$$L_{\mathbf{x}} \leq \left| \left\{ i \in [m] : \left| \mathbf{a}_i^\top \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right| \leq \eta \right\} \right| + \left| \left\{ i \in [m] : \left| \mathbf{a}_i^\top \left(\mathbf{x} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right) \right| > \frac{\eta}{2} \right\} \right|, \quad \forall \eta > 0, \quad (4.9)$$

and separately bound the two terms over a wide enough range of η . Finally, we choose η as a function of $\mathbf{x} \in \mathcal{X}$ to render a signal-dependent bound. See Appendix D.1 for further details.

4.3 Non-Uniform Instance Optimality

We have the following non-uniform instance optimality in the recovery of a fixed $\mathbf{x} \in \mathbb{S}^{n-1}$. Unlike Theorem 4.1, the above result does not lose logarithmic factor in the model error term.

Theorem 4.2. *Fix $\mathbf{x} \in \mathbb{S}^{n-1}$. Under Gaussian matrix $\mathbf{A}$, if $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - \exp(-cs \log \frac{en}{s})$, for any $t \geq 0$,*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2\left(\frac{2}{5}\right)^t + \frac{C_1 s}{m} \log\left(\frac{en}{s}\right) \log^{1/2}\left(\frac{m}{s}\right) + C_2 \delta_s(\mathbf{x}).$$

Proof. The proof is given in Appendix D.2. □

Remark 4.3. *Previous works handle model error by treating the approximately s -sparse signals in $\sqrt{s}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$, with the near-optimal uniform error rate being $\tilde{O}((s/m)^{1/3})$; see (1.10). Ignoring logarithmic factors, our Theorem 4.1 improves on (1.10) over all signals $\mathbf{x}$ obeying $\tau_s(\mathbf{x}) \lesssim (\frac{s}{m})^{1/3}$, and likewise our non-uniform Theorem 4.2 gives a faster rate for $\mathbf{x}$ satisfying $\delta_s(\mathbf{x}) \lesssim (\frac{s}{m})^{1/3}$.*

Compared to the proof of Theorem 4.1, the main distinction lies in the establishment of the RAIC, especially in bounding the gradient mismatch term (4.5). Indeed we can reuse the previous strategy: for the fixed $\mathbf{x}$, a simple Chernoff bound yields $L_{\mathbf{x}} \lesssim s \log \frac{en}{s} + m \delta_s(\mathbf{x})$, and then Lemma C.9 gives the bound $O(\delta_s(\mathbf{x}) \log^{1/2}(e/\delta_s(\mathbf{x})))$. However, this exhibits an additional logarithmic

factor compared to the desired $O(\delta_s(\mathbf{x}))$ in (D.17). The removal of this logarithmic factor requires a more fine-grained analysis, which first decomposes the gradient into the components along the orthogonal directions $\mathbf{x} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}$, $\mathbf{x} + \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}$, and $\text{span}(\mathbf{x}, \mathbf{x}_{[s]})^\perp$, and then establishes concentration inequalities on each of the three components; see Appendix D.2. Note that similar argument has appeared in [47] for a different purpose.

5 Sparse ReLU Regression

Sparse ReLU regression is the problem of recovering $\mathbf{x} \in \mathbb{R}^n$ with an s -sparse prior from $\mathbf{y} = \text{ReLU}(\mathbf{A}\mathbf{x})$, where $\text{ReLU}(a) = \max\{a, 0\} = \frac{a+|a|}{2}$ is the ReLU activation in neural networks. Note that the parameter estimation problem in ReLU regression was considered in [61, 41, 49].

5.1 Algorithm

Our algorithm is a slight variant of the projected gradient descent in [61].

IHT with the ReLU-based ℓ_2 loss. We adopt the ReLU-based ℓ_2 loss

$$\mathcal{L}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{m} \sum_{i=1}^m [\text{ReLU}(\mathbf{a}_i^\top \mathbf{u}) - y_i]^2$$

to fit ReLUs. Note that one can choose a “generalized gradient” of $\mathcal{L}_{\mathbf{x}}(\mathbf{u})$ as

$$\partial \mathcal{L}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{m} \sum_{i=1}^m [\text{ReLU}(\mathbf{a}_i^\top \mathbf{u}) - y_i] [1 + \text{sign}(\mathbf{a}_i^\top \mathbf{u})] \mathbf{a}_i,$$

and therefore we substitute $y_i = \text{ReLU}(\mathbf{a}_i^\top \mathbf{x})$ to introduce

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{m} \sum_{i=1}^m [\text{ReLU}(\mathbf{a}_i^\top \mathbf{u}) - \text{ReLU}(\mathbf{a}_i^\top \mathbf{x})] [1 + \text{sign}(\mathbf{a}_i^\top \mathbf{u})] \mathbf{a}_i. \quad (5.1)$$

Using a fixed step size of $\eta = 1$, we shall consider the **IHT**

$$\mathbf{x}_{t+1} = H_s(\mathbf{x}_t - \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)), \quad t \geq 0. \quad (5.2)$$

Initialization. We consider the projection-based estimator of [59, 71]. By the rotational invariance of $\mathbf{a}_i$,

$$\mathbb{E}(y_i \mathbf{a}_i) = \mathbb{E}[\text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i] = \mathbb{E} \left[\text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \right] \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \frac{\mathbf{x}}{2},$$

the idea of constructing $\mathbf{x}_0$ is to project the average $\frac{1}{m} \sum_{i=1}^m 2y_i \mathbf{a}_i$ onto Σ_s^n :

$$\mathbf{x}_0 = H_s \left(\frac{1}{m} \sum_{i=1}^m 2y_i \mathbf{a}_i \right).$$

In fact, [61] leveraged the same initialization.

An additional projection. In the same spirit as BTAF (Algorithm 1) for SPR, we modify [61] by incorporating an additional projection into (5.2) to ensure the boundedness of the iterates. We shall use $2\|\mathbf{x}_0\|_2$ as a crude upper bound on $\|\mathbf{x}\|_2$ and then utilize the projection onto $\mathbb{B}_2^n(2\|\mathbf{x}\|_2)$, changing (5.2) to

$$\mathbf{x}_{t+1} = P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}(H_s(\mathbf{x}_t - \mathbf{h}_s(\mathbf{x}_t))).$$

Putting the pieces together, we formally introduce the full procedure, referred to as bounded ReLU iterative hard thresholding (BReLUHT), as Algorithm 3.

Algorithm 3 Bounded ReLU Iterative Hard Thresholding (BReLUHT)

Input: Matrix $\mathbf{A}$, measurements $\mathbf{y} = \text{ReLU}(\mathbf{A}\mathbf{x})$, sparsity level s

1: **Step 1: Initialization**

2: Compute $\mathbf{x}_0 = H_s(\frac{1}{m} \sum_{i=1}^m 2y_i \mathbf{a}_i)$

3: **Step 2: Refinement**

4: **for** $t = 0, 1, 2, \dots$ **do**

5: $\mathbf{x}_{t+1} = P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}\left(H_s\left(\mathbf{x}_t - \frac{1}{m} \sum_{i=1}^m [\text{ReLU}(\mathbf{a}_i^\top \mathbf{x}_t) - y_i][1 + \text{sign}(\mathbf{a}_i^\top \mathbf{x}_t)]\mathbf{a}_i\right)\right),$

6: **end for**

Output: Sequence $\{\mathbf{x}_t\}_{t \geq 0}$

For $\mathbf{x} = 0$, BReLUHT returns $\mathbf{x}_t = 0$ and achieves exact recovery. As such, we restrict our attention to nonzero $\mathbf{x}$ in the subsequent analysis. We will not discuss the technical details in this section as they are largely analogous to the SPR analysis.

5.2 Instance Optimality

Our main result establishes the instance optimality of BReLUHT, providing the analogue of (1.2) for sparse ReLU regression.

Theorem 5.1. *Under Gaussian matrix $\mathbf{A}$, if $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - C \exp(-cs \log \frac{en}{s})$, $\{\mathbf{x}_t\}_{t \geq 0}$ generated by Algorithm 3 satisfies*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{c_1 \|\mathbf{x}\|_2}{2^t} + C_2 \tau_s(\mathbf{x}), \quad \forall t \geq 0, \forall \mathbf{x} \in \mathbb{R}^n.$$

In particular, if s is even and $s = 2s_0$, then

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{c_3 \|\mathbf{x}\|_2}{2^t} + \frac{C_4 \|\mathbf{x} - \mathbf{x}_{[s_0]}\|_1}{\sqrt{s_0}}, \quad \forall t \geq 0, \forall \mathbf{x} \in \mathbb{R}^n.$$

Proof. The proof is provided in Appendix E.1. □

5.3 Non-Uniform Instance Optimality

We have the following result for the non-uniform recovery of a fixed $\mathbf{x}$, which finds the analogue of (1.3) in sparse ReLU regression.

Theorem 5.2. *Under Gaussian matrix $\mathbf{A}$, for any fixed $\mathbf{x} \in \mathbb{R}^n$, if $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - C \exp(-cs \log \frac{en}{s})$, $\{\mathbf{x}_t\}_{t \geq 0}$ generated by Algorithm 3 satisfies*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{c_1 \|\mathbf{x}\|_2}{2^t} + C_2 \delta_s(\mathbf{x}), \quad \forall t \geq 0.$$

Proof. The proof appears in Appendix E.2. □

6 Conclusion

Instance optimality is a standard notion in compressed sensing that characterizes the robustness of the algorithms to model error. Its non-uniform counterpart also connects naturally to oracle inequalities in statistics. In this work, we developed a general framework to establish instance optimality for sparse recovery problems with nonlinear measurements. The main ingredient is the signal-dependent RAIC for a suitable gradient map, which leads to instance optimality of IHT. We established such RAICs for sparse phase retrieval, one-bit compressed sensing and sparse ReLU regression with Gaussian designs and obtained instance optimal guarantees. In fact, our framework also applies to phase-only compressed sensing and establishes instance optimal guarantee comparable to [20]; see [14, Section 2]. There remain a number of interesting questions for future research, such as extension to non-Gaussian designs, sharp oracle inequalities under nonlinear observations, the tuning of the sparsity level, among many others.

References

- [1] R. Adamczak. A tail inequality for suprema of unbounded empirical processes with applications to markov chains. *Electronic Journal of Probability*, 13:1000–1034, 2008.
- [2] P. C. Bellec, G. Lecué, and A. B. Tsybakov. Slope meets lasso: improved oracle bounds and optimality. *The Annals of Statistics*, 46(6B):3603–3642, 2018.
- [3] A. Bhandari, F. Krahmer, and R. Raskar. On unlimited sampling and reconstruction. *IEEE Transactions on Signal Processing*, 69:3827–3839, 2020.
- [4] P. J. Bickel, Y. Ritov, and A. B. Tsybakov. Simultaneous analysis of lasso and dantzig selector. *The Annals of Statistics*, 37(4):1705–1732, 2009.
- [5] T. Blumensath and M. E. Davies. Iterative hard thresholding for compressed sensing. *Applied and Computational Harmonic Analysis*, 27(3):265–274, 2009.
- [6] S. Boucheron, G. Lugosi, and P. Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, February 2013.
- [7] P. T. Boufounos and R. G. Baraniuk. 1-bit compressive sensing. In *2008 42nd Annual Conference on Information Sciences and Systems*, pages 16–21. IEEE, 2008.
- [8] A. Bourrier, M. E. Davies, T. Peleg, P. Pérez, and R. Gribonval. Fundamental performance limits for ideal decoders in high-dimensional linear inverse problems. *IEEE Transactions on Information Theory*, 60(12):7928–7946, 2014.
- [9] F. Bunea, A. B. Tsybakov, and M. H. Wegkamp. Sparsity oracle inequalities for the lasso. *Electronic Journal of Statistics*, 1:169–194, 2007.
- [10] T. T. Cai and A. Zhang. Sparse representation of a polytope and recovery of sparse signals and low-rank matrices. *IEEE Transactions on Information Theory*, 60(1):122–132, 2013.
- [11] T. T. Cai, X. Li, and Z. Ma. Optimal rates of convergence for noisy sparse phase retrieval via thresholded Wirtinger flow. *The Annals of Statistics*, 44(5):2221–2251, 2016.

- [12] E. J. Candes, T. Strohmer, and V. Voroninski. Phaselift: Exact and stable signal recovery from magnitude measurements via convex programming. *Communications on Pure and Applied Mathematics*, 66(8):1241–1274, 2013.
- [13] E. J. Candes, X. Li, and M. Soltanolkotabi. Phase retrieval via wirtinger flow: Theory and algorithms. *IEEE Transactions on Information Theory*, 61(4):1985–2007, 2015.
- [14] J. Chen and A. Maleki. Supplementary material for “instance optimal sparse recovery from nonlinear observations: a unified framework”. 2026. URL <https://junrenchen58.github.io/files/iop-supp.pdf>.
- [15] J. Chen and A. Mazumdar. Finite-sample performance of gradient descent in logistic regression with gaussian design. *arXiv preprint arXiv:2606.21683*, 2026.
- [16] J. Chen and M. Yuan. Optimal quantized compressed sensing via projected gradient descent. *arXiv preprint arXiv:2407.04951*, 2024.
- [17] J. Chen and M. Yuan. One-bit phase retrieval: Optimal rates and efficient algorithms. *IEEE Transactions on Information Theory*, 72(7):5251–5292, 2026.
- [18] J. Chen, L. Ding, D. Xia, and M. Yuan. Low-rank tensor estimation from nonlinear observations: A unified framework. *arXiv preprint arXiv:2510.16965*, 2026.
- [19] J. Chen, A. Mazumdar, and M. Yuan. Near-optimal lower bounds on one-bit compressed sensing of approximately sparse signals. *arXiv preprint arXiv:2607.06750*, 2026.
- [20] J. Chen, M. K. Ng, and J. Scarlett. Robust instance optimal phase-only compressed sensing. *Information and Inference: A Journal of the IMA*, 15(2):iaag014, 2026.
- [21] G. Chinot, F. Kuchelmeister, M. Löffler, and S. van de Geer. Adaboost and robust one-bit compressed sensing. *Mathematical Statistics and Learning*, 5(1):117–158, 2022.
- [22] A. Cohen, W. Dahmen, and R. DeVore. Compressed sensing and best k -term approximation. *Journal of the American Mathematical Society*, 22(1):211–231, 2009.
- [23] C. Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. iii. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970.
- [24] S. Dirksen and S. Mendelson. Non-gaussian hyperplane tessellations and robust one-bit compressed sensing. *Journal of the European Mathematical Society*, 23(9):2913–2947, 2021.
- [25] S. Dirksen and N. Q. Strachan. A resolution of the gaussian hyperplane tessellation conjecture on the sphere. *arXiv preprint arXiv:2508.05194*, 2025.
- [26] S. Dirksen, S. Mendelson, and A. Stollenwerk. Sharp estimates on random hyperplane tessellations. *SIAM Journal on Mathematics of Data Science*, 4(4):1396–1419, 2022.
- [27] S. Dirksen, S. Mendelson, and A. Stollenwerk. Fast metric embedding into the hamming cube. *SIAM Journal on Computing*, 53(2):315–345, 2024.
- [28] A. Elsenner and S. van de Geer. Sharp oracle inequalities for stationary points of nonconvex penalized m-estimators. *IEEE Transactions on Information Theory*, 65(3):1452–1472, 2018.

- [29] S. Foucart. Hard thresholding pursuit: an algorithm for compressive sensing. *SIAM Journal on Numerical Analysis*, 49(6):2543–2563, 2011.
- [30] S. Foucart and T. Needham. Sparse recovery from saturated measurements. *Information and Inference: A Journal of the IMA*, 6(2):196–212, 2017.
- [31] S. Foucart and H. Rauhut. A mathematical introduction to compressive sensing. In *Applied and Numerical Harmonic Analysis*, 2013.
- [32] M. P. Friedlander, H. Jeong, Y. Plan, and Ö. Yilmaz. Nbiht: An efficient algorithm for 1-bit compressed sensing with optimal error decay rate. *IEEE Transactions on Information Theory*, 68(2):1157–1177, 2021.
- [33] B. Gao, Y. Wang, and Z. Xu. Stable signal recovery from phaseless measurements. *Journal of Fourier Analysis and Applications*, 22(4):787–808, 2016.
- [34] M. Genzel and A. Stollenwerk. A unified approach to uniform signal recovery from nonlinear observations. *Foundations of Computational Mathematics*, 23(3):899–972, 2023.
- [35] M. X. Goemans and D. P. Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *Journal of the ACM (JACM)*, 42(6):1115–1145, 1995.
- [36] L. Jacques and T. Feuillen. The importance of phase in complex compressive sensing. *IEEE Transactions on Information Theory*, 67(6):4150–4161, 2021.
- [37] L. Jacques, J. N. Laska, P. T. Boufounos, and R. G. Baraniuk. Robust 1-bit compressive sensing via binary stable embeddings of sparse vectors. *IEEE Transactions on Information Theory*, 59(4):2082–2102, 2013.
- [38] K. Jaganathan, Y. C. Eldar, and B. Hassibi. Phase retrieval: An overview of recent developments. *Optical Compressive Imaging*, pages 279–312, 2016.
- [39] G. Jagatap and C. Hegde. Sample-efficient algorithms for recovering structured signals from magnitude-only measurements. *IEEE Transactions on Information Theory*, 65(7):4434–4456, 2019.
- [40] H. C. Jung, J. Maly, L. Palzer, and A. Stollenwerk. Quantized compressed sensing by rectified linear units. *IEEE Transactions on Information Theory*, 67(6):4125–4149, 2021.
- [41] S. M. M. Kalan, M. Soltanolkotabi, and A. S. Avestimehr. Fitting relus via sgd and quantized sgd. *arXiv preprint arXiv:1901.06587*, 2019.
- [42] N. Keriven and R. Gribonval. Instance optimal decoding and the restricted isometry property. In *Journal of Physics: Conference Series*, volume 1131, page 012002. IOP Publishing, 2018.
- [43] V. Koltchinskii, K. Lounici, and A. B. Tsybakov. Nuclear-norm penalization and optimal rates for noisy low-rank matrix completion. *The Annals of Statistics*, 39(5):2302–2329, 2011.
- [44] X. Li and V. Voroninski. Sparse signal recovery from quadratic measurements via convex programming. *SIAM Journal on Mathematical Analysis*, 45(5):3019–3033, 2013.
- [45] S. Mallat. *A Wavelet Tour of Signal Processing: The Sparse Way*. Academic Press, 3rd edition, 2009.

- [46] P. Massart. About the constants in talagrand’s concentration inequalities for empirical processes. *The Annals of Probability*, 28(2):863–884, 2000.
- [47] N. Matsumoto and A. Mazumdar. Binary iterative hard thresholding converges with optimal number of measurements for 1-bit compressed sensing. *Journal of the ACM*, 71(5):1–64, 2024.
- [48] N. Matsumoto and A. Mazumdar. Learning sparse generalized linear models with binary outcomes via iterative hard thresholding. In *Proceedings of Thirty Eighth Conference on Learning Theory*, volume 291, pages 3933–4032. PMLR, 2025.
- [49] A. Mazumdar and A. S. Rawat. Learning and recovery in the relu model. In *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 108–115. IEEE, 2019.
- [50] S. Mendelson. Upper bounds on product and multiplier empirical processes. *Stochastic Processes and their Applications*, 126(12):3652–3680, 2016.
- [51] D. Needell and J. A. Tropp. Cosamp: Iterative signal recovery from incomplete and inaccurate samples. *Applied and Computational Harmonic Analysis*, 26(3):301–321, 2009.
- [52] A. V. Oppenheim and J. S. Lim. The importance of phase in signals. *Proceedings of the IEEE*, 69(5):529–541, 1981.
- [53] S. Oymak and B. Recht. Near-optimal bounds for binary embeddings of arbitrary sets. *arXiv preprint arXiv:1512.04433*, 2015.
- [54] H. B. Petersen and P. Jung. Robust instance-optimal recovery of sparse signals at unknown noise levels. *Information and Inference: A Journal of the IMA*, 11(3):845–887, 2022.
- [55] Y. Plan and R. Vershynin. Robust 1-bit compressed sensing and sparse logistic regression: A convex programming approach. *IEEE Transactions on Information Theory*, 59(1):482–494, 2012.
- [56] Y. Plan and R. Vershynin. One-bit compressed sensing by linear programming. *Communications on Pure and Applied Mathematics*, 66(8):1275–1297, 2013.
- [57] Y. Plan and R. Vershynin. Dimension reduction by random hyperplane tessellations. *Discrete & Computational Geometry*, 51(2):438–461, 2014.
- [58] Y. Plan and R. Vershynin. The generalized lasso with non-linear observations. *IEEE Transactions on Information Theory*, 62(3):1528–1537, 2016.
- [59] Y. Plan, R. Vershynin, and E. Yudovina. High-dimensional estimation with geometric constraints. *Information and Inference: A Journal of the IMA*, 6(1):1–40, 2017.
- [60] Y. Shechtman, Y. C. Eldar, O. Cohen, H. N. Chapman, J. Miao, and M. Segev. Phase retrieval with application to optical imaging: a contemporary overview. *IEEE Signal Processing Magazine*, 32(3):87–109, 2015.
- [61] M. Soltanolkotabi. Learning relus via gradient descent. *Advances in Neural Information Processing Systems*, 30, 2017.

- [62] M. Soltanolkotabi. Structured signal recovery from quadratic measurements: Breaking sample complexity barriers via nonconvex optimization. *IEEE Transactions on Information Theory*, 65(4):2374–2400, 2019.
- [63] B. Stucky and S. Van De Geer. Sharp oracle inequalities for square root regularization. *Journal of Machine Learning Research*, 18(67):1–29, 2017.
- [64] Y. Traonmilin and R. Gribonval. Stable recovery of low-dimensional cones in hilbert spaces: One rip to rule them all. *Applied and Computational Harmonic Analysis*, 45(1):170–205, 2018.
- [65] R. Vershynin. *High-dimensional probability: An introduction with applications in data science*, volume 47. Cambridge University Press, 2018.
- [66] M. J. Wainwright. *High-dimensional statistics: A non-asymptotic viewpoint*, volume 48. Cambridge University Press, 2019.
- [67] G. Wang, G. B. Giannakis, and Y. C. Eldar. Solving systems of random quadratic equations via truncated amplitude flow. *IEEE Transactions on Information Theory*, 64(2):773–794, 2017.
- [68] G. Wang, L. Zhang, G. B. Giannakis, M. Akçakaya, and J. Chen. Sparse phase retrieval via truncated amplitude flow. *IEEE Transactions on Signal Processing*, 66(2):479–491, 2017.
- [69] P. Wojtaszczyk. Stability and instance optimality for gaussian measurements in compressed sensing. *Foundations of Computational Mathematics*, 10(1):1–13, 2010.
- [70] Y. Xia and Z. Xu. Instance optimality in phase retrieval. *Applied and Computational Harmonic Analysis*, 80:101818, 2026.
- [71] C. Xu and L. Jacques. Quantized compressive sensing with rip matrices: The benefit of dithering. *Information and Inference: A Journal of the IMA*, 9(3):543–586, 2020.
- [72] H. Zhang, Y. Zhou, Y. Liang, and Y. Chi. A nonconvex approach for phase retrieval: Reshaped wirtinger flow and incremental algorithms. *Journal of Machine Learning Research*, 18, 2017.

A Proofs for Convergence of IHT and NIHT

We start with a useful lemma which controls the “post-hard-thresholding error” by top- $2s$ ℓ_2 norm.

Lemma A.1. *For any $\mathbf{u} \in \mathbb{R}^n$ and any $\mathbf{v} \in \Sigma_s^n$, we have $\|H_s(\mathbf{u}) - \mathbf{v}\|_2 \leq 2\|\mathbf{u} - \mathbf{v}\|_{2,2s}$.*

Proof. For $\mathcal{T} \subset [n]$, recall that $\mathbf{u}_{\mathcal{T}}$ is obtained from $\mathbf{u}$ by setting the entries not in $\mathcal{T}$ as 0. Let $\mathcal{T}_1, \mathcal{T}_2$ be the support sets of $H_s(\mathbf{u})$ and $\mathbf{v}$, respectively, then

$$\|H_s(\mathbf{u}) - \mathbf{v}\|_2 \leq \|H_s(\mathbf{u}) - \mathbf{u}_{\mathcal{T}_1 \cup \mathcal{T}_2}\|_2 + \|\mathbf{v} - \mathbf{u}_{\mathcal{T}_1 \cup \mathcal{T}_2}\|_2 \leq 2\|\mathbf{v} - \mathbf{u}_{\mathcal{T}_1 \cup \mathcal{T}_2}\|_2,$$

where the second inequality holds because $H_s(\mathbf{u})$ is the best s -sparse approximation to $\mathbf{u}_{\mathcal{T}_1 \cup \mathcal{T}_2}$. The result then follows from $\|\mathbf{v} - \mathbf{u}_{\mathcal{T}_1 \cup \mathcal{T}_2}\|_2 = \|(\mathbf{u} - \mathbf{v})_{\mathcal{T}_1 \cup \mathcal{T}_2}\|_2 \leq \|\mathbf{u} - \mathbf{v}\|_{2,2s}$. $\square$

A.1 Proofs of Theorems 2.1–2.2 (Convergence of IHT and NIHT)

Proof. We prove the two results simultaneously. Let $q = 1$ for IHT and $q = 2$ for NIHT, and write $\rho := 2q\mu_1$, $b := 2q\mu_2 + 3q\|\mathbf{x} - \mathbf{x}_{[s]}\|_2$. Under the assumptions of the corresponding theorem, $\rho < 1$ and $R_{\mathbf{x}}^{\text{loc}} > b/(1 - \rho)$. We define a sequence $\{f_t\}_{t \geq 0}$ by

$$f_0 = \|\mathbf{x}_0 - \mathbf{x}\|_2, \quad f_{t+1} = \rho f_t + b, \quad t \geq 0.$$

It is not hard to find that, for any $t \geq 0$,

$$\begin{aligned} f_t &= \rho^t \|\mathbf{x}_0 - \mathbf{x}\|_2 + (1 - \rho^t) \frac{b}{1 - \rho} \\ &\leq \rho^t \|\mathbf{x}_0 - \mathbf{x}\|_2 + \frac{b}{1 - \rho}. \end{aligned} \tag{A.1}$$

In light of $\|\mathbf{x}_0 - \mathbf{x}\|_2 \leq R_{\mathbf{x}}^{\text{loc}}$ and $b/(1 - \rho) < R_{\mathbf{x}}^{\text{loc}}$, we further obtain

$$f_t \leq \max \left\{ \|\mathbf{x}_0 - \mathbf{x}\|_2, \frac{b}{1 - \rho} \right\} \leq R_{\mathbf{x}}^{\text{loc}}, \quad \forall t \geq 0.$$

Therefore, all that remains is to prove $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq f_t$ for any $t \geq 0$, and we use induction to achieve this. We first derive a common one-step estimate. For IHT, triangle inequality and Lemma A.1 give

$$\begin{aligned} \|\mathbf{x}_{t+1} - \mathbf{x}\|_2 &= \|H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)) - \mathbf{x}\|_2 \\ &\stackrel{(a)}{\leq} \|H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)) - \mathbf{x}_{[s]}\|_2 + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \\ &\stackrel{(b)}{\leq} 2\|\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t) - \mathbf{x}_{[s]}\|_{2s,2} + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \\ &\stackrel{(c)}{\leq} 2\|\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t) - \mathbf{x}\|_{2s,2} + 3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2. \end{aligned}$$

For NIHT, using additionally $\mathbf{x} \in \mathbb{S}^{n-1}$, we have

$$\begin{aligned} \|\mathbf{x}_{t+1} - \mathbf{x}\|_2 &= \left\| \frac{H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t))}{\|H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t))\|_2} - \mathbf{x} \right\|_2 \\ &\stackrel{(a)}{\leq} 2\|H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)) - \mathbf{x}\|_2 \\ &\stackrel{(b)}{\leq} 2\|H_s(\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)) - \mathbf{x}_{[s]}\|_2 + 2\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \\ &\stackrel{(c)}{\leq} 4\|\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t) - \mathbf{x}_{[s]}\|_{2s,2} + 2\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \\ &\stackrel{(d)}{\leq} 4\|\mathbf{x}_t - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t) - \mathbf{x}\|_{2s,2} + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2. \end{aligned}$$

In the first display, (a) and (c) are due to triangle inequality, and (b) follows from Lemma A.1. In the second display, (a) is a standard bound that can be found in [20, Equation (2.12)] for instance, (b) and (d) are due to triangle inequality, and (c) follows from Lemma A.1. Consequently, for either algorithm,

$$\|\mathbf{x}_{t+1} - \mathbf{x}\|_2 \leq 2q\|\mathbf{x}_t - \mathbf{x} - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)\|_{2s,2} + 3q\|\mathbf{x} - \mathbf{x}_{[s]}\|_2. \tag{A.2}$$

We now prove $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq f_t$ by induction. The base case holds trivially. Suppose that $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq f_t$. Since $f_t \leq R_{\mathbf{x}}^{\text{loc}}$, we have $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq R_{\mathbf{x}}^{\text{loc}}$. Moreover, $\mathbf{x}_t \in \Sigma_s^n$ for [IHT](#), while $\mathbf{x}_t \in \Sigma_s^{n,*}$ for [NIHT](#). Thus, in light of (2.1) and (2.3), we have $\mathbf{x}_t \in \mathcal{U}$, and the assumed RAIC yields

$$\|\mathbf{x}_t - \mathbf{x} - \eta \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)\|_{2s,2} \leq \mu_1 \|\mathbf{x}_t - \mathbf{x}\|_2 + \mu_2.$$

Substituting this into (A.2) and using $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq f_t$, we obtain

$$\|\mathbf{x}_{t+1} - \mathbf{x}\|_2 \leq 2q\mu_1 \|\mathbf{x}_t - \mathbf{x}\|_2 + 2q\mu_2 + 3q\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq \rho f_t + b = f_{t+1}.$$

This completes the induction. Finally, substituting $q = 1$ and $q = 2$ into (A.1) yields (2.2) and (2.4), respectively, and completes the proof. $\square$

B Gaussian Width and Covering Number

Our subsequent proofs build on some concentration lemmas and existing results involving Gaussian width and covering number of $\mathcal{U} \subset \mathbb{R}^n$. Here, we shall pause to provide a brief introduction. With $\mathbf{g} \sim N(0, \mathbf{I}_n)$, the Gaussian width of $\mathcal{U} \subset \mathbb{R}^n$ is defined as

$$\omega(\mathcal{U}) = \mathbb{E} \sup_{\mathbf{u} \in \mathcal{U}} \mathbf{g}^\top \mathbf{u}.$$

Also, the covering number of $\mathcal{U}$ at radius r , denoted by $\mathcal{N}(\mathcal{U}, r)$, is the minimal number of radius- r ℓ_2 balls needed to cover $\mathcal{U}$. As an equivalent notion, $\log \mathcal{N}(\mathcal{U}, r)$ is referred to as the metric entropy of $\mathcal{U}$ at radius r . In the following, we collect the Gaussian width and metric entropy estimates used in this paper.

Lemma B.1. *Let $\{\mathbf{e}_j\}_{j=1}^n$ be the canonical basis in $\mathbb{R}^n$, $s \in [n]$, $r \in (0, 2)$, and $\mathcal{X}_1 := \{\mathbf{u} \in \mathbb{S}^{n-1} : \tau_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\}$ for some $c_* \in (0, 1)$. Then for some universal constant C ,*

$$\omega(\{\mathbf{e}_j\}_{j=1}^n) \leq C\sqrt{\log n}; \tag{B.1}$$

$$\omega(\Sigma_s^{n,*}) \leq \omega(\sqrt{s}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}) \leq C\sqrt{s \log \frac{en}{s}}; \tag{B.2}$$

$$\log \mathcal{N}(\Sigma_s^{n,*}, r) \leq s \log \left(\frac{12n}{sr} \right); \tag{B.3}$$

$$\omega(\mathcal{X}_1) \leq C\sqrt{s \log \frac{en}{s}}. \tag{B.4}$$

Proof. For the first three inequalities, see, e.g., [65, 55, 56]. We now prove (B.4). For $\mathbf{w} \in \mathcal{X}_1$,

$$\|\mathbf{w}\|_1 = \|\mathbf{w}_{[s]}\|_1 + \sum_{i \geq 1} \|\mathbf{w}_{[(i+1)s]} - \mathbf{w}_{[is]}\|_1 \leq \sqrt{s} \|\mathbf{w}_{[s]}\|_2 + \sum_{i \geq 1} \sqrt{s} \|\mathbf{w}_{[(i+1)s]} - \mathbf{w}_{[is]}\|_2 \leq (1 + c_*)\sqrt{s},$$

and hence $\mathcal{X}_1 \subset \mathbb{B}_1^n((1 + c_*)\sqrt{s}) \cap \mathbb{S}^{n-1}$. Thus, (B.2) gives $\omega(\mathcal{X}_1) \leq \omega(\mathbb{B}_1^n((1 + c_*)\sqrt{s}) \cap \mathbb{S}^{n-1}) \lesssim \sqrt{s \log(en/s)}$. $\square$

C Proofs for Sparse Phase Retrieval

Throughout this appendix, we work with $\mathbf{h}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{u}| - |\mathbf{a}_i^\top \mathbf{x}|) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i$ and consider $\mathbf{x} \neq 0$.

C.1 Proof of Theorem 3.1 (Instance Optimal SPR)

We let $\mathcal{X} := \{\mathbf{u} \in \mathbb{R}^n : \tau_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\}$ for some sufficiently small $c_* > 0$. For convenience, we shorten the sparse restricted eigenvalue of a matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$ as

$$\Gamma_s(\mathbf{M}) := \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \|\mathbf{M}\mathbf{u}\|_2. \quad (\text{C.1})$$

We note the following bound for a Gaussian matrix.

Lemma C.1 (See, e.g., [65, Section 9]). $\mathbf{A} \sim N^{m \times n}(0, 1)$ satisfies

$$\mathbb{P}\left(\Gamma_s\left(\frac{\mathbf{A}}{\sqrt{m}}\right) \leq 1 + \sqrt{\frac{Cs \log(en/s)}{m}}\right) \geq 1 - 2 \exp(-s \log(en/s)). \quad (\text{C.2})$$

C.1.1 Proof of the RAIC in (3.6)

For some $c > 0$, we seek to establish (3.6). By definition, this is equivalent to

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 4\tau_s(\mathbf{x}), \quad \forall \mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2), \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\}. \quad (\text{C.3})$$

We start with the decomposition

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \|\mathbf{u} - \mathbf{x}_{[s]} - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \quad (\text{C.4})$$

(i) **Bounding** $\|\mathbf{u} - \mathbf{x}_{[s]} - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$

Noticing that $\mathbf{u}$ and $\mathbf{x}_{[s]}$ are both s -sparse, we control the first term $\|\mathbf{u} - \mathbf{x}_{[s]} - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$ by using the RAIC established in [18].

Lemma C.2. If $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - c_1 \exp(-c_2 m)$,

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2,2s} \leq \frac{1}{4} \|\mathbf{u} - \mathbf{x}\|_2, \quad \forall \mathbf{u}, \mathbf{x} \in \Sigma_s^n \setminus \{0\} \text{ obeying } \|\mathbf{u} - \mathbf{x}\|_2 \leq \tilde{c} \|\mathbf{x}\|_2$$

holds for some small enough universal constant $\tilde{c} > 0$.

Proof. By Definition 2.1, the claim is equivalent to

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; \tilde{c}\|\mathbf{x}\|_2), \mathbf{x}, \frac{1}{4} \|\mathbf{u} - \mathbf{x}\|_2\right), \quad \forall \mathbf{x} \in \Sigma_s^n \setminus \{0\}.$$

The result then directly follows from [18, Theorem B.9] with $\mathcal{C} = \Sigma_s^n$, $\mathcal{K} = \Sigma_{2s}^{n,*}$, together with the estimates (B.2)–(B.3). $\square$

For $\mathbf{x} \in \mathcal{X} \setminus \{0\}$ and $\mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2)$, note that

$$\|\mathbf{u} - \mathbf{x}_{[s]}\|_2 \leq \|\mathbf{u} - \mathbf{x}\|_2 + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq c\|\mathbf{x}\|_2 + \tau_s(\mathbf{x}) \leq (c + c_*) \|\mathbf{x}\|_2.$$

Therefore, by setting $c \leq \frac{\tilde{c}}{4}$ and $c_* \leq \frac{\tilde{c}}{4}$ for the $\tilde{c}$ in Lemma C.2, we obtain the uniform bound

$$\|\mathbf{u} - \mathbf{x}_{[s]} - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}_{[s]}\|_2}{4} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + \frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{4}. \quad (\text{C.5})$$

(ii) **Bounding** $\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$

For all $\mathbf{x} \in \mathcal{X} \setminus \{0\}$ and $\mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2)$,

$$\begin{aligned}
\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} &= \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \langle \mathbf{w}, \mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u}) \rangle \\
&= \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{x}| - |\mathbf{a}_i^\top \mathbf{x}_{[s]}|) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i^\top \mathbf{w} \\
&\stackrel{(a)}{\leq} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m |\mathbf{a}_i^\top (\mathbf{x} - \mathbf{x}_{[s]})| |\mathbf{a}_i^\top \mathbf{w}| \\
&\stackrel{(b)}{\leq} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m \sum_{j \geq 1} |\mathbf{a}_i^\top (\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]})| |\mathbf{a}_i^\top \mathbf{w}| \\
&\stackrel{(c)}{\leq} \sum_{j \geq 1} \|\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}\|_2 \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \sup_{\mathbf{v} \in \Sigma_s^{n,*}} \frac{1}{m} \sum_{i=1}^m |\mathbf{v}^\top \mathbf{a}_i \mathbf{a}_i^\top \mathbf{w}| \\
&\stackrel{(d)}{\leq} \sum_{j \geq 1} \|\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}\|_2 \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{\|\mathbf{A}\mathbf{v}\|_2}{\sqrt{m}} \sup_{\mathbf{v} \in \Sigma_s^{n,*}} \frac{\|\mathbf{A}\mathbf{w}\|_2}{\sqrt{m}} \\
&\stackrel{(e)}{\leq} \sum_{j \geq 1} \|\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}\|_2 \cdot \Gamma_{2s}\left(\frac{\mathbf{A}}{\sqrt{m}}\right) \cdot \Gamma_s\left(\frac{\mathbf{A}}{\sqrt{m}}\right) \\
&\stackrel{(f)}{\leq} 2 \sum_{j \geq 1} \|\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}\|_2 = 2\tau_s(\mathbf{x}), \tag{C.6}
\end{aligned}$$

where (a) is due to triangle inequality, in (b) we use $\mathbf{x} - \mathbf{x}_{[s]} = \sum_{j \geq 1} (\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]})$ and triangle inequality, in (c) we suppose $\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]} \neq 0$ and utilize $\frac{\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}}{\|\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}\|_2} \in \Sigma_s^{n,*}$, (d) is due to Cauchy–Schwarz inequality, (e) follows from the definition of $\Gamma_s(\cdot)$ in (C.1), and (f) holds with the promised probability in light of Lemma C.1.

Substituting (C.5) and (C.6) into (C.4) and using $\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq \tau_s(\mathbf{x})$, we arrive at (C.3).

C.1.2 Proof of the Initialization Guarantee (3.7)

Since $\mathbf{x}_0 \in \Sigma_s^n$ by construction, to ensure (2.12), it remains to show

$$\text{dist}(\mathbf{x}_0, \mathbf{x}) \leq c\|\mathbf{x}\|_2, \quad \forall \mathbf{x} \in \mathcal{X} \tag{C.7}$$

for the constant c in (C.3). For convenience we let $\bar{\mathbf{x}} := \mathbf{x}/\|\mathbf{x}\|_2$ and define

$$\tilde{\mathbf{S}}_{\mathbf{x}} := \frac{1}{m} \sum_{i=1}^m T_{2\|\mathbf{x}\|_2}(y_i) \mathbf{a}_i \mathbf{a}_i^\top \tag{C.8}$$

as a surrogate of the data matrix $\hat{\mathbf{S}}_{\mathbf{x}} = \frac{1}{m} \sum_{i=1}^m T_{2\lambda_{\mathbf{x}}}(y_i) \mathbf{a}_i \mathbf{a}_i^\top$. The following lemma computes $\mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]$.

Lemma C.3. *For any $\mathbf{x} \in \mathbb{R}^n$, we have $\mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}] = \|\mathbf{x}\|_2 (c_b \mathbf{I}_n + c_\diamond \bar{\mathbf{x}} \bar{\mathbf{x}}^\top)$ where $c_b \approx 0.7809$, $c_\diamond \approx 0.6899$ are universal constants.*

Proof. The proof is a straightforward calculation based on the rotational invariance of $\mathbf{a}_i$. We omit the details. $\square$

(iii) A Deterministic Initialization Error Bound

We then provide a deterministic bound on the initialization error for $\mathbf{x} \in \mathcal{X} \setminus \{0\}$, i.e., the nonzero signals with small enough $\frac{\tau_s(\mathbf{x})}{\|\mathbf{x}\|_2}$. In fact, the bound is valid for a larger set of signals with small $\frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}\|_2}$; see (C.9) below. The proof is analogous to existing analysis (e.g., [39]) but needs additional technicalities to handle model error.

Lemma C.4. *Let $\mathbf{x}$ be nonzero. There exist universal constants $C > c > 0$ such that the following hold. For any $\bar{c} \in (0, c)$, if*

$$\frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}\|_2} \leq \bar{c}, \quad (\text{C.9})$$

$$\frac{|\lambda_{\mathbf{x}} - \|\mathbf{x}\|_2|}{\|\mathbf{x}\|_2} \leq \bar{c}, \quad (\text{C.10})$$

$$\Phi(\mathbf{x}) := \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top (\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]) \mathbf{u} \right| \leq \bar{c}, \quad (\text{C.11})$$

$$\Psi(\mathbf{x}) := \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \{\mathbf{e}_j\}_{j=1}^n} \left| \mathbf{u}^\top (\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]) \mathbf{u} \right| \leq \frac{\bar{c}}{s}, \quad (\text{C.12})$$

then $\text{dist}(\mathbf{x}_0, \mathbf{x}) \leq C\sqrt{\bar{c}}\|\mathbf{x}\|_2$.

Proof. We start with

$$\text{dist}(\mathbf{x}_0, \mathbf{x}) \leq \text{dist}(\mathbf{x}_0, \mathbf{x}_{[s]}) + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \stackrel{(\text{C.9})}{\leq} \text{dist}(\mathbf{x}_0, \mathbf{x}_{[s]}) + c_0\|\mathbf{x}\|_2, \quad (\text{C.13})$$

and notice that

$$\begin{aligned} \text{dist}(\mathbf{x}_0, \mathbf{x}_{[s]}) &= \text{dist}(\lambda_{\mathbf{x}} \mathbf{v}_{\mathbf{x}}, \mathbf{x}_{[s]}) \\ &\leq \text{dist}(\lambda_{\mathbf{x}} \mathbf{v}_{\mathbf{x}}, \|\mathbf{x}\|_2 \mathbf{v}_{\mathbf{x}}) + \text{dist}\left(\|\mathbf{x}\|_2 \mathbf{v}_{\mathbf{x}}, \frac{\|\mathbf{x}\|_2 \mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2}\right) + \text{dist}\left(\frac{\|\mathbf{x}\|_2 \mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2}, \mathbf{x}_{[s]}\right) \\ &\leq |\lambda_{\mathbf{x}} - \|\mathbf{x}\|_2| + \|\mathbf{x}\|_2 \text{dist}\left(\mathbf{v}_{\mathbf{x}}, \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2}\right) + \|\mathbf{x}\|_2 \left\| \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}\|_2} \right\|_2, \end{aligned} \quad (\text{C.14})$$

where in the last inequality we use triangle inequality and $\text{dist}(\mathbf{u}, \mathbf{v}) \leq \|\mathbf{u} - \mathbf{v}\|_2$. We are left to bound the second and third terms in (C.14).

Bounding the third term in (C.14). By triangle inequality,

$$\left\| \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}\|_2} \right\|_2 \leq \left\| \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 + \left\| \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}\|_2} \right\|_2, \quad (\text{C.15})$$

and moreover,

$$\left\| \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}\|_2} \right\|_2 = \left| 1 - \frac{\|\mathbf{x}_{[s]}\|_2}{\|\mathbf{x}\|_2} \right| \leq \frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}\|_2} \stackrel{(\text{C.9})}{\leq} \bar{c}. \quad (\text{C.16})$$

For the first term on the right-hand side of (C.15), by triangle inequality,

$$\left\| \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 \leq \left\| \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}\|_2} - \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 + \left\| \frac{\mathbf{x}_{\mathcal{I}_{\mathbf{x}}}}{\|\mathbf{x}_{[s]}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2$$

$$= \frac{||\mathbf{x}_{[s]}||_2 - \|\mathbf{x}_{\mathcal{I}_x}\|_2 + \|\mathbf{x}_{\mathcal{I}_x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}_{[s]}\|_2} \leq \frac{2\|\mathbf{x}_{\mathcal{I}_x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}_{[s]}\|_2}. \quad (\text{C.17})$$

We let $I_0 \subset [n]$ be a size- s index set such that $\text{supp}(\mathbf{x}_{[s]}) \subset I_0$ and notice

$$\|\mathbf{x}_{\mathcal{I}_x} - \mathbf{x}_{[s]}\|_2 \leq \|\mathbf{x}_{\mathcal{I}_x \setminus I_0}\|_2 + \|\mathbf{x}_{I_0 \setminus \mathcal{I}_x}\|_2, \quad (\text{C.18})$$

where $\|\mathbf{x}_{\mathcal{I}_x \setminus I_0}\|_2$ can be bounded by

$$\|\mathbf{x}_{\mathcal{I}_x \setminus I_0}\|_2 \leq \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \stackrel{(\text{C.9})}{\leq} \bar{c}\|\mathbf{x}\|_2. \quad (\text{C.19})$$

To bound $\|\mathbf{x}_{I_0 \setminus \mathcal{I}_x}\|_2$, with no loss of generality we suppose $I_0 \setminus \mathcal{I}_x \neq \emptyset$, which together with $|I_0| = |\mathcal{I}_x| = s$ leads to

$$|\mathcal{I}_x \setminus I_0| = |I_0 \setminus \mathcal{I}_x| \geq 1.$$

In turn, for any $j \in I_0 \setminus \mathcal{I}_x$, there exists $k_i \in \mathcal{I}_x \setminus I_0$ such that $k_i \neq k_{i'}$ if $i \neq i'$, and by the construction of $\mathcal{I}_x$,

$$[\hat{\mathbf{S}}_{\mathbf{x}}]_{j,j} \leq [\hat{\mathbf{S}}_{\mathbf{x}}]_{k_j,k_j}, \quad \forall j \in I_0 \setminus \mathcal{I}_x. \quad (\text{C.20})$$

Moreover, (C.12) along with Lemma C.3 implies

$$[\hat{\mathbf{S}}_{\mathbf{x}}]_{j,j} - \|\mathbf{x}\|_2 \left(c_b + \frac{c_\diamond x_j^2}{\|\mathbf{x}\|_2^2} \right) \geq -\frac{\bar{c}\|\mathbf{x}\|_2}{s}, \quad (\text{C.21})$$

$$[\hat{\mathbf{S}}_{\mathbf{x}}]_{k_j,k_j} - \|\mathbf{x}\|_2 \left(c_b + \frac{c_\diamond x_{k_j}^2}{\|\mathbf{x}\|_2^2} \right) \leq \frac{\bar{c}\|\mathbf{x}\|_2}{s}. \quad (\text{C.22})$$

Combining (C.20)–(C.22) yields

$$x_j^2 \leq x_{k_j}^2 + \frac{2\bar{c}}{c_\diamond s} \|\mathbf{x}\|_2^2, \quad \forall j \in I_0 \setminus \mathcal{I}_x, \quad (\text{C.23})$$

and therefore

$$\begin{aligned} \|\mathbf{x}_{I_0 \setminus \mathcal{I}_x}\|_2 &= \left(\sum_{j \in I_0 \setminus \mathcal{I}_x} x_j^2 \right)^{1/2} \\ &\stackrel{(a)}{\leq} \left(\sum_{j \in I_0 \setminus \mathcal{I}_x} \left[x_{k_j}^2 + \frac{2\bar{c}\|\mathbf{x}\|_2^2}{c_\diamond s} \right] \right)^{1/2} \\ &\stackrel{(b)}{\leq} \left(\|\mathbf{x} - \mathbf{x}_{[s]}\|_2^2 + \frac{2\bar{c}\|\mathbf{x}\|_2^2}{c_\diamond} \right)^{1/2} \\ &\stackrel{(c)}{\leq} \left(c_0^2 + \frac{2\bar{c}}{c_\diamond} \right)^{1/2} \|\mathbf{x}\|_2, \end{aligned} \quad (\text{C.24})$$

where (a) follows from Equation (C.23), (b) holds because $k_i \neq k_{i'}$ for $i \neq i'$ and $|I_0 \setminus \mathcal{I}_x| \leq s$, and (c) is due to Equation (C.9).

Substituting (C.19) and (C.24) into (C.18) yields

$$\|\mathbf{x}_{\mathcal{I}_x} - \mathbf{x}_{[s]}\|_2 \lesssim \sqrt{\bar{c}}\|\mathbf{x}\|_2. \quad (\text{C.25})$$

In light of (C.17), along with $\|\mathbf{x}_{[s]}\|_2 \geq (1 - \bar{c})\|\mathbf{x}\|_2$ from (C.9) and taking sufficiently small c (and hence $\bar{c}$), we obtain

$$\left\| \frac{\mathbf{x}_{\mathcal{I}_x}}{\|\mathbf{x}_{\mathcal{I}_x}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 \lesssim \frac{\sqrt{\bar{c}}}{1 - \bar{c}} \lesssim \sqrt{\bar{c}}. \quad (\text{C.26})$$

Substituting this and (C.16) into (C.15) leads to

$$\left\| \frac{\mathbf{x}_{\mathcal{I}_x}}{\|\mathbf{x}_{\mathcal{I}_x}\|_2} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}\|_2} \right\|_2 \lesssim \sqrt{\bar{c}}. \quad (\text{C.27})$$

Bounding the second term in (C.14). By Algorithm 1 and Lemma C.3, $\mathbf{v}_x$ and $\mathbf{x}_{\mathcal{I}_x}/\|\mathbf{x}_{\mathcal{I}_x}\|_2$ are the leading eigenvectors of $[\hat{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x}$ and $[\mathbb{E}\tilde{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x}$, respectively. Therefore,

$$\|[\hat{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x} - [\mathbb{E}\tilde{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x}\|_{op} \stackrel{(a)}{\leq} \sup_{\substack{\mathcal{I} \subset [n] \\ |\mathcal{I}| \leq s}} \|[\hat{\mathbf{S}}_x]_{\mathcal{I}, \mathcal{I}} - [\mathbb{E}\tilde{\mathbf{S}}_x]_{\mathcal{I}, \mathcal{I}}\|_{op} = \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top (\hat{\mathbf{S}}_x - \mathbb{E}[\tilde{\mathbf{S}}_x]) \mathbf{u} \right| \stackrel{(b)}{\leq} \bar{c} \|\mathbf{x}\|_2,$$

where (a) holds because $|\mathcal{I}_x| \leq s$, (b) is due to (C.11).

By Lemma C.3, the first two eigenvalues of

$$[\mathbb{E}\tilde{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x} = \left[c_b \|\mathbf{x}\|_2 \mathbf{I}_n + c_\diamond \frac{\mathbf{x}\mathbf{x}^\top}{\|\mathbf{x}\|_2} \right]_{\mathcal{I}_x, \mathcal{I}_x} = c_b \|\mathbf{x}\|_2 [\mathbf{I}_n]_{\mathcal{I}_x, \mathcal{I}_x} + \frac{c_\diamond \mathbf{x}_{\mathcal{I}_x} \mathbf{x}_{\mathcal{I}_x}^\top}{\|\mathbf{x}\|_2}$$

are $c_b \|\mathbf{x}\|_2 + \frac{c_\diamond \|\mathbf{x}_{\mathcal{I}_x}\|_2^2}{\|\mathbf{x}\|_2}$ and $c_b \|\mathbf{x}\|_2$ (or 0 if $s = 1$), thus the eigenvalue gap is at least $\frac{c_\diamond \|\mathbf{x}_{\mathcal{I}_x}\|_2^2}{\|\mathbf{x}\|_2}$. Moreover, by (C.25), $\|\mathbf{x}_{[s]}\|_2 \geq (1 - \bar{c})\|\mathbf{x}\|_2$, and sufficiently small $\bar{c}$,

$$\|\mathbf{x}_{\mathcal{I}_x}\|_2 \geq \|\mathbf{x}_{[s]}\|_2 - \|\mathbf{x}_{[s]} - \mathbf{x}_{\mathcal{I}_x}\|_2 \geq (1 - O(\sqrt{\bar{c}}))\|\mathbf{x}\|_2 \geq \frac{\|\mathbf{x}\|_2}{2}.$$

Hence, the eigengap between the first two eigenvalues of $[\mathbb{E}\tilde{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x}$ is at least $\frac{c_\diamond \|\mathbf{x}_{\mathcal{I}_x}\|_2^2}{\|\mathbf{x}\|_2} \geq \frac{c_\diamond \|\mathbf{x}\|_2}{4}$. By sufficiently small $\bar{c}$ and C.11,

$$\|[\hat{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x} - [\mathbb{E}\tilde{\mathbf{S}}_x]_{\mathcal{I}_x, \mathcal{I}_x}\|_{op} \leq \frac{c_\diamond \|\mathbf{x}\|_2}{8}.$$

Hence, Davis-Kahan's theorem [23], along with the simple bound $\text{dist}(\mathbf{u}, \mathbf{v}) \leq \sqrt{2} \|\mathbf{u}\mathbf{u}^\top - \mathbf{v}\mathbf{v}^\top\|_{op}$ for $\mathbf{u}, \mathbf{v} \in \mathbb{S}^{n-1}$, yields

$$\text{dist} \left(\mathbf{v}_x, \frac{\mathbf{x}_{\mathcal{I}_x}}{\|\mathbf{x}_{\mathcal{I}_x}\|_2} \right) \leq \sqrt{2} \left\| \mathbf{v}_x \mathbf{v}_x^\top - \frac{\mathbf{x}_{\mathcal{I}_x} \mathbf{x}_{\mathcal{I}_x}^\top}{\|\mathbf{x}_{\mathcal{I}_x}\|_2^2} \right\|_{op} \lesssim \bar{c}. \quad (\text{C.28})$$

By substituting $|\lambda_x - \|\mathbf{x}\|_2| \leq \bar{c} \|\mathbf{x}\|_2$, (C.27), (C.28), and (C.14) into (C.13), we reach $\text{dist}(\mathbf{x}_0, \mathbf{x}) \lesssim \sqrt{\bar{c}} \|\mathbf{x}\|_2$, completing the proof. $\square$

Therefore, to achieve (C.7), it is sufficient to establish (C.9)–(C.12) with small enough $\bar{c}$ for all $\mathbf{x} \in \mathcal{X}$. By the definition of $\mathcal{X}$, (C.9) holds as long as c_* is small enough. We proceed to establish (C.10)–(C.12) separately.

(iv) Establishing (C.10) for $\mathbf{x} \in \mathcal{X} \setminus \{0\}$

The following lemma, which follows from an ℓ_1 embedding result due to [57], guarantees (C.10) under $m \gtrsim s \log \frac{en}{s}$.

Lemma C.5. *If $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - 2 \exp(-cs \log \frac{en}{s})$,*

$$|\lambda_{\mathbf{x}} - \|\mathbf{x}\|_2| \leq C \sqrt{\frac{s \log(en/s)}{m}} \cdot \|\mathbf{x}\|_2, \quad \forall \mathbf{x} \in \mathcal{X}.$$

Proof. We only need to prove

$$Z := \sup_{\mathbf{x} \in \mathcal{X}} \left| \frac{\lambda_{\mathbf{x}}}{\|\mathbf{x}\|_2} - 1 \right| = O\left(\sqrt{\frac{s \log(en/s)}{m}}\right).$$

By homogeneity, we let $\mathcal{X}^* = \mathcal{X} \cap \mathbb{S}^{n-1}$ to obtain

$$Z = \sup_{\mathbf{x} \in \mathcal{X}} \left| \sqrt{\frac{\pi}{2}} \frac{1}{m} \sum_{i=1}^m \left| \mathbf{a}_i^\top \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \right| - 1 \right| = \sup_{\mathbf{x} \in \mathcal{X}^*} \left| \sqrt{\frac{\pi}{2}} \frac{1}{m} \sum_{i=1}^m |\mathbf{a}_i^\top \mathbf{x}| - 1 \right|.$$

By [57, Lemma 2.1],

$$\mathbb{P}\left(Z > \frac{4\omega(\mathcal{X}^*)}{\sqrt{m}} + t\right) \leq 2 \exp\left(-\frac{mt^2}{2}\right), \quad \forall t > 0. \quad (\text{C.29})$$

By (B.4), we set $t = \sqrt{s \log(en/s)/m}$ in (C.29) to conclude the proof. $\square$

(v) Establishing (C.11)–(C.12) for $\mathbf{x} \in \mathcal{X} \setminus \{0\}$

To ensure (C.11)–(C.12) for $\mathbf{x} \in \mathcal{X} \setminus \{0\}$, we shall establish upper bound on $\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \Phi(\mathbf{x})$ and $\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \Psi(\mathbf{x})$.

Lemma C.6. *If $m \gtrsim s \log \frac{en}{s}$, then*

$$\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \Psi(\mathbf{x}) \leq \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \Phi(\mathbf{x}) \lesssim \log\left(\frac{em}{s \log(en/s)}\right) \sqrt{\frac{s \log(en/s)}{m}} \quad (\text{C.30})$$

holds with probability at least $1 - C \exp(-cs \log \frac{en}{s})$.

Before proving Lemma C.6, we record a few auxiliary results. The first is an elementary contraction bound for products.

Lemma C.7 (Product contraction). *Let $\mathcal{P}, \mathcal{Q} \subset [0, 1]^m$ be countable sets, and let $\{\varepsilon_i\}_{i=1}^m$ be independent Rademacher random variables. Then*

$$\mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}, q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i p_i q_i \right| \lesssim \mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}} \left| \sum_{i=1}^m \varepsilon_i p_i \right| + \mathbb{E}_\varepsilon \sup_{q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i q_i \right|.$$

Proof. For any $p, q \in \mathbb{R}$, we have $pq = \frac{1}{4}[(p+q)^2 - (p-q)^2]$. Therefore, by triangle inequality,

$$\mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}, q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i p_i q_i \right| \leq \frac{1}{4} \mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}, q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i (p_i + q_i)^2 \right| + \frac{1}{4} \mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}, q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i (p_i - q_i)^2 \right|.$$

Since $p_i + q_i \in [0, 2]$ and $p_i - q_i \in [-1, 1]$, the maps $a \mapsto a^2$ on the two ranges admit globally Lipschitz extensions, vanishing at zero, with Lipschitz constants 4 and 2, respectively. Thus, the contraction principle (see, e.g., [65, Chapter 6]) gives

$$\begin{aligned} \mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}, q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i p_i q_i \right| &\lesssim \mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}, q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i (p_i + q_i) \right| + \mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}, q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i (p_i - q_i) \right| \\ &\leq 2\mathbb{E}_\varepsilon \sup_{p \in \mathcal{P}} \left| \sum_{i=1}^m \varepsilon_i p_i \right| + 2\mathbb{E}_\varepsilon \sup_{q \in \mathcal{Q}} \left| \sum_{i=1}^m \varepsilon_i q_i \right|, \end{aligned}$$

which proves the claim. $\square$

The following Massart's concentration inequality follows by taking $\eta = 0.1$ in [1, Theorem 2] and absorbing the resulting numerical constants into C . Here, $\|f\|_\infty := \sup_x |f(x)|$ denotes the supremum norm, where the supremum is taken over the underlying measurable space.

Lemma C.8 (Massart's inequality; see [46] and [1, Theorem 2]). *Let $X_1, \dots, X_m$ be independent random variables taking values in a measurable space, and let $\mathcal{F}$ be a countable class of measurable functions such that $\mathbb{E}f(X_i) = 0$, $\|f\|_\infty \leq b$ ($\forall f \in \mathcal{F}$), $i \in [m]$. Define*

$$Z := \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^m f(X_i) \right|, \quad \sigma^2 := \sup_{f \in \mathcal{F}} \sum_{i=1}^m \mathbb{E}f^2(X_i).$$

Then, for every $t \geq 0$, with probability at least $1 - \exp(-t)$,

$$Z \leq 1.1\mathbb{E}Z + C(\sigma\sqrt{t} + bt),$$

where $C > 0$ is a universal constant.

The third lemma provides a uniform upper bound on Gaussian empirical quadratic forms over all size- ηm subsets of the samples. This concentration bound is recurring in subsequent analysis.

Lemma C.9. [24, Theorem 2.10] *Let $\mathbf{a}_1, \dots, \mathbf{a}_m$ be i.i.d. copies of $N(0, \mathbf{I}_n)$, and let $\mathcal{U} \subset \mathbb{R}^n$. For any $\eta \in \{\frac{1}{m}, \frac{2}{m}, \dots, \frac{m-1}{m}, 1\}$, the event*

$$\sup_{\mathbf{u} \in \mathcal{U}} \max_{\substack{\mathcal{G} \subset [m] \\ |\mathcal{G}| = \eta m}} \left(\frac{1}{\eta m} \sum_{i \in \mathcal{G}} |\mathbf{a}_i^\top \mathbf{u}|^2 \right)^{1/2} \lesssim \frac{\omega(\mathcal{U})}{\sqrt{\eta m}} + \left(\sup_{\mathbf{u} \in \mathcal{U}} \|\mathbf{u}\|_2 \right) \sqrt{\log \frac{e}{\eta}}$$

holds with probability at least $1 - 2\exp(-c\eta m \log(\frac{e}{\eta}))$.

We are now ready to prove Lemma C.6.

Proof of Lemma C.6. We start with

$$\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \Phi(\mathbf{x}) \leq \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top [\hat{\mathbf{S}}_{\mathbf{x}} - \tilde{\mathbf{S}}_{\mathbf{x}}] \mathbf{u} \right| + \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top (\tilde{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]) \mathbf{u} \right|. \quad (\text{C.31})$$

The first term can be bounded by

$$\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top [\hat{\mathbf{S}}_{\mathbf{x}} - \tilde{\mathbf{S}}_{\mathbf{x}}] \mathbf{u} \right|$$

$$\begin{aligned}
&\stackrel{(a)}{\leq} \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \frac{1}{m} \sum_{i=1}^m \left[\mathsf{T}_{2\lambda_{\mathbf{x}}}(|\mathbf{a}_i^\top \mathbf{x}|) - \mathsf{T}_{2\|\mathbf{x}\|_2}(|\mathbf{a}_i^\top \mathbf{x}|) \right] (\mathbf{a}_i^\top \mathbf{u})^2 \right| \\
&\stackrel{(b)}{\leq} \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \frac{2}{m} \sum_{i=1}^m \left| \frac{\lambda_{\mathbf{x}}}{\|\mathbf{x}\|_2} - 1 \right| (\mathbf{a}_i^\top \mathbf{u})^2 \\
&\stackrel{(c)}{\lesssim} \sqrt{\frac{s \log(en/s)}{m}} \cdot \left[\Gamma_s \left(\frac{\mathbf{A}}{\sqrt{m}} \right) \right]^2 \stackrel{(d)}{\lesssim} \sqrt{\frac{s \log(en/s)}{m}},
\end{aligned} \tag{C.32}$$

where in (a) we substitute (3.3) and (C.8), (b) follows from triangle inequality and

$$\sup_{a \in \mathbb{R}} |\mathsf{T}_{\xi_1}(a) - \mathsf{T}_{\xi_2}(a)| \leq |\xi_1 - \xi_2|,$$

(c) is due to the event in Lemma C.5, and (d) is due to the event in Lemma C.1.

To bound the second term in (C.31), we utilize the homogeneity to restrict to $\mathcal{X}^* = \mathcal{X} \cap \mathbb{S}^{n-1}$:

$$\begin{aligned}
&\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \mathbf{u}^\top (\tilde{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]) \mathbf{u} \right| \\
&= \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \frac{1}{m} \sum_{i=1}^m \mathsf{T}_2(|\mathbf{a}_i^\top \bar{\mathbf{x}}|) (\mathbf{a}_i^\top \mathbf{u})^2 - \mathbb{E} \left[\mathsf{T}_2(|\mathbf{a}_i^\top \bar{\mathbf{x}}|) (\mathbf{a}_i^\top \mathbf{u})^2 \right] \right| \\
&= \sup_{\mathbf{w} \in \mathcal{X}^*} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \frac{1}{m} \sum_{i=1}^m \mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) (\mathbf{a}_i^\top \mathbf{u})^2 - \mathbb{E} \left[\mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) (\mathbf{a}_i^\top \mathbf{u})^2 \right] \right|.
\end{aligned} \tag{C.33}$$

The difficulty in (C.33) is that $(\mathbf{a}_i^\top \mathbf{u})^2$ is unbounded. We truncate this factor, use symmetrization and contraction for the resulting bounded process, and then invoke Lemma C.9 to control the discarded tail uniformly over $\mathbf{u}$. Specifically, let

$$\tau = C_0 \log \left(\frac{em}{s \log(en/s)} \right), \tag{C.34}$$

where $C_0 > 0$ is a sufficiently large universal constant. By triangle inequality, the process in (C.33) is bounded by $Z_{\text{bd}} + Z_{\text{tail}}$, where

$$\begin{aligned}
Z_{\text{bd}} &:= \sup_{\mathbf{w} \in \mathcal{X}^*} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \frac{1}{m} \sum_{i=1}^m \mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau) - \mathbb{E} \left[\mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau) \right] \right|, \\
Z_{\text{tail}} &:= \sup_{\mathbf{w} \in \mathcal{X}^*} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \frac{1}{m} \sum_{i=1}^m \mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) \left[(\mathbf{a}_i^\top \mathbf{u})^2 - ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau) \right] - \mathbb{E} \left[\mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) \left((\mathbf{a}_i^\top \mathbf{u})^2 - ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau) \right) \right] \right|.
\end{aligned}$$

Bounding Z_{bd} . Since the indexed functions are continuous in $(\mathbf{w}, \mathbf{u})$, and their expectations are continuous by bounded convergence, we may restrict the two index sets to countable dense subsets without changing the suprema. Let $\{\varepsilon_i\}_{i=1}^m$ be independent Rademacher random variables, independent of $\{\mathbf{a}_i\}_{i=1}^m$. By symmetrization (see, e.g., [65, Chapter 6]) and Lemma C.7, applied conditionally on $\{\mathbf{a}_i\}_{i=1}^m$ with $p_i = \frac{\mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|)}{2}$, $q_i = \frac{(\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau}{\tau}$, we obtain

$$\mathbb{E} Z_{\text{bd}} \lesssim \frac{1}{m} \mathbb{E} \sup_{\mathbf{w} \in \mathcal{X}^*} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \sum_{i=1}^m \varepsilon_i \mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau) \right|$$

$$\lesssim \frac{\tau}{m} \mathbb{E} \sup_{\mathbf{w} \in \mathcal{X}^*} \left| \sum_{i=1}^m \varepsilon_i \mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) \right| + \frac{1}{m} \mathbb{E} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \sum_{i=1}^m \varepsilon_i ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau) \right|.$$

The maps $a \mapsto \mathsf{T}_2(|a|)$ and $a \mapsto a^2 \wedge \tau$ vanish at zero and are 1-Lipschitz and $2\sqrt{\tau}$ -Lipschitz, respectively. Thus, another application of the contraction principle in [65, Chapter 6] yields

$$\begin{aligned} \mathbb{E} Z_{\text{bd}} &\lesssim \frac{\tau}{m} \mathbb{E} \sup_{\mathbf{w} \in \mathcal{X}^*} \left| \sum_{i=1}^m \varepsilon_i \mathbf{a}_i^\top \mathbf{w} \right| + \frac{\sqrt{\tau}}{m} \mathbb{E} \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \left| \sum_{i=1}^m \varepsilon_i \mathbf{a}_i^\top \mathbf{u} \right| \\ &\stackrel{(a)}{=} \frac{\tau \omega(\mathcal{X}^*) + \sqrt{\tau} \omega(\Sigma_s^{n,*})}{\sqrt{m}} \stackrel{(b)}{\lesssim} \tau \sqrt{\frac{s \log(en/s)}{m}}, \end{aligned}$$

where (a) follows from $\sum_{i=1}^m \varepsilon_i \mathbf{a}_i \sim N(0, m\mathbf{I}_n)$ and the symmetry of $\mathcal{X}^*$ and $\Sigma_s^{n,*}$, while (b) follows from (B.4) and (B.2).

We next pass from the expectation to a high-probability bound. Apply Lemma C.8 to the centered functions indexed by $(\mathbf{w}, \mathbf{u}) \in \mathcal{X}^* \times \Sigma_s^{n,*}$. Since $0 \leq \mathsf{T}_2(|a|) \leq 2$, every centered function is bounded in absolute value by 2τ . Moreover, for every $\mathbf{w} \in \mathcal{X}^*$ and $\mathbf{u} \in \Sigma_s^{n,*}$,

$$\text{Var} \left(\mathsf{T}_2(|\mathbf{a}_i^\top \mathbf{w}|) ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau) \right) \leq \mathbb{E} \left[\mathsf{T}_2^2(|\mathbf{a}_i^\top \mathbf{w}|) ((\mathbf{a}_i^\top \mathbf{u})^2 \wedge \tau)^2 \right] \leq 4\mathbb{E}(\mathbf{a}_i^\top \mathbf{u})^4 = 12.$$

Therefore, we take $t = s \log(en/s)$ in Lemma C.8. With probability at least $1 - \exp(-s \log(en/s))$,

$$Z_{\text{bd}} \lesssim \mathbb{E} Z_{\text{bd}} + \sqrt{\frac{s \log(en/s)}{m}} + \frac{\tau s \log(en/s)}{m} \lesssim \tau \sqrt{\frac{s \log(en/s)}{m}}, \quad (\text{C.35})$$

where the last inequality uses $m \gtrsim s \log(en/s)$ and $\tau \geq 1$.

Bounding Z_{tail} . using $0 \leq \mathsf{T}_2(|a|) \leq 2$ and $a^2 - (a^2 \wedge \tau) = (a^2 - \tau)_+ \leq a^2 \mathbb{1}(a^2 > \tau)$, we have

$$Z_{\text{tail}} \leq 2 \underbrace{\sup_{\mathbf{u} \in \Sigma_s^{n,*}} \frac{1}{m} \sum_{i=1}^m (\mathbf{a}_i^\top \mathbf{u})^2 \mathbb{1}((\mathbf{a}_i^\top \mathbf{u})^2 > \tau)}_{:= \Xi_1} + 2 \underbrace{\sup_{\mathbf{u} \in \Sigma_s^{n,*}} \mathbb{E} \left[(\mathbf{a}_i^\top \mathbf{u})^2 \mathbb{1}((\mathbf{a}_i^\top \mathbf{u})^2 > \tau) \right]}_{:= \Xi_2}. \quad (\text{C.36})$$

Bounding Ξ_1 . Let $k = \lceil s \log \frac{en}{s} \rceil$. Under a sufficiently large implied constant in $m \gtrsim s \log(en/s)$, we have $k \leq m$. Applying Lemma C.9 with $\eta = k/m$ and $\mathcal{U} = \Sigma_s^{n,*}$, and then using (B.2), we obtain, with probability at least $1 - 2 \exp(-cs \log(en/s))$,

$$\sup_{\mathbf{u} \in \Sigma_s^{n,*}} \max_{\substack{\mathcal{G} \subset [m] \\ |\mathcal{G}|=k}} \frac{1}{m} \sum_{i \in \mathcal{G}} (\mathbf{a}_i^\top \mathbf{u})^2 \lesssim \frac{\omega^2(\Sigma_s^{n,*})}{m} + \frac{k}{m} \log \frac{em}{k} \lesssim \frac{s \log(en/s)}{m} \log \left(\frac{em}{s \log(en/s)} \right). \quad (\text{C.37})$$

For any $\mathbf{u} \in \Sigma_s^{n,*}$, let $\mathcal{I}_{\mathbf{u}} := \{i \in [m] : (\mathbf{a}_i^\top \mathbf{u})^2 > \tau\}$, then

$$\Xi_1 \leq \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \frac{1}{m} \sum_{i \in \mathcal{I}_{\mathbf{u}}} (\mathbf{a}_i^\top \mathbf{u})^2. \quad (\text{C.38})$$

We claim that, on the event (C.37), $|\mathcal{I}_{\mathbf{u}}| < k$ for every $\mathbf{u} \in \Sigma_s^{n,*}$. Indeed, if $|\mathcal{I}_{\mathbf{u}}| \geq k$, we can choose a size- k set $\mathcal{G} \subset \mathcal{I}_{\mathbf{u}}$, which together with (C.34) gives

$$\frac{1}{m} \sum_{i \in \mathcal{G}} (\mathbf{a}_i^\top \mathbf{u})^2 > \frac{k\tau}{m} \geq \frac{C_0 s \log(en/s)}{m} \log \left(\frac{em}{s \log(en/s)} \right),$$

contradicting (C.37) when C_0 is sufficiently large.

In view of (C.38), since all the summands are nonnegative, we may enlarge $\mathcal{I}_{\mathbf{u}}$ to a size- k subset of $[m]$ and conclude that

$$\Xi_1 \leq \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \max_{\substack{\mathcal{G} \subset [m] \\ |\mathcal{G}|=k}} \frac{1}{m} \sum_{i \in \mathcal{G}} (\mathbf{a}_i^\top \mathbf{u})^2 \stackrel{(C.37)}{\lesssim} \frac{s \log(en/s)}{m} \log \left(\frac{em}{s \log(en/s)} \right). \quad (C.39)$$

Bounding Ξ_2 . By rotational invariance and elementary Gaussian calculation,

$$\Xi_2 := \mathbb{E}[g^2 \mathbb{1}(g^2 > \tau)] \stackrel{(a)}{\lesssim} (\sqrt{\tau} + 1) \exp(-\tau/2) \stackrel{(b)}{\lesssim} \frac{s \log(en/s)}{m} \log \left(\frac{em}{s \log(en/s)} \right), \quad (C.40)$$

where (a) follows from integration by parts, and (b) holds because

$$\exp(-\tau/2) = \left(\frac{em}{s \log(en/s)} \right)^{-C_0/2} \lesssim \frac{s \log(en/s)}{m},$$

and $\sqrt{\tau} + 1 \lesssim \log \left(\frac{em}{s \log(en/s)} \right)$ when C_0 is a fixed sufficiently large constant.

Therefore, substituting (C.39) and (C.40) into (C.36) yields

$$Z_{\text{tail}} \lesssim \frac{s \log(en/s)}{m} \log \left(\frac{em}{s \log(en/s)} \right) \lesssim \log \left(\frac{em}{s \log(en/s)} \right) \sqrt{\frac{s \log(en/s)}{m}}. \quad (C.41)$$

Combining (C.33), (C.35), and (C.41), we conclude that the second term in (C.31) is bounded by

$$C \log \left(\frac{em}{s \log(en/s)} \right) \sqrt{\frac{s \log(en/s)}{m}}$$

with probability at least $1 - C \exp(-cs \log(en/s))$. Combining this with (C.32) proves (C.30). $\square$

We are ready to combine the previous components to establish (C.7), and therefore conclude with (3.7).

Lemma C.10 (Uniform initialization). *Under small enough c_* , if*

$$m \gtrsim s^3 \log \frac{en}{s} \log^2 \left(\frac{em}{s \log(en/s)} \right), \quad (C.42)$$

then (C.7) holds with probability at least $1 - C \exp(-cs \log \frac{en}{s})$.

Proof. By Lemma C.4, it is sufficient to verify (C.9)–(C.12) with small enough $\bar{c}$ for all $\mathbf{x} \in \mathcal{X} \setminus \{0\}$. (C.9) is guaranteed by using small enough c_* in $\mathcal{X}$. Under $m \gtrsim s \log(en/s)$, Lemma C.5 implies (C.10) over all $\mathbf{x} \in \mathcal{X} \setminus \{0\}$. To ensure (C.11)–(C.12) over $\mathbf{x} \in \mathcal{X} \setminus \{0\}$, in light of Lemma C.6, with the promised probability we only need to guarantee

$$\log \left(\frac{em}{s \log(en/s)} \right) \sqrt{\frac{s \log(en/s)}{m}} \leq \frac{c}{s}$$

for small enough c , which is exactly (C.42) with large enough hidden constant. $\square$

C.1.3 Projection Has No Effect on $\mathbf{x} \in \mathcal{X}$

As discussed in the proof outline in Section 3.2, the previous analysis already yields the following: if $\{\mathbf{x}_t\}_{t \geq 0}$ is obtained by running

$$\mathbf{x}_{t+1} = H_s(\mathbf{x}_t - \mathbf{h}_{\mathbf{x}}(\mathbf{x}_t)), \quad t \geq 0$$

with $\mathbf{x}_0$ obeying (C.7) satisfies (3.8), then (3.8) holds. Now we want to show that (3.8) *remains true for Algorithm 1 which incorporates the additional projection onto $\mathbb{B}_2^n(2\lambda_{\mathbf{x}})$* . To this end, it is enough to show that *the projection has no impact on the iterates satisfying (3.8)*. More specifically, we proceed to show the following: if (3.8) holds, then

$$\|\mathbf{x}_t\|_2 \leq 2\lambda_{\mathbf{x}}, \quad \forall t \geq 0, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (\text{C.43})$$

(vi) Establishing (C.43)

Under small enough c, c_* , (3.8) yields

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq c\|\mathbf{x}\|_2 + 22c_*\|\mathbf{x}\|_2 < \frac{\|\mathbf{x}\|_2}{2},$$

and therefore $\|\mathbf{x}_t\|_2 \leq \frac{3}{2}\|\mathbf{x}\|_2$. On the other hand, (C.10) with small c_1 leads to $\lambda_{\mathbf{x}} \geq \frac{4\|\mathbf{x}\|_2}{5}$, and hence

$$2\lambda_{\mathbf{x}} \geq \frac{8}{5}\|\mathbf{x}\|_2 \geq \frac{3}{2}\|\mathbf{x}\|_2 > \|\mathbf{x}_t\|_2, \quad \forall \mathbf{x} \in \mathcal{X}.$$

Combining the previous two displays yields (C.43).

C.1.4 A Separate Argument for $\mathbf{x} \in \mathcal{X}^c$

We show that $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$ can guarantee the desired bound for $\mathbf{x} \in \mathcal{X}^c = \{\mathbf{u} \in \mathbb{R}^n : \tau_s(\mathbf{u}) > c_*\|\mathbf{u}\|_2\}$.

Lemma C.11. *If $m \gtrsim s \log(\frac{en}{s})$, then with probability at least $1 - 2 \exp(-s \log(\frac{en}{s}))$,*

$$\text{dist}(\mathbf{x}_t, \mathbf{x}) \leq \left(\frac{5}{c_*} + 4\right) \tau_s(\mathbf{x}), \quad \forall \mathbf{x} \in \mathcal{X}^c.$$

Proof. For all $\mathbf{x} \in \mathcal{X}^c$,

$$\begin{aligned} \lambda_{\mathbf{x}} &= \sqrt{\frac{\pi}{2}} \frac{\|\mathbf{A}\mathbf{x}\|_1}{m} \leq \sqrt{\frac{\pi}{2}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\sqrt{m}} = \sqrt{\frac{\pi}{2}} \frac{\|\mathbf{A}\mathbf{x}_{[s]} + \sum_{i \geq 1} \mathbf{A}(\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]})\|_2}{\sqrt{m}} \\ &\stackrel{(a)}{\leq} \sqrt{\frac{\pi}{2}} \frac{\|\mathbf{A}\mathbf{x}_{[s]}\|_2}{\sqrt{m}} + \sqrt{\frac{\pi}{2}} \sum_{i \geq 1} \frac{\|\mathbf{A}(\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]})\|_2}{\sqrt{m}} \\ &\stackrel{(b)}{\leq} \sqrt{\frac{\pi}{2}} \cdot \frac{3}{2} \left(\|\mathbf{x}_{[s]}\|_2 + \tau_s(\mathbf{x}) \right) \\ &\stackrel{(c)}{\leq} \frac{3}{2} \sqrt{\frac{\pi}{2}} \left(\frac{1}{c_*} + 1 \right) \tau_s(\mathbf{x}), \end{aligned} \quad (\text{C.44})$$

where (a) is due to triangle inequality, (b) is due to $\Gamma_s(\frac{\mathbf{A}}{\sqrt{m}}) \leq \frac{3}{2}$ from Lemma C.1, (c) holds because $\mathbf{x} \in \mathcal{X}^c$, which gives $\|\mathbf{x}_{[s]}\|_2 \leq \|\mathbf{x}\|_2 \leq \frac{1}{c_*} \tau_s(\mathbf{x})$. Due to the projection onto $\mathbb{B}_2^n(2\lambda_{\mathbf{x}})$, $\|\mathbf{x}_t\|_2 \leq 2\lambda_{\mathbf{x}}$ always holds, and therefore

$$\text{dist}(\mathbf{x}_t, \mathbf{x}) \leq \|\mathbf{x}\|_2 + 2\lambda_{\mathbf{x}} \leq \left(\frac{1}{c_*} + 3\sqrt{\frac{\pi}{2}}(c_*^{-1} + 1)\right) \tau_s(\mathbf{x})$$

for all $\mathbf{x} \in \mathcal{X}^c$. This leads to the desired claim. $\square$

C.2 Proof of Theorem 3.2 (Non-Uniform Instance Optimality)

We focus on nonzero $\mathbf{x}$. Following the unified framework in Section 2.2.2, the proof consists of the following three steps:

1. **(Decomposition of $\mathbb{R}^n$)** We choose $\mathcal{X} = \{\mathbf{u} \in \mathbb{R}^n : \delta_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\}$ for some small enough universal constant c_* ;
2. **(Instance Optimality if $\mathbf{x} \in \mathcal{X}$)**

RAIC. For some universal constant $c > 0$, we establish

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC} \left(\Sigma_{2s}^{n,*}; \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2), \mathbf{x}, \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \right) \quad (\text{C.45})$$

Under small enough c_* , this renders (2.9)–(2.11) with

$$\mu_1 = \frac{1}{4}, \quad \mu_2 = 0, \quad e(\mathbf{x}) = 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \text{and} \quad R_{\mathbf{x}}^{\text{loc}} = c\|\mathbf{x}\|_2.$$

Initialization. Under $m \gtrsim s^2$ (up to logarithmic factor), we further show that, w.h.p.,

$$\mathbf{x}_0 \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2). \quad (\text{C.46})$$

(Recall that we assume $\text{dist}(\mathbf{x}_0, \mathbf{x}) = \|\mathbf{x}_0 - \mathbf{x}\|_2$ with no loss of generality.) In turn, suppose that $\{\mathbf{x}_t\}$ is generated by running Algorithm 1 without $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$, Lemma 2.1 guarantees

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{c\|\mathbf{x}\|_2}{2^t} + C\|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall t \geq 0. \quad (\text{C.47})$$

Incorporating $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$. We further show that $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$ has no effect on the iterates, hence the iterates of Algorithm 1 satisfy (3.8).

3. **(Instance Optimality if $\mathbf{x} \in \mathcal{X}^c$)** In this case, the desired error bound (3.13) reduces to a $\Theta(1)$ bound, and a simple argument suffices.

C.2.1 Proof of the RAIC in (C.45) if $\mathbf{x} \in \mathcal{X}$

If $\mathbf{x} \in \mathcal{X} \setminus \{0\}$, we shall first prove the RAIC in (C.45), i.e.,

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall \mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2). \quad (\text{C.48})$$

We notice that (C.4) and (C.5) remain valid under our current definition of $\mathcal{X}$. This yields

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + \frac{5\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{4} + \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \quad (\text{C.49})$$

for all $\mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2)$. To bound $\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$,

$$\begin{aligned} \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} &= \left\| \frac{1}{m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{x}| - |\mathbf{a}_i^\top \mathbf{x}_{[s]}|) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i \right\|_{2s,2} \\ &= \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m (|\mathbf{a}_i^\top \mathbf{x}| - |\mathbf{a}_i^\top \mathbf{x}_{[s]}|) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i^\top \mathbf{w} \end{aligned}$$

$$\begin{aligned}
&\stackrel{(a)}{\leq} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m |\mathbf{a}_i^\top (\mathbf{x} - \mathbf{x}_{[s]})| |\mathbf{a}_i^\top \mathbf{w}| \\
&\stackrel{(b)}{=} \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m |\mathbf{a}_i^\top \mathbf{x}_e| |\mathbf{a}_i^\top \mathbf{w}| \\
&\stackrel{(c)}{\leq} \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \frac{\|\mathbf{A}\mathbf{x}_e\|_2}{\sqrt{m}} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{\|\mathbf{A}\mathbf{w}\|_2}{\sqrt{m}} \\
&\stackrel{(d)}{\leq} 2\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \frac{\|\mathbf{A}\mathbf{x}_e\|_2}{\sqrt{m}}, \tag{C.50}
\end{aligned}$$

where (a) is due to triangle inequality, in (b) we suppose $\mathbf{x} \neq \mathbf{x}_{[s]}$ and let $\mathbf{x}_e = \frac{\mathbf{x} - \mathbf{x}_{[s]}}{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}$, (c) is due to Cauchy–Schwarz inequality, and (d) holds because of $\Gamma_{2s}(\frac{\mathbf{A}}{\sqrt{m}}) \leq 2$ from Lemma C.1.

Moreover, since $\mathbf{A}\mathbf{x}_e \sim N(0, \mathbf{I}_m)$, Equation (3.7) of [65] yields

$$\mathbb{P}\left(\frac{\|\mathbf{A}\mathbf{x}_e\|_2}{\sqrt{m}} \leq 2\right) \geq 1 - 2\exp(-c'm). \tag{C.51}$$

In turn, $\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \leq 4\|\mathbf{x} - \mathbf{x}_{[s]}\|_2$. Combining with (C.49) yields (C.48).

C.2.2 Proof of the Initialization Guarantee (C.46) if $\mathbf{x} \in \mathcal{X}$

We shall use Lemma C.4 again. Since (C.9) is ensured by small enough c_* in the definition of $\mathcal{X}$, it remains to guarantee (C.10)–(C.12).

Establishing (C.10)

Under $m \gtrsim \log n$, (C.10) is guaranteed by the following Lemma C.12 which gives a norm estimate bound tighter than Lemma C.5.

Lemma C.12. *For any $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$,*

$$\mathbb{P}\left(|\lambda_{\mathbf{x}} - \|\mathbf{x}\|_2| \leq C\sqrt{\frac{\log n}{m}}\|\mathbf{x}\|_2\right) \geq 1 - C_1n^{-1}.$$

Proof. This is a simple outcome of $\|\lambda_{\mathbf{x}}/\|\mathbf{x}\|_2 - 1\|_{\psi_2} = O(m^{-1/2})$ and the sub-Gaussian tail bound (see, e.g., [65, Section 2]). We omit further details. $\square$

Establishing (C.11)–(C.12)

It remains to bound $\Phi(\mathbf{x})$ and $\Psi(\mathbf{x})$ for a fixed $\mathbf{x} \in \mathcal{X} \setminus \{0\}$, in order to guarantee (C.11) and (C.12) with small enough $\bar{c}$. As we shall see, the transition to a fixed $\mathbf{x}$ allows for essentially tighter bounds than Lemma C.6. We first introduce a concentration inequality for product process.

Lemma C.13 ([50, Theorem 1.13], see also [34, Page 938]). *Let $\mathbf{a}_1, \dots, \mathbf{a}_m$ be i.i.d. copies of a random vector $\mathbf{a}$, and let $\{g_{\mathbf{u}}(\mathbf{a})\}_{\mathbf{u} \in \mathcal{U}}$ and $\{h_{\mathbf{v}}(\mathbf{a})\}_{\mathbf{v} \in \mathcal{V}}$ be real-valued stochastic processes indexed by $\mathcal{U}, \mathcal{V} \subset \mathbb{R}^n$, respectively. Assume that there exist $K_1, K_2, r_1, r_2 \geq 0$ such that*

$$\|g_{\mathbf{u}}(\mathbf{a}) - g_{\mathbf{u}'}(\mathbf{a})\|_{\psi_2} \leq K_1\|\mathbf{u} - \mathbf{u}'\|_2, \quad \|g_{\mathbf{u}}(\mathbf{a})\|_{\psi_2} \leq r_1, \quad \forall \mathbf{u}, \mathbf{u}' \in \mathcal{U},$$

and

$$\|h_{\mathbf{v}}(\mathbf{a}) - h_{\mathbf{v}'}(\mathbf{a})\|_{\psi_2} \leq K_2 \|\mathbf{v} - \mathbf{v}'\|_2, \quad \|h_{\mathbf{v}}(\mathbf{a})\|_{\psi_2} \leq r_2, \quad \forall \mathbf{v}, \mathbf{v}' \in \mathcal{V}.$$

Then there exist universal constants $C > c > 0$ such that, for every $t \geq 1$,

$$\begin{aligned} & \sup_{\mathbf{u} \in \mathcal{U}} \sup_{\mathbf{v} \in \mathcal{V}} \left| \frac{1}{m} \sum_{i=1}^m g_{\mathbf{u}}(\mathbf{a}_i) h_{\mathbf{v}}(\mathbf{a}_i) - \mathbb{E}[g_{\mathbf{u}}(\mathbf{a}) h_{\mathbf{v}}(\mathbf{a})] \right| \\ & \leq C \left(\frac{(K_1 \omega(\mathcal{U}) + tr_1) \cdot (K_2 \omega(\mathcal{V}) + tr_2)}{m} + \frac{r_1 K_2 \omega(\mathcal{V}) + r_2 K_1 \omega(\mathcal{U}) + tr_1 r_2}{\sqrt{m}} \right). \end{aligned}$$

holds with probability at least $1 - 2 \exp(-ct^2)$.

The following lemma shows that $m \gtrsim s^2 \log n$ guarantees (C.11) and (C.12) with small enough $\bar{c}$, with the promised probability.

Lemma C.14. *Let $\mathbf{x} \in \mathcal{X} \setminus \{0\}$. Under small enough c_* , if $m \gtrsim s \log \frac{en}{s}$, then*

$$\Phi(\mathbf{x}) \lesssim \sqrt{\frac{s \log(en/s)}{m}} \quad \text{and} \quad \Psi(\mathbf{x}) \lesssim \sqrt{\frac{\log n}{m}}$$

hold with probability at least $1 - C_1 n^{-1} - C_2 \exp(-c_3 s \log(en/s))$.

Proof. We seek to bound $\|\mathbf{x}\|_2^{-1} \sup_{\mathbf{u} \in \mathcal{U}_1} |\mathbf{u}^\top (\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]) \mathbf{u}|$, where $\mathcal{U}_1 = \Sigma_s^{n,*}$ or $\{\mathbf{e}_j\}_{j=1}^n$. These correspond to $\Phi(\mathbf{x})$ and $\Psi(\mathbf{x})$ in (C.11)–(C.12), respectively. For the fixed $\mathbf{x}$ and $\bar{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|_2$, let

$$F(\mathbf{u}) := \frac{1}{m} \sum_{i=1}^m [\mathbb{T}_2(|\mathbf{a}_i^\top \bar{\mathbf{x}}|)(\mathbf{a}_i^\top \mathbf{u})^2 - \mathbb{E} \mathbb{T}_2(|\mathbf{a}_i^\top \bar{\mathbf{x}}|)(\mathbf{a}_i^\top \mathbf{u})^2].$$

Revisiting the arguments in (C.31) and (C.33) yields

$$\frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \mathcal{U}_1} |\mathbf{u}^\top (\hat{\mathbf{S}}_{\mathbf{x}} - \mathbb{E}[\tilde{\mathbf{S}}_{\mathbf{x}}]) \mathbf{u}| \leq \frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \mathcal{U}_1} |\mathbf{u}^\top (\hat{\mathbf{S}}_{\mathbf{x}} - \tilde{\mathbf{S}}_{\mathbf{x}}) \mathbf{u}| + \sup_{\mathbf{u} \in \mathcal{U}_1} |F(\mathbf{u})| := \Xi_3 + \Xi_4$$

for $\mathcal{U}_1 = \Sigma_s^{n,*}$ and $\{\mathbf{e}_j\}_{j=1}^n$.

Bounding Ξ_3 . We revisit the arguments in (C.32), but now we consider a fixed $\mathbf{x}$ and therefore can tighten the bound $|\frac{\lambda_{\mathbf{x}}}{\|\mathbf{x}\|_2} - 1| \lesssim \sqrt{\frac{s \log(en/s)}{m}}$ used in (c) therein to $|\frac{\lambda_{\mathbf{x}}}{\|\mathbf{x}\|_2} - 1| \lesssim \sqrt{\frac{\log(n)}{m}}$ from Lemma C.12. This yields

$$\frac{1}{\|\mathbf{x}\|_2} \sup_{\mathbf{u} \in \mathcal{U}_1} |\mathbf{u}^\top (\hat{\mathbf{S}}_{\mathbf{x}} - \tilde{\mathbf{S}}_{\mathbf{x}}) \mathbf{u}| \leq \left| \frac{\lambda_{\mathbf{x}}}{\|\mathbf{x}\|_2} - 1 \right| \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \frac{2}{m} \sum_{i=1}^m (\mathbf{a}_i^\top \mathbf{u})^2 \lesssim \sqrt{\frac{\log(n)}{m}}$$

with the promised probability.

Bounding Ξ_4 . Since $\bar{\mathbf{x}}$ is fixed, we can apply Lemma C.13 to

$$\Xi_4 := \sup_{\mathbf{u} \in \mathcal{U}_1} |F(\mathbf{u})| \leq \sup_{\mathbf{u}, \mathbf{v} \in \mathcal{U}_1} \left| \frac{1}{m} \sum_{i=1}^m [\mathbb{T}_2(|\mathbf{a}_i^\top \bar{\mathbf{x}}|) \mathbf{a}_i^\top \mathbf{u} \mathbf{a}_i^\top \mathbf{v} - \mathbb{E} \mathbb{T}_2(|\mathbf{a}_i^\top \bar{\mathbf{x}}|) \mathbf{a}_i^\top \mathbf{u} \mathbf{a}_i^\top \mathbf{v}] \right|$$

with “ $\mathcal{U} = \{\bar{\mathbf{x}}\} \times \mathcal{U}_1$, $g_{\mathbf{u}}(\mathbf{a}_i) = \mathbf{T}_2(|\mathbf{a}_i^\top \bar{\mathbf{x}}|)\mathbf{a}_i^\top \mathbf{u}$, $K_1 = O(1)$, $r_1 = O(1)$, $\mathcal{V} = \mathcal{U}_1$, $h_{\mathbf{v}}(\mathbf{a}_i) = \mathbf{a}_i^\top \mathbf{v}$, $K_2 = O(1)$, $r_2 = O(1)$ ” yields

$$\mathbb{P}\left(\sup_{\mathbf{u} \in \mathcal{U}_1} |F(\mathbf{u})| \lesssim \frac{\omega^2(\mathcal{U}_1) + u}{m} + \sqrt{\frac{\omega^2(\mathcal{U}_1) + u}{m}}\right) \geq 1 - 2\exp(-c_1 u)$$

for any $u \geq 1$. Under $m \gtrsim s^2 \log n$, we set $u \asymp s \log(en/s)$ for $\mathcal{U}_1 = \Sigma_s^{n,*}$, and $u \asymp \log n$ for $\mathcal{U}_1 = \{\mathbf{e}_j\}_{j=1}^n$, along with the estimates (B.1) and (B.2) to yield

$$\mathbb{P}\left(\sup_{\mathbf{u} \in \Sigma_s^{n,*}} |F(\mathbf{u})| \lesssim \sqrt{\frac{s \log(en/s)}{m}}, \sup_{\mathbf{u} \in \{\mathbf{e}_j\}_{j=1}^n} |F(\mathbf{u})| \lesssim \sqrt{\frac{\log n}{m}}\right) \geq 1 - 2n^{-1}.$$

Combining these pieces, we obtain $\Phi(\mathbf{x}) \lesssim \sqrt{\frac{s \log(en/s)}{m}}$ and $\Psi(\mathbf{x}) \lesssim \sqrt{\frac{\log n}{m}}$, as claimed. $\square$

We now invoke Lemma C.4 to establish (C.46) for the case of $\mathbf{x} \in \mathcal{X}$. Specifically, if $m \gtrsim s^2 \log n$ (which is dictated by $\sqrt{\frac{\log n}{m}} \lesssim \frac{1}{2}$), then (C.10)–(C.12) hold for the fixed $\mathbf{x} \in \mathcal{X}$ with the promised probability. Then, Lemma C.4 yields (C.46).

C.2.3 The Projection Has No Impact if $\mathbf{x} \in \mathcal{X}$

We are ready to complete the analysis for the case of $\mathbf{x} \in \mathcal{X}$. Suppose $\{\mathbf{x}_t\}_{t \geq 0}$ is generated by Algorithm 1 without $P_{\mathbb{B}_2^n(2\lambda_{\mathbf{x}})}$, then the previous analysis guarantees (C.47) with high probability. To establish that the iterates of the original Algorithm 1 satisfy (C.47), we only need to show that the projection onto $\mathbb{B}_2^n(2\lambda_{\mathbf{x}})$ does not affect the iterates. In fact, it is enough to show the following: if $\{\mathbf{x}_t\}_{t \geq 0}$ satisfies (C.47), then $\|\mathbf{x}_t\|_2 \leq 2\lambda_{\mathbf{x}}$ holds for any $t \geq 0$. To this end, we can set c_*, c sufficiently small, so that (C.47) ensures

$$\|\mathbf{x}_t\|_2 \leq \|\mathbf{x}\|_2 + \|\mathbf{x}_t - \mathbf{x}\|_2 \leq c\|\mathbf{x}\|_2 + O(c_*)\|\mathbf{x}\|_2 + \|\mathbf{x}\|_2 < \frac{3}{2}\|\mathbf{x}\|_2.$$

Also, on the event of Lemma C.12, $\lambda_{\mathbf{x}} \geq \frac{3}{4}\|\mathbf{x}\|_2$ and hence $2\lambda_{\mathbf{x}} \geq \frac{3}{2}\|\mathbf{x}\|_2$. This delivers $\|\mathbf{x}_t\|_2 \leq 2\lambda_{\mathbf{x}}$ for all $t \geq 0$.

C.2.4 A Separate Argument if $\mathbf{x} \in \mathcal{X}^c$

We assume that we are on the event of Lemma C.12. Under the sample complexity $m \gtrsim s^2 \log n$, $\|\mathbf{x}_t\|_2 \leq 2\lambda_{\mathbf{x}} \leq 3\|\mathbf{x}\|_2$ holds for any $t \geq 0$. Thus, $\text{dist}(\mathbf{x}_t, \mathbf{x}) \leq \|\mathbf{x}_t\|_2 + \|\mathbf{x}\|_2 \leq 4\|\mathbf{x}\|_2 \leq \frac{4\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{c_*}$. This addresses the case of $\mathbf{x} \in \mathcal{X}^c$.

D Proofs for One-Bit Compressed Sensing

In this appendix, we work with $\mathbf{h}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{u}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}))\mathbf{a}_i$. We allow $\mathbf{u}, \mathbf{x}$ to take values in $\mathbb{R}^n \setminus \{0\}$, and notice that

$$\mathbf{h}_{\mathbf{u}}(\mathbf{x}) = \mathbf{h}_{\mathbf{u}/\|\mathbf{u}\|_2}\left(\frac{\mathbf{x}}{\|\mathbf{x}\|_2}\right), \quad \forall \mathbf{u}, \mathbf{x} \in \mathbb{R}^n \setminus \{0\}.$$

D.1 Proof of Theorem 4.1 (Instance Optimal 1bCS)

We set $\mathcal{X} = \{\mathbf{u} \in \mathbb{S}^{n-1} : \tau_s(\mathbf{u}) \leq c_*\}$ for some small enough constant c_* . In the outline of proof in Section 4.2, it remains to prove (4.2). By definition, this is equivalent to

$$\|\mathbf{u} - \mathbf{x} - \sqrt{2\pi}\mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{10} + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) + e(\mathbf{x}), \quad \forall \mathbf{u} \in \Sigma_s^{n,*}, \mathbf{x} \in \mathcal{X},$$

where $e(\mathbf{x}) \asymp \tau_s(\mathbf{x}) \log\left(\frac{1}{\tau_s(\mathbf{x})}\right) := e_0(\mathbf{x})$. (D.1)

D.1.1 Proof of the RAIC in (D.1)

By $\mathbf{x} = \mathbf{x}_{[s]} + \mathbf{x} - \mathbf{x}_{[s]}$,

$$\begin{aligned} \left\| \mathbf{u} - \mathbf{x} - \sqrt{2\pi}\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \right\|_{2s,2} &\leq \underbrace{\left\| \mathbf{u} - \mathbf{x}_{[s]} - \sqrt{2\pi}\mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u}) \right\|_{2s,2}}_{:=\Xi_5} \\ &\quad + \underbrace{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \sqrt{2\pi} \left\| \mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u}) \right\|_{2s,2}}_{:=\Xi_6} \end{aligned} \quad (D.2)$$

(i) **Bounding $\Xi_5 := \left\| \mathbf{u} - \mathbf{x}_{[s]} - \sqrt{2\pi}\mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u}) \right\|_{2s,2}$**

By triangle inequality,

$$\Xi_5 \leq \left\| \mathbf{x}_{[s]} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 + \left\| \mathbf{u} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} - \sqrt{2\pi}\mathbf{h}_{\mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2}(\mathbf{u}) \right\|_{2s,2} := \left\| \mathbf{x}_{[s]} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 + \Xi'_5,$$

where the first term is controlled by

$$\left\| \mathbf{x}_{[s]} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 = |1 - \|\mathbf{x}_{[s]}\|_2| \leq \|\mathbf{x} - \mathbf{x}_{[s]}\|_2.$$

In light of $\mathbf{u}, \mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2 \in \Sigma_s^{n,*}$, Ξ'_5 can be controlled by the RAIC in [47, Theorem 3.3]. For a formulation of RAIC more aligned with our setting, we refer to [16].

Lemma D.1 ([16, Theorem 5], see also the restatement in [18, Theorem B.11]). *If $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - \exp(-cs \log \frac{en}{s})$,*

$$\|\mathbf{u} - \mathbf{x} - \sqrt{2\pi} \cdot \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{1}{10} \|\mathbf{u} - \mathbf{x}\|_2 + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right), \quad \forall \mathbf{u}, \mathbf{x} \in \Sigma_s^{n,*}.$$

Proof. By Definition 2.1, the desired claim is equivalent to

$$\sqrt{2\pi}\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^{n,*}, \mathbf{x}, \frac{1}{10}\|\mathbf{u} - \mathbf{x}\|_2 + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right)\right), \quad \forall \mathbf{x} \in \Sigma_s^{n,*}.$$

Note that the general RAIC in [18, Definition B.1] is consistent with Definition 2.1. Therefore, we can apply [18, Theorem B.11] with $\mathcal{C} = \Sigma_s^n$ to obtain the following: if $m \gtrsim \frac{s}{\delta} \log(\frac{12n}{s\delta})$ for some small enough δ , then with probability at least $1 - C \exp(-cs \log \frac{en}{s})$,

$$\sqrt{2\pi}\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^{n,*}, \mathbf{x}, O\left(\sqrt{\delta\|\mathbf{u} - \mathbf{x}\|_2} + \delta\sqrt{\log(1/\delta)}\right)\right), \quad \forall \mathbf{x} \in \Sigma_s^{n,*}.$$

Therefore, we can set $\delta \asymp \frac{s}{m} \log \frac{mn}{s^2}$ to establish the RAIC with error function $\sqrt{\frac{C_2 s}{m} \log(\frac{mn}{s^2})} \|\mathbf{u} - \mathbf{x}\|_2 + \frac{C_3 s}{m} \log(\frac{mn}{s^2}) \sqrt{\log \frac{m}{s}}$. This is then relaxed to claimed error function by

$$\sqrt{\frac{C_2 s}{m} \log(\frac{mn}{s^2})} \|\mathbf{u} - \mathbf{x}\|_2 \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{10} + O\left(\frac{s}{m} \log \frac{mn}{s^2}\right).$$

The proof is complete. $\square$

On the event of Lemma D.1, for all $\mathbf{u} \in \Sigma_s^{n,*}$ and $\mathbf{x} \in \mathcal{X}$,

$$\begin{aligned} \Xi'_5 &\stackrel{(a)}{\leq} \frac{1}{10} \left\| \mathbf{u} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) \\ &\stackrel{(b)}{\leq} \frac{1}{10} \|\mathbf{u} - \mathbf{x}\|_2 + \frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{5} + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right), \end{aligned}$$

where (a) is due to the RAIC in Lemma D.1, (b) is due to triangle inequality. Combining these bounds, we arrive at

$$\Xi_5 \leq \frac{1}{10} \|\mathbf{u} - \mathbf{x}\|_2 + \frac{6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{5} + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right). \quad (\text{D.3})$$

(ii) Bounding $\Xi_6 := \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$

By (4.1),

$$\begin{aligned} \Xi_6 &:= \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \\ &= \left\| \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{u}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x})) \mathbf{a}_i - \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{u}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i \right\|_{2s,2} \\ &= \left\| \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i \right\|_{2s,2} \\ &= \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i^\top \mathbf{u} \end{aligned} \quad (\text{D.4})$$

The key is to establish uniform and signal-dependent bound on

$$L_{\mathbf{x}} := |\{i \in [m] : \text{sign}(\mathbf{a}_i^\top \mathbf{x}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})\}|,$$

the number of the hyperplanes $\{\mathbf{u} : \mathbf{a}_i^\top \mathbf{u} = 0\}$, $i \in [m]$ that separate $\mathbf{x}$ and $\mathbf{x}_{[s]}$. We establish the following novel hyperplane tessellation result. It will be proved along the subsequent analysis.

Lemma D.2 (Instance-dependent hyperplane tessellation). *Let $\mathcal{X} = \{\mathbf{u} \in \mathbb{S}^{n-1} : \sum_{i \geq 1} \|\mathbf{u}_{[(i+1)s]}\|_2 \leq c_*\}$ for some small universal constant c_* . If $m \gtrsim s \log \frac{mn}{s^2}$, then with probability at least $1 - C \exp(-cs \log \frac{en}{s})$, $L_{\mathbf{x}} = |\{i \in [m] : \text{sign}(\mathbf{a}_i^\top \mathbf{x}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})\}|$ satisfies*

$$\frac{L_{\mathbf{x}}}{m} \lesssim \frac{s}{m} \log\left(\frac{mn}{s^2}\right) + \tau_s(\mathbf{x}) \log^{1/2} \frac{1}{\tau_s(\mathbf{x})}, \quad \forall \mathbf{x} \in \mathcal{X}.$$

We shall note a uniform bound on $I_{\mathbf{x}}(\eta) := |\{i \in [m] : |\mathbf{a}_i^\top \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}| \leq \eta\}|$ over $\mathbf{x} \in \mathcal{X}$.

Lemma D.3. [20, Lemma B.1] If $m \gtrsim s \log \frac{mn}{s^2}$, then for any $\eta \in \frac{1}{m}\mathbb{Z}_s$,

$$\mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} I_{\mathbf{x}}(\eta) \leq \eta m\right) \geq 1 - \exp(-c\eta m). \quad (\text{D.5})$$

Proof. While Lemma B.1 in [20] is stated for m complex Gaussian vectors, its proof yields the identical statement for real Gaussian vectors a_i 's, up to minor changes. Since $\mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2 \in \Sigma_s^{n,*}$, it follows that

$$\sup_{\mathbf{x} \in \mathcal{X}} I_{\mathbf{x}}(\eta) \leq \sup_{\mathbf{x} \in \Sigma_s^{n,*}} I_{\mathbf{x}}(\eta).$$

Then by Lemma B.1 in [20]—along with the estimates (B.3) and (B.2)—we conclude that (D.5) holds if $m \gtrsim \frac{s}{\eta} \log \frac{n}{s\eta}$, which can be ensured by $\eta \gtrsim \frac{s}{m} \log \frac{mn}{s^2}$. Under $m \gtrsim s \log \frac{mn}{s^2}$, this is satisfied by $\eta \in \frac{\mathbb{Z}_s}{m}$; see (D.7). $\square$

Proof of Lemma D.2. To start the proof, note that for any $\eta > 0$ and $\mathbf{x} \in \mathcal{X}$,

$$L_{\mathbf{x}} \leq \underbrace{\left|\left\{i \in [m] : \left|\mathbf{a}_i^\top \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}\right| \leq \eta\right\}\right|}_{I_{\mathbf{x}}(\eta)} + \underbrace{\left|\left\{i \in [m] : \left|\mathbf{a}_i^\top \left(\mathbf{x} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}\right)\right| > \frac{\eta}{2}\right\}\right|}_{J_{\mathbf{x}}(\eta)}. \quad (\text{D.6})$$

To see this, if some i is not counted in either $I_{\mathbf{x}}(\eta)$ or $J_{\mathbf{x}}(\eta)$, that is,

$$\left|\mathbf{a}_i^\top \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}\right| > \eta \quad \text{and} \quad \left|\mathbf{a}_i^\top \left(\mathbf{x} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}\right)\right| \leq \frac{\eta}{2},$$

then it follows that

$$\text{sign}(\mathbf{a}_i^\top \mathbf{x}) = \text{sign}\left(\mathbf{a}_i^\top \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}\right) = \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]}),$$

and this index i does not contribute to $L_{\mathbf{x}}$. In addition, for each $\mathbf{x} \in \mathcal{X}$, $I_{\mathbf{x}}(\eta)$ is non-decreasing while $J_{\mathbf{x}}(\eta)$ is non-increasing in η , that is,

$$I_{\mathbf{x}}(\eta_1) \leq I_{\mathbf{x}}(\eta_2), \quad J_{\mathbf{x}}(\eta_1) \geq J_{\mathbf{x}}(\eta_2), \quad \forall \eta_1 \leq \eta_2.$$

To get signal-dependent bound, we will choose different η for different $\mathbf{x}$. Thus, we shall control $I_{\mathbf{x}}(\eta)$ and $J_{\mathbf{x}}(\eta)$ for η in the following range:

$$\eta \in \frac{\mathbb{Z}_s}{m}, \quad \text{where } \mathbb{Z}_s = \left\{l \in \mathbb{Z} : C^* s \log \frac{mn}{s^2} \leq l \leq m\right\} \quad (\text{D.7})$$

for some large enough C^* .

By using Lemma D.3 and taking a union bound, the event

$$I_{\mathbf{x}}(\eta) \leq \eta m, \quad \forall \eta \in \frac{\mathbb{Z}_s}{m}, \quad \forall \mathbf{x} \in \mathcal{X} \quad (\text{D.8})$$

holds with probability at least

$$1 - \sum_{\eta \in \mathbb{Z}_s/m} \exp(-c\eta m) \geq 1 - \sum_{\mathbb{Z} \ni i \geq C^* s \log \frac{mn}{s^2}} \exp(-ci) \geq 1 - C \exp\left(-cs \log \frac{mn}{s^2}\right).$$

We proceed on the event (D.8).

We then turn to bounding $J_{\mathbf{x}}(\eta)$, again for all $\eta \in \frac{\mathbb{Z}_s}{m}$. To this end, we instead look at the quantity

$$E_s(\eta, \mathbf{x}) := \max_{|S|=\eta m} \left(\frac{1}{\eta m} \sum_{i \in S} \left| \mathbf{a}_i^\top \left(\mathbf{x} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right) \right|^2 \right)^{1/2}, \quad \eta \in \frac{\mathbb{Z}_s}{m}.$$

For any $\mathbf{x}$, it is not hard to see that

$$E_s(\eta, \mathbf{x}) \leq \frac{\eta}{2} \implies J_{\mathbf{x}}(\eta) \leq \eta m. \quad (\text{D.9})$$

(In fact, $J_{\mathbf{x}}(\eta) > \eta m$ necessarily implies $E_s(\eta, \mathbf{x}) > \frac{\eta}{2}$.) Therefore, we seek to establish a signal-dependent bound on $E_s(\eta, \mathbf{x})$, which holds uniformly for all $\mathbf{x} \in \mathcal{X}$ and all $\eta \in \frac{\mathbb{Z}_s}{m}$.

Bounding $E_s(\eta, \mathbf{x})$. Applying Lemma C.9 to $\mathcal{U} = \Sigma_{2s}^{n,*}$ and using (B.2), we obtain

$$\mathbb{P} \left(\sup_{\mathbf{u} \in \Sigma_{2s}^{n,*}} \max_{\substack{S \subset [m] \\ |S|=\eta m}} \left(\frac{1}{\eta m} \sum_{i \in S} |\mathbf{a}_i^\top \mathbf{u}|^2 \right)^{1/2} \lesssim \sqrt{\frac{s \log(en/s)}{\eta m}} + \sqrt{\log \frac{e}{\eta}} \right) \geq 1 - 2 \exp(-c\eta m)$$

for each $\eta \in \frac{\mathbb{Z}_s}{m}$. Then taking a union bound over $\eta \in \frac{\mathbb{Z}_s}{m}$,

$$\sup_{\mathbf{u} \in \Sigma_{2s}^{n,*}} \max_{\substack{\mathcal{G} \subset [m] \\ |\mathcal{G}|=\eta m}} \left(\frac{1}{\eta m} \sum_{i \in \mathcal{G}} |\mathbf{a}_i^\top \mathbf{u}|^2 \right)^{1/2} \lesssim \sqrt{\frac{s \log(en/s)}{\eta m}} + \sqrt{\log \frac{e}{\eta}}, \quad \forall \eta \in \frac{\mathbb{Z}_s}{m} \quad (\text{D.10})$$

holds with probability at least $1 - 2 \sum_{\eta \in \mathbb{Z}_s/m} \exp(-c\eta m) \geq 1 - C \exp(-cs \log \frac{mn}{s})$. We proceed on the event (D.10). For any $S \subset [m]$, we let $\mathbf{A}_S \in \mathbb{R}^{|S| \times n}$ denote the submatrix of $\mathbf{A}$ constituted by the rows in S only. Uniformly for all $\eta \in \frac{\mathbb{Z}_s}{m}$ and $\mathbf{x} \in \mathcal{X}$,

$$\begin{aligned} E_s(\eta, \mathbf{x}) &= \max_{|S|=\eta m} \frac{\|\mathbf{A}_S(\mathbf{x} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2})\|_2}{\sqrt{\eta m}} \\ &\stackrel{(a)}{\leq} \max_{|S|=\eta m} \frac{\|\mathbf{A}_S(\mathbf{x}_{[s]} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2})\|_2}{\sqrt{\eta m}} + \max_{|S|=\eta m} \frac{\|\mathbf{A}_S(\mathbf{x} - \mathbf{x}_{[s]})\|_2}{\sqrt{\eta m}} \\ &\stackrel{(b)}{\leq} |1 - \|\mathbf{x}_{[s]}\|_2| \max_{|S|=\eta m} \frac{\|\mathbf{A}_S \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}\|_2}{\sqrt{\eta m}} + \max_{|S|=\eta m} \sum_{i \geq 1} \frac{\|\mathbf{A}_S(\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]})\|_2}{\sqrt{\eta m}} \\ &\stackrel{(c)}{\leq} |1 - \|\mathbf{x}_{[s]}\|_2| \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \max_{|S|=\eta m} \frac{\|\mathbf{A}_S \mathbf{u}\|_2}{\sqrt{\eta m}} + \sum_{i \geq 1} \|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2 \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \max_{|S|=\eta m} \frac{\|\mathbf{A}_S \mathbf{u}\|_2}{\sqrt{\eta m}} \\ &\stackrel{(d)}{\leq} 2 \sum_{i \geq 1} \|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2 \sup_{\mathbf{u} \in \Sigma_s^{n,*}} \max_{|S|=\eta m} \frac{\|\mathbf{A}_S \mathbf{u}\|_2}{\sqrt{\eta m}} \\ &\stackrel{(e)}{\lesssim} \sqrt{\log \frac{e}{\eta}} \cdot \tau_s(\mathbf{x}), \end{aligned}$$

where (a) is due to triangle inequality, in (b) we use $\mathbf{x} - \mathbf{x}_{[s]} = \sum_{i \geq 1} (\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]})$ and triangle inequality, in (c) we use $\frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2}, \frac{\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}}{\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2} \in \Sigma_s^{n,*}$, (d) holds because $|1 - \|\mathbf{x}_{[s]}\|_2| \leq \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq \tau_s(\mathbf{x})$, (e) is due to the event in Equation (D.10).

In light of (D.9), for any $\mathbf{x} \in \mathcal{X}$ and $\eta \in \frac{\mathbb{Z}_s}{m}$, there exists some small enough universal constant c_0 such that

$$\tau_s(\mathbf{x}) \leq \frac{c_0 \eta}{\sqrt{\log(e/\eta)}} \implies |J_{\mathbf{x}}(\eta)| \leq \eta m. \quad (\text{D.11})$$

Therefore, for each $\mathbf{x} \in \mathcal{X}$, we shall set

$$\eta = \eta_{\mathbf{x}} := \max \left\{ \frac{\lceil C^* s \log \frac{mn}{s^2} \rceil}{m}, C_0 \tau_s(\mathbf{x}) \log^{1/2} \frac{1}{\tau_s(\mathbf{x})} \right\} \quad (\text{D.12})$$

for some large enough C_0 such that $m\eta_{\mathbf{x}} \in \mathbb{Z}$ and the left-hand side of (D.11) holds. In turn,

$$J_{\mathbf{x}}(\eta_{\mathbf{x}}) \leq \eta_{\mathbf{x}} m, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (\text{D.13})$$

By (D.8), we also have

$$I_{\mathbf{x}}(\eta_{\mathbf{x}}) \leq \eta_{\mathbf{x}} m, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (\text{D.14})$$

Combining (D.6), (D.13) and (D.14) yields

$$L_{\mathbf{x}} \leq 2\eta_{\mathbf{x}} m, \quad \forall \mathbf{x} \in \mathcal{X}, \quad \text{where } \eta_{\mathbf{x}} \text{ is defined in (D.12)}. \quad (\text{D.15})$$

The proof of Lemma D.2 is now complete. $\square$

We are now ready to establish a final bound on Ξ_6 in (D.2). Using the expression in (D.4), it holds for all $\mathbf{x} \in \mathcal{X}$ that

$$\begin{aligned} \Xi_6 &\stackrel{(a)}{\leq} \sup_{\mathbf{u} \in \Sigma_{2s}^{n,*}} \frac{1}{2m} \sum_{i=1}^m |\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})| |\mathbf{a}_i^\top \mathbf{u}| \\ &\stackrel{(b)}{\leq} \sup_{\mathbf{u} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{\substack{i \in [m]: \text{sign}(\mathbf{a}_i^\top \mathbf{x}) \\ \neq \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})}} |\mathbf{a}_i^\top \mathbf{u}| \\ &\stackrel{(c)}{\leq} \sup_{\mathbf{u} \in \Sigma_{2s}^{n,*}} \max_{\substack{I \subset [m] \\ |I|=2\eta_{\mathbf{x}} m}} \frac{1}{m} \sum_{i \in I} |\mathbf{a}_i^\top \mathbf{u}| \\ &\stackrel{(d)}{\leq} 4\eta_{\mathbf{x}} \sup_{\mathbf{u} \in \Sigma_{2s}^{n,*}} \max_{\substack{I \subset [m] \\ |I|=2\eta_{\mathbf{x}} m}} \left(\frac{\sum_{i \in I} |\mathbf{a}_i^\top \mathbf{u}|^2}{2\eta_{\mathbf{x}} m} \right)^{1/2} \\ &\stackrel{(e)}{\lesssim} \eta_{\mathbf{x}} \sqrt{\log \frac{e}{\eta_{\mathbf{x}}}} \\ &\stackrel{(f)}{\lesssim} \frac{s}{m} \log \left(\frac{mn}{s^2} \right) \log^{1/2} \left(\frac{m}{s} \right) + e_0(\mathbf{x}), \end{aligned} \quad (\text{D.16})$$

where (a) follows from triangle inequality, in (b) we restrict to the nonzero summands, (c) is due to the event in Equation (D.15), (d) is due to Cauchy–Schwarz inequality, (e) holds because of the event in (D.10), in (f) we substitute $\eta_{\mathbf{x}}$ in (D.12) and $e_0(\mathbf{x})$ in (D.1).

Substituting (D.3) and (D.16) into (D.2), along with $\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq e_0(\mathbf{x})$ for $\mathbf{x} \in \mathcal{X}$, we arrive at (D.1).

D.2 Proof of Theorem 4.2 (Non-Uniform Instance Optimality)

Following the unified framework in Section 2.2.2, the proof consists of the following three steps:

1. **(Decomposition of $\mathbb{S}^{n-1}$)** For some small enough universal constant $c_* > 0$, we set $\mathcal{X} = \{\mathbf{u} \in \mathbb{S}^{n-1} : \delta_s(\mathbf{u}) \leq c_*\}$.
2. **(Instance optimality if $\mathbf{x} \in \mathcal{X}$)** We establish the RAIC

$$\sqrt{2\pi}\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^{n,*}, \mathbf{x}, \frac{\|\mathbf{u} - \mathbf{x}\|_2}{10} + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) + C_2 \delta_s(\mathbf{x})\right) \quad (\text{D.17})$$

This renders (2.13)–(2.16) with $\mu_1 = \frac{1}{10}$, $R_{\mathbf{x}}^{\text{loc}} = \infty$, $\mathbf{x}_0 = \mathbf{e}_1$, and hence Theorem 2.2 yields the desired estimate.

3. **(Instance optimality if $\mathbf{x} \in \mathcal{X}^c$)** If $\mathbf{x} \in \mathcal{X}^c$, the crude bound $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2$ suffices: $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{2\delta_s(\mathbf{x})}{c_*}$. Note that this step is already complete.

It remains to establish (D.17), which by definition is equivalent to

$$\|\mathbf{u} - \mathbf{x} - \sqrt{2\pi}\mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{1}{10}\|\mathbf{u} - \mathbf{x}\|_2 + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) + C_2 \|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall \mathbf{u} \in \Sigma_s^{n,*}. \quad (\text{D.18})$$

D.2.1 Proof of the RAIC in (D.18)

We start with the decomposition in (D.2) and use the bound on T_1 in (D.3)³ and (D.4), we obtain

$$\begin{aligned} \left\| \mathbf{u} - \mathbf{x} - \sqrt{2\pi}\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \right\|_{2s,2} &\leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{10} + \frac{C_1 s}{m} \log\left(\frac{mn}{s^2}\right) \log^{1/2}\left(\frac{m}{s}\right) \\ &+ \frac{11\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{5} + \sqrt{2\pi} \left\| \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i \right\|_{2s,2}. \end{aligned} \quad (\text{D.19})$$

All that remains is to show that

$$\left\| \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i \right\|_{2s,2} \lesssim \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \frac{s \log(en/s)}{m} \quad (\text{D.20})$$

holds with the promised probability. With no loss of generality, we assume $\mathbf{x} \neq \mathbf{x}_{[s]}$. By $\text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]}) = \text{sign}(\mathbf{a}_i^\top \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2})$, we seek to perform a sharp analysis with two unit vectors, $\mathbf{x}$ and $\mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2$. We shall define (β_1, β_2) , an orthogonal basis of $\text{span}(\mathbf{x}, \mathbf{x}_{[s]})$, as

$$\beta_1 := \frac{\mathbf{x} - \mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2}{\|\mathbf{x} - \mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2\|_2}, \quad \text{and} \quad \beta_2 := \frac{\mathbf{x} + \mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2}{\|\mathbf{x} + \mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2\|_2}.$$

We also introduce the shorthand

$$E^{(i)} := \{\text{sign}(\mathbf{a}_i^\top \mathbf{x}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})\}, \quad i \in [m].$$

Under these conventions, for $i \in [m]$,

$$\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]}) = \text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]}/\|\mathbf{x}_{[s]}\|_2)$$

³Note that (D.3) remains valid under the current definition of $\mathcal{X}$.

$$= 2\mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top (\mathbf{x} - \mathbf{x}_{[s]} / \|\mathbf{x}_{[s]}\|_2)) = 2\mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1),$$

and $\mathbf{a}_i$ can be decomposed into

$$\mathbf{a}_i = (\mathbf{a}_i^\top \boldsymbol{\beta}_1) \boldsymbol{\beta}_1 + (\mathbf{a}_i^\top \boldsymbol{\beta}_2) \boldsymbol{\beta}_2 + \underbrace{[\mathbf{a}_i - (\mathbf{a}_i^\top \boldsymbol{\beta}_1) \boldsymbol{\beta}_1 - (\mathbf{a}_i^\top \boldsymbol{\beta}_2) \boldsymbol{\beta}_2]}_{:= \mathbf{a}_i^\perp}.$$

By the preceding two identities,

$$\begin{aligned} \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i &= \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) \mathbf{a}_i \\ &= \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) |\mathbf{a}_i^\top \boldsymbol{\beta}_1| \boldsymbol{\beta}_1 + \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) (\mathbf{a}_i^\top \boldsymbol{\beta}_2) \boldsymbol{\beta}_2 + \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) \mathbf{a}_i^\perp. \end{aligned}$$

Taking $\|\cdot\|_{2s,2}$ norm and using triangle inequality,

$$\begin{aligned} &\left\| \frac{1}{2m} \sum_{i=1}^m (\text{sign}(\mathbf{a}_i^\top \mathbf{x}) - \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) \mathbf{a}_i \right\|_{2s,2} \\ &\leq \left| \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) |\mathbf{a}_i^\top \boldsymbol{\beta}_1| \right| \\ &\quad + \left| \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) (\mathbf{a}_i^\top \boldsymbol{\beta}_2) \right| \\ &\quad + \left\| \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) \mathbf{a}_i^\perp \right\|_{2s,2} := \Xi_7 + \Xi_8 + \Xi_9. \end{aligned} \tag{D.21}$$

Bounding Ξ_7 and Ξ_8

The main tool to bound Ξ_7, Ξ_8 is the following moment-based Bernstein's inequality.

Lemma D.4. [6, Theorem 2.10] *Let $X_1, \dots, X_n$ be independent random variables, and assume that for some $v, c > 0$, $\sum_{i=1}^n \mathbb{E}|X_i|^q \leq \frac{q!}{2} v c^{q-2}$ holds for all integers $q \geq 2$, then*

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mathbb{E}X_i)\right| \geq \sqrt{2vt} + ct\right) \leq 2\exp(-t), \quad \forall t > 0.$$

The moment bounds and expectations of $\mathbb{1}(E^{(i)}) |\mathbf{a}_i^\top \boldsymbol{\beta}_1|$ and $\mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) (\mathbf{a}_i^\top \boldsymbol{\beta}_2)$ have been established in [16].

Lemma D.5. [16, Facts 1, 2] *For some universal constant C ,*

$$\begin{aligned} \mathbb{E}(\mathbb{1}(E^{(i)}) |\mathbf{a}_i^\top \boldsymbol{\beta}_1|^p) &\leq C \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \frac{p!}{2}, \\ \mathbb{E}(\mathbb{1}(E^{(i)}) |\mathbf{a}_i^\top \boldsymbol{\beta}_2|^p) &\leq C \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \frac{p!}{2} \end{aligned}$$

hold for any integer $p \geq 2$. Moreover,

$$\mathbb{E}(\mathbb{1}(E^{(i)}) |\mathbf{a}_i^\top \boldsymbol{\beta}_1|) = \frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\sqrt{2\pi}}, \quad \text{and} \quad \mathbb{E}(\mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) \mathbf{a}_i^\top \boldsymbol{\beta}_2) = 0.$$

Therefore, Lemma D.4 with $X_i = \mathbb{1}(E^{(i)})|\mathbf{a}_i^\top \boldsymbol{\beta}_1|$ yields

$$\mathbb{P}\left(\left|\frac{1}{m} \sum_{i=1}^m \{\mathbb{1}(E^{(i)})|\mathbf{a}_i^\top \boldsymbol{\beta}_1| - \mathbb{E}[\mathbb{1}(E^{(i)})|\mathbf{a}_i^\top \boldsymbol{\beta}_1|]\}\right| \lesssim \sqrt{\frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2^2 t}{m}} + \frac{t}{m}\right) \leq 2 \exp(-t)$$

for any $t > 0$. Setting $t = s \log(\frac{en}{s})$ and using triangle inequality along with $\mathbb{E}[\mathbb{1}(E^{(i)})|\mathbf{a}_i^\top \boldsymbol{\beta}_1|] = \frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\sqrt{2\pi}}$, we obtain that

$$\mathbb{P}\left(\Xi_7 \lesssim \frac{s \log(en/s)}{m} + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2\right) \geq 1 - 2 \exp(-s \log \frac{en}{s}). \quad (\text{D.22})$$

Analogously, a straightforward application of Lemma D.4 with $X_i = \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1)(\mathbf{a}_i^\top \boldsymbol{\beta}_2)$ yields

$$\Xi_8 \lesssim \sqrt{\frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 s \log(en/s)}{m}} + \frac{s \log(en/s)}{m} \lesssim \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \frac{s \log(en/s)}{m} \quad (\text{D.23})$$

with probability at least $1 - 2 \exp(-s \log \frac{en}{s})$.

Bounding Ξ_9

By rotational invariance, $\mathbf{a}_i^\perp$ is independent of $(\mathbf{a}_i^\top \boldsymbol{\beta}_1, \mathbf{a}_i^\top \boldsymbol{\beta}_2)$. To bound

$$\Xi_9 = \left\| \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) \mathbf{a}_i^\perp \right\|_{2s,2} = \sup_{\mathbf{u} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) \langle \mathbf{a}_i^\perp, \mathbf{u} \rangle,$$

we first treat the randomness of $(\mathbf{a}_i^\perp)_{i=1}^m$ and then deal with the randomness of $(\mathbf{a}_i^\top \boldsymbol{\beta}_1, \mathbf{a}_i^\top \boldsymbol{\beta}_2)_{i=1}^m$. Conditioning on $(\mathbf{a}_i^\top \boldsymbol{\beta}_1, \mathbf{a}_i^\top \boldsymbol{\beta}_2)_{i=1}^m$, then $\mathbb{1}(E^{(i)})$'s are deterministic, and a calculation based on [65, Proposition 2.6.1] yields

$$\left\| \frac{1}{m} \sum_{i=1}^m \mathbb{1}(E^{(i)}) \text{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1) \mathbf{a}_i^\perp \right\|_{\psi_2} \lesssim \frac{\sqrt{\sum_{i=1}^m \mathbb{1}(E^{(i)})}}{m}.$$

In turn, [65, Exercise 8.6.5] (along with (B.2)) yields

$$\mathbb{P}\left(\Xi_9 \lesssim \frac{\sqrt{[\sum_{i=1}^m \mathbb{1}(E^{(i)})] s \log(en/s)}}{m} \middle| (\mathbf{a}_i^\top \boldsymbol{\beta}_1, \mathbf{a}_i^\top \boldsymbol{\beta}_2)_{i=1}^m\right) \geq 1 - 2 \exp\left(-s \log \frac{en}{s}\right). \quad (\text{D.24})$$

We now bound $\sum_{i=1}^m \mathbb{1}(E^{(i)}) = L_{\mathbf{x}}$. In light of (e.g., [35, Lemma 3.2])

$$\mathbb{P}(\text{sign}(\mathbf{a}_i^\top \mathbf{x}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{x}_{[s]})) = \frac{\arccos(\langle \mathbf{x}, \mathbf{x}_{[s]} \rangle / \|\mathbf{x}_{[s]}\|_2)}{\pi} \leq \frac{1}{2} \left\| \mathbf{x} - \frac{\mathbf{x}_{[s]}}{\|\mathbf{x}_{[s]}\|_2} \right\|_2 \leq \|\mathbf{x} - \mathbf{x}_{[s]}\|_2,$$

by letting

$$\eta = \max \left\{ \frac{\lceil C_* s \log \frac{en}{s} \rceil}{m}, \tilde{C} \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \right\} \quad \text{for some } C_*, \tilde{C} \geq 2$$

such that ηm is an integer, Chernoff bound yields

$$\mathbb{P}(L_{\mathbf{x}} \leq \eta m) \geq \mathbb{P}\left(\text{Binomial}\left(m, \frac{\eta}{2}\right) \leq \eta m\right) \geq 1 - \exp(-c'\eta m) \geq 1 - \exp\left(-c''s \log \frac{en}{s}\right).$$

Substituting this into (D.24) yields that

$$\Xi_9 \lesssim \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \frac{s \log(en/s)}{m} \quad (\text{D.25})$$

holds with the promised probability. Combining (D.22), (D.23), (D.25), and (D.21), we arrive at (D.20) and complete the proof.

E Proofs for Sparse ReLU Regression

We proceed to the proofs of sparse ReLU regression. Throughout this section, we define $\mathbf{h}_{\mathbf{x}}(\mathbf{u})$ as in (5.1), i.e., $\mathbf{h}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{m} \sum_{i=1}^m [\text{ReLU}(\mathbf{a}_i^\top \mathbf{u}) - \text{ReLU}(\mathbf{a}_i^\top \mathbf{x})][1 + \text{sign}(\mathbf{a}_i^\top \mathbf{u})]\mathbf{a}_i$. As the algorithm achieves exact recovery for $\mathbf{x} = 0$, the subsequent analysis treats nonzero $\mathbf{x}$ only.

E.1 Proof of Theorem 5.1 (Instance Optimal Sparse ReLU Regression)

Following the unified framework in Section 2.2.1, the main steps are outlined in the following:

1. **(Decomposition of $\mathbb{R}^n$)** Set $\mathcal{X} = \{\mathbf{u} \in \mathbb{R}^n : \tau_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\}$ for some small enough universal constant $c_* > 0$;
2. **(Instance Optimality for $\mathbf{x} \in \mathcal{X}$)**

RAIC. We establish the following signal-dependent RAIC: for some universal constant $c > 0$,

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2), \mathbf{x}, \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 6\tau_s(\mathbf{x})\right), \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\}. \quad (\text{E.1})$$

By definition, this reads

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 6\tau_s(\mathbf{x}), \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\}, \quad \forall \mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2). \quad (\text{E.2})$$

By the conventions in Section 2.2, this renders $\mu_1 = \frac{1}{4}$, $\mu_2 = 0$, $e(\mathbf{x}) = 6\tau_s(\mathbf{x})$, $R_{\mathbf{x}}^{\text{loc}} = c\|\mathbf{x}\|_2$, thus satisfying (2.9)–(2.11) as long as c_* in the definition of $\mathcal{X}$ is chosen small enough.

Initialization. We then establish

$$\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{\|\mathbf{x}_0 - \mathbf{x}\|_2}{\|\mathbf{x}\|_2} \leq C' \sqrt{\frac{s \log(en/s)}{m}} + 3c_*. \quad (\text{E.3})$$

Combining with $\mathbf{x}_0 \in \Sigma_s^n$, we arrive at (2.12) under $m \gtrsim s \log \frac{en}{s}$ and small enough c_* . Therefore, by Theorem 2.3, $\{\mathbf{x}_t\}_{t \geq 0}$ produced by Algorithm 3 without $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$ satisfies

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{c\|\mathbf{x}\|_2}{2^t} + 30\tau_s(\mathbf{x}), \quad \forall t \geq 0, \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\}. \quad (\text{E.4})$$

Incorporating $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$. Built upon (E.4), we further show that $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$ has no effect on the iterates of $\mathbf{x} \in \mathcal{X} \setminus \{0\}$, thus (E.4) is satisfied by the original Algorithm 3.

3. **(Instance Optimality for $\mathbf{x} \in \mathcal{X}^c$)** By using $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$ and the definition of $\mathcal{X}$, a simple argument suffices.

E.1.1 Proof of the RAIC in (E.2)

We start with the following RAIC over exactly sparse signals.

Lemma E.1. [18, Theorem B.10] If $m \gtrsim s \log \frac{en}{s}$, then with probability at least $1 - c_1 \exp(-c_2 m)$,

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2,2s} \leq \frac{1}{4} \|\mathbf{u} - \mathbf{x}\|_2, \quad \forall \mathbf{u}, \mathbf{x} \in \Sigma_s^n \setminus \{0\} \text{ obeying } \|\mathbf{u} - \mathbf{x}\|_2 \leq \tilde{c} \|\mathbf{x}\|_2$$

holds for some small enough universal constant $\tilde{c} > 0$.

Proof. By Definition 2.1, the desired claim is equivalent to

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; \tilde{c} \|\mathbf{x}\|_2), \mathbf{x}, \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4}\right), \quad \forall \mathbf{x} \in \Sigma_s^n \setminus \{0\}.$$

Therefore, the result follows from [18, Theorem B.10] with $\mathcal{K} = \Sigma_{2s}^{n,*}$, $\mathcal{C} = \Sigma_s^n$. $\square$

We now seek an extension to $\mathbf{x} \in \mathcal{X} \setminus \{0\}$. We set c in (E.2) as $c = \frac{\tilde{c}}{2}$, where $\tilde{c}$ is the universal constant in Lemma E.1. For any $\mathbf{x} \in \mathcal{X} \setminus \{0\}$ and $\mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c \|\mathbf{x}\|_2)$, triangle inequality gives

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \|\mathbf{u} - \mathbf{x}_{[s]} - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}. \quad (\text{E.5})$$

By $\mathbf{u}, \mathbf{x}_{[s]} \in \Sigma_s^n$ and $\|\mathbf{u} - \mathbf{x}_{[s]}\|_2 \leq \|\mathbf{u} - \mathbf{x}\|_2 + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq (c + c_*) \|\mathbf{x}\|_2 \leq \tilde{c} \|\mathbf{x}\|_2$ (the last inequality is ensured by setting $c_* \leq \frac{\tilde{c}}{2}$), Lemma E.1 gives

$$\|\mathbf{u} - \mathbf{x}_{[s]} - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}_{[s]}\|_2}{4} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + \frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{4}. \quad (\text{E.6})$$

It remains to bound $\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$, which can be accomplished by

$$\begin{aligned} \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} &= \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \langle \mathbf{w}, \mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u}) \rangle \\ &= \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m [\text{ReLU}(\mathbf{a}_i^\top \mathbf{x}_{[s]}) - \text{ReLU}(\mathbf{a}_i^\top \mathbf{x})][1 + \text{sign}(\mathbf{a}_i^\top \mathbf{u})] \mathbf{a}_i^\top \mathbf{w} \\ &\stackrel{(a)}{\leq} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{2}{m} \sum_{i=1}^m |\text{ReLU}(\mathbf{a}_i^\top \mathbf{x}_{[s]}) - \text{ReLU}(\mathbf{a}_i^\top \mathbf{x})| |\mathbf{a}_i^\top \mathbf{w}| \\ &\stackrel{(b)}{\leq} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{2}{m} \sum_{i=1}^m |\mathbf{a}_i^\top (\mathbf{x} - \mathbf{x}_{[s]})| |\mathbf{a}_i^\top \mathbf{w}| \\ &\stackrel{(c)}{\leq} 4 \sum_{j \geq 1} \|\mathbf{x}_{[(j+1)s]} - \mathbf{x}_{[js]}\|_2 = 4\tau_s(\mathbf{x}), \end{aligned} \quad (\text{E.7})$$

where in (a) we use triangle inequality and $|1 + \text{sign}(\mathbf{a}_i^\top \mathbf{u})| \leq 2$, (b) holds because of $|\text{ReLU}(a) - \text{ReLU}(b)| \leq |a - b|$, (c) is due to the bound derived in (C.6). Putting the pieces together, along with $\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq \tau_s(\mathbf{x})$, we arrive at (E.2).

E.1.2 Proof of the Initialization Guarantee (E.3)

In view of $\mathbf{x}_0 = H_s(\frac{1}{m} \sum_{i=1}^m 2y_i \mathbf{a}_i) = H_s(\frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i)$,

$$\begin{aligned}
\sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{\|\mathbf{x}_0 - \mathbf{x}\|_2}{\|\mathbf{x}\|_2} &= \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \left\| H_s \left(\frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \frac{\mathbf{x}}{\|\mathbf{x}\|_2}) \mathbf{a}_i \right) - \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \right\|_2 \\
&\stackrel{(a)}{=} \sup_{\mathbf{x} \in \mathcal{X}^*} \left\| H_s \left(\frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i \right) - \mathbf{x} \right\|_2 \\
&\stackrel{(b)}{\leq} \sup_{\mathbf{x} \in \mathcal{X}^*} \left\| H_s \left(\frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i \right) - \mathbf{x}_{[s]} \right\|_2 + \sup_{\mathbf{x} \in \mathcal{X}^*} \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \\
&\stackrel{(c)}{\leq} \sup_{\mathbf{x} \in \mathcal{X}^*} 2 \left\| \frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i - \mathbf{x}_{[s]} \right\|_{2s,2} + c_* \\
&\stackrel{(d)}{\leq} \sup_{\mathbf{x} \in \mathcal{X}^*} 2 \left\| \frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i - \mathbf{x} \right\|_{2s,2} + 3c_*, \tag{E.8}
\end{aligned}$$

where (a) is due to $\frac{\mathbf{x}}{\|\mathbf{x}\|_2} \in \mathcal{X}^* = \mathcal{X} \cap \mathbb{S}^{n-1}$, (b) is due to triangle inequality, (c) follows from Lemma A.1 and the definition of $\mathcal{X}$, (d) follows from triangle inequality and again the definition of $\mathcal{X}$.

Moreover, the first term is bounded by

$$\begin{aligned}
&\sup_{\mathbf{x} \in \mathcal{X}^*} \left\| \frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i - \mathbf{x} \right\|_{2s,2} \\
&= \sup_{\mathbf{x} \in \mathcal{X}^*} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{2}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \mathbf{w} - \mathbf{x}^\top \mathbf{w} \\
&\stackrel{(a)}{=} \sup_{\mathbf{x} \in \mathcal{X}^*} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m \left[2 \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \mathbf{w} - \mathbb{E}(2 \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \mathbf{w}) \right] \\
&\stackrel{(b)}{\lesssim} \sqrt{\frac{s \log(en/s)}{m}}, \tag{E.9}
\end{aligned}$$

where (a) is due to $\mathbb{E}(2 \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \mathbf{w}) = \mathbf{x}^\top \mathbf{w}$, (b) follows from Lemma C.13, the estimates (B.2) and (B.4), and $m \gtrsim s \log \frac{en}{s}$, and it holds with probability at least $1 - 2 \exp(-c' s \log \frac{en}{s})$. The claim follows.

E.1.3 The Projection Has No Effect on $\mathbf{x} \in \mathcal{X} \setminus \{0\}$

Suppose that $\{\mathbf{x}_t\}_{t=0}^\infty$ is generated by running Algorithm 3 without $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$, then the RAIC and initialization bound guarantees (E.4), which along with $\tau_s(\mathbf{x}) \leq c_* \|\mathbf{x}\|_2$ implies

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq (c + 30c_*) \|\mathbf{x}\|_2 \implies \|\mathbf{x}_t\|_2 \leq (1 + c + 30c_*) \|\mathbf{x}\|_2, \quad \forall t \geq 0, \mathbf{x} \in \mathcal{X} \setminus \{0\}.$$

On the other hand, under $m \gtrsim s \log \frac{en}{s}$,

$$(E.3) \implies \sup_{\mathbf{x} \in \mathcal{X} \setminus \{0\}} \frac{\|\mathbf{x}_0 - \mathbf{x}\|_2}{\|\mathbf{x}\|_2} \leq 4c_* \implies \|\mathbf{x}_0\|_2 \geq (1 - 4c_*) \|\mathbf{x}\|_2, \quad \forall \mathbf{x} \in \mathcal{X} \setminus \{0\}.$$

Thus, so long as c and c_* are small enough,

$$\|\mathbf{x}_t\|_2 \leq 2\|\mathbf{x}_0\|_2, \quad \forall t \geq 0, \mathbf{x} \in \mathcal{X} \setminus \{0\}.$$

This shows that adding the projection $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$ does not affect $\{\mathbf{x}_t\}_{t \geq 0}$ satisfying (E.4), and therefore $\{\mathbf{x}_t\}_{t \geq 0}$ obtained from Algorithm 3 also satisfies (E.4).

E.1.4 A Separate Argument for $\mathbf{x} \in \mathcal{X}^c$

Finally, we handle $\mathbf{x} \in \mathcal{X}^c$ by a different argument based on $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$. For all $\mathbf{x} \in \mathcal{X}^c$, due to the projection, $\{\mathbf{x}_t\}_{t \geq 0}$ (the iterates of Algorithm 3) satisfy

$$\begin{aligned} \|\mathbf{x}_t\|_2 &\leq 2\|\mathbf{x}_0\|_2 = \left\| \frac{4}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i \right\|_{s,2} = \sup_{\mathbf{w} \in \Sigma_s^{n,*}} \frac{4}{m} \sum_{i=1}^m \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \mathbf{w} \\ &\stackrel{(a)}{\leq} \sup_{\mathbf{w} \in \Sigma_s^{n,*}} \frac{4}{m} \sum_{i=1}^m |\mathbf{a}_i^\top \mathbf{x}| |\mathbf{a}_i^\top \mathbf{w}| \stackrel{(b)}{\leq} \frac{4\|\mathbf{A}\mathbf{x}\|_2}{\sqrt{m}} \sup_{\mathbf{w} \in \Sigma_s^{n,*}} \frac{\|\mathbf{A}\mathbf{w}\|_2}{\sqrt{m}} \stackrel{(c)}{\leq} \frac{6\|\mathbf{A}\mathbf{x}\|_2}{\sqrt{m}} \stackrel{(d)}{\leq} 9\left(\frac{1}{c_*} + 1\right) \tau_s(\mathbf{x}), \end{aligned}$$

where (a) is due to triangle inequality, (b) is due to Cauchy–Schwarz inequality, (c) is due to Lemma C.1, and (d) is obtained by using the bound established in (C.44). Therefore, by triangle inequality and the definition of $\mathcal{X}^c$,

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \|\mathbf{x}_t\|_2 + \|\mathbf{x}\|_2 \leq \left(\frac{10}{c_*} + 9\right) \tau_s(\mathbf{x}), \quad \forall t \geq 0, \mathbf{x} \in \mathcal{X}^c,$$

as desired.

E.2 Proof of Theorem 5.2 (Non-Uniform Instance Optimality)

Following the unified framework in Section 2.2.2, the proof consists of the following steps:

1. **(Decomposition of $\mathbb{R}^n$)** Let $\mathcal{X} = \{\mathbf{u} \in \mathbb{R}^n : \delta_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\}$ for some small enough universal constant $c_* > 0$.
2. **(Instance Optimality if $\mathbf{x} \in \mathcal{X}$)** For the fixed $\mathbf{x}$, we establish the RAIC

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \text{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2), \mathbf{x}, \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 10\|\mathbf{x} - \mathbf{x}_{[s]}\|_2\right). \quad (\text{E.10})$$

By the notation in Section 2.2, the RAIC renders $\mu_1 = \frac{1}{4}$, $\mu_2 = 0$, $e(\mathbf{x}) = 10\|\mathbf{x} - \mathbf{x}_{[s]}\|_2$, and $R_{\mathbf{x}}^{\text{loc}} = c\|\mathbf{x}\|_2$, thus (2.9)–(2.11) are satisfied by the $\mathbf{x} \in \mathcal{X}$ as long as c_* is sufficiently small. Then, we reuse the initialization bound from Theorem 5.1 to ensure (2.12), i.e.,

$$\mathbf{x}_0 \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2). \quad (\text{E.11})$$

As such, Theorem 2.1 concludes that Algorithm 3 without $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$ satisfies the desired bound. Finally, we incorporate $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$ by showing that it has no impact.

3. **(Instance Optimality if $\mathbf{x} \in \mathcal{X}^c$)** This case is handled by a different argument.

E.2.1 Proof of the RAIC (E.10) if $\mathbf{x} \in \mathcal{X}$

Following the proof outline in Section 5.3, the main bulk of technical work lies in the proof of the RAIC in (E.10). Particularly, for a fixed nonzero $\mathbf{x} \in \mathcal{X} = \{\mathbf{u} \in \mathbb{R}^n : \delta_s(\mathbf{u}) \leq c_* \|\mathbf{u}\|_2\}$, we seek to show that

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + 10\|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall \mathbf{u} \in \Sigma_s^n \cap \mathbb{B}_2^n(\mathbf{x}; c\|\mathbf{x}\|_2) \quad (\text{E.12})$$

holds for some universal constant $c > 0$. To this end, we reuse (E.5) and (E.6) to obtain

$$\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u} - \mathbf{x}\|_2}{4} + \frac{5\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{4} + \|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2},$$

and the only difference arises in bounding $\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2}$:

$$\|\mathbf{h}_{\mathbf{x}}(\mathbf{u}) - \mathbf{h}_{\mathbf{x}_{[s]}}(\mathbf{u})\|_{2s,2} \stackrel{(a)}{\leq} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{2}{m} \sum_{i=1}^m |\mathbf{a}_i^\top (\mathbf{x} - \mathbf{x}_{[s]})| |\mathbf{a}_i^\top \mathbf{w}| \stackrel{(b)}{\leq} 8\|\mathbf{x} - \mathbf{x}_{[s]}\|_2,$$

where (a) holds due to the first two inequalities in (E.7), (b) is by (C.50) and (C.51). Combining these pieces establishes (E.12), which is exactly the desired RAIC (E.10).

E.2.2 Proof of the Initialization Guarantee (E.11) if $\mathbf{x} \in \mathcal{X}$

Since $\mathbf{x}_0 \in \Sigma_s^n$ by construction, we only need to further ensure $\|\mathbf{x}_0 - \mathbf{x}\|_2 \leq c\|\mathbf{x}\|_2$. For any fixed $\mathbf{x} \in \mathbb{R}^n$ (need not be in $\mathcal{X}$), with probability at least $1 - 2\exp(-cs \log \frac{en}{s})$ we have

$$\begin{aligned} \frac{\|\mathbf{x}_0 - \mathbf{x}\|_2}{\|\mathbf{x}\|_2} &\stackrel{(a)}{\leq} \sup_{\mathbf{w} \in \Sigma_{2s}^{n,*}} \frac{2}{m} \sum_{i=1}^m \left[2 \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \mathbf{w} - \mathbb{E}(2 \text{ReLU}(\mathbf{a}_i^\top \mathbf{x}) \mathbf{a}_i^\top \mathbf{w}) \right] + \frac{3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}\|_2} \\ &\stackrel{(b)}{\leq} C_1 \sqrt{\frac{s \log(en/s)}{m}} + \frac{3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}\|_2}, \end{aligned} \quad (\text{E.13})$$

where (a) is obtained by adapting the argument in Equations (E.8) and (E.9), (b) holds with probability at least $1 - 2\exp(-cs \log \frac{en}{s})$ due to Lemma C.13 (note that $\mathbf{x}$ is fixed). Since $\frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{\|\mathbf{x}\|_2} \leq c_*$ when $\mathbf{x} \in \mathcal{X}$, $\|\mathbf{x}_0 - \mathbf{x}\|_2 \leq c\|\mathbf{x}\|_2$ is satisfied under $m \gtrsim s \log \frac{en}{s}$ and small enough c_* .

E.2.3 The Projection Has No Effect if $\mathbf{x} \in \mathcal{X}$

Suppose that $\{\mathbf{x}_t\}_{t \geq 0}$ is obtained by Algorithm 3 without $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$, then under the established components Theorem 2.1 yields

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \frac{c\|\mathbf{x}\|_2}{2^t} + 46\|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall t \geq 0. \quad (\text{E.14})$$

This implies $\|\mathbf{x}_t\|_2 \leq \|\mathbf{x}_t - \mathbf{x}\|_2 + \|\mathbf{x}\|_2 \leq (1 + c + 46c_*)\|\mathbf{x}\|_2$ for any $t \geq 0$. On the other hand, (2.12) with $R_{\mathbf{x}}^{\text{loc}} = c\|\mathbf{x}\|_2$ yields $\|\mathbf{x}_0\|_2 \geq (1 - c)\|\mathbf{x}\|_2$. Hence, under small enough c and c_* , we have $\|\mathbf{x}_t\|_2 \leq 2\|\mathbf{x}_0\|_2$ for any $t \geq 0$. Hence, adding $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$ has no effect on the iterates. We conclude that (E.14) holds for Algorithm 3.

E.2.4 Separate Argument if $\mathbf{x} \in \mathcal{X}^c$

All that remains is to address the case of $\mathbf{x} \in \mathcal{X}^c$, where we continue to assume the high-probability event (E.13). This event implies

$$\|\mathbf{x}_0\|_2 \leq \|\mathbf{x}\|_2 + \|\mathbf{x} - \mathbf{x}_0\|_2 \leq \left(1 + C_1 \sqrt{\frac{s \log(en/s)}{m}}\right) \|\mathbf{x}\|_2 + 3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq 2\|\mathbf{x}\|_2 + 3\|\mathbf{x} - \mathbf{x}_{[s]}\|_2.$$

Due to $P_{\mathbb{B}_2^n(2\|\mathbf{x}_0\|_2)}$, the iterates of Algorithm 3 satisfy $\|\mathbf{x}_t\|_2 \leq 2\|\mathbf{x}_0\|_2 \leq 4\|\mathbf{x}\|_2 + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2$. Combining with $\|\mathbf{x}\|_2 \leq \frac{\|\mathbf{x} - \mathbf{x}_{[s]}\|_2}{c_*}$, we conclude that

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \|\mathbf{x}_t\|_2 + \|\mathbf{x}\|_2 \leq 5\|\mathbf{x}\|_2 + 6\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq (5c_*^{-1} + 6)\|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall t \geq 0,$$

as desired.